

PARENTHOOD TIMING AND GENDER INEQUALITY

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Abstract

I develop a new methodology delivering the first causal estimates of the separate effects of parenthood and its timing, using quasi-experimental variation in intrauterine insemination (IUI) success. Women whose first IUI fails often undergo further procedures, generating sequences of fertility shocks that affect whether and when they conceive. In Dutch administrative data, motherhood persistently reduces earnings by 18% and hours by 16% relative to childlessness, causing up to half of post-childbirth gender inequality. A 2.3-year delay—though unintended—not only postpones these losses but reduces them in the four years after childbirth: by 12% for earnings and 23% for hours.

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1 Introduction

The differential impact of parenthood on women’s and men’s careers is widely viewed as a key driver of gender inequality in the labor market (Goldin, 2014; Bertrand, 2020; Cortés & Pan, 2023; Olivetti et al., 2024). Assessing this impact is challenging because women choose whether and when to have children, and these decisions may be correlated with career outcomes independent of parenthood. A growing literature addresses this challenge by exploiting quasi-experimental variation in fertility, showing that women’s employment, hours worked, and earnings fall substantially and persistently following childbirth (Lundborg et al., 2017; Gallen et al., 2024; Bensnes et al., 2025).

Yet our understanding of how fertility affects women’s careers remains incomplete because existing evidence does not distinguish between two margins — whether women become mothers and when they do so. Existing studies typically compare women who conceive following a plausibly random fertility shock, such as successful in vitro fertilization or contraceptive failure, with those who do not. The latter group includes both women who remain childless and women who become mothers later. These estimates therefore reflect a weighted average of two effects: the effect of motherhood relative to childlessness and the effect of earlier relative to later motherhood.

The timing of motherhood can itself affect women’s careers, but the direction of this effect is theoretically ambiguous. Postponing childbearing mechanically eliminates motherhood-related losses during the delay and may reduce them after childbirth, since women enter motherhood with more experience and better jobs. Yet it may also shift the career interruption to a stage when earnings and promotion potential are greatest, raising its cost. Separately quantifying the effects of becoming a mother and of becoming one earlier versus later is therefore essential for understanding fertility decisions and designing policies that influence whether and when women have children.

This paper provides the first causal evidence on the separate effects of motherhood and its timing. To do so, I develop a new methodology that disentangles these effects and apply it to a common but previously unstudied source of fertility variation: success in intrauterine insemination (IUI). The approach exploits the fact that women whose first IUI attempt fails often try again. By leveraging variation across the entire sequence of IUI attempts, the method separately quantifies the effect of motherhood relative to childlessness and the effect of later relative to earlier motherhood. I apply the method to Dutch administrative data. I first show that success at both initial and subsequent IUI attempts is uncorrelated with prior labor-market outcomes, conditional on age, supporting the assumption that the resulting fertility variation is plausibly random. Using this variation, I find that motherhood substantially and persistently reduces women’s earnings and work hours, while later childbearing—even when unintended—mitigates these losses. My results show that both the incidence and timing of motherhood shape gender inequality. They indicate that women face two distinct career trade-offs: whether to become mothers and when to do so.

The method builds on the idea that success at each IUI attempt contains a random component. Ideally, this repeated variation would help recover average outcomes under motherhood beginning at the first attempt, motherhood beginning later, and childlessness, allowing the incidence and timing effects to be assessed separately. The obstacle is selection, which operates along two mar-

gins: women choose how many attempts to pursue, and some conceive naturally regardless of their IUI outcomes. Both margins may be correlated with career outcomes independently of parenthood. Consequently, directly comparing women with different IUI and conception histories would reintroduce the selection problem that random attempt success is meant to address.

To overcome this challenge, I classify women into latent types along the two margins of selection: how many attempts they would pursue if all failed, and whether they would eventually become mothers regardless of IUI success. I first show that women who remain childless after a given IUI sequence reveal the average childless outcome for their type, while continuation and natural-conception rates following each failure identify the distribution of types. Together, these objects recover the average childless outcome for women whose motherhood depends on eventual IUI success. I next show that their average outcome under motherhood is not point identified without additional assumptions, but can be bounded using the outcomes of women whose first attempt succeeds and the group’s population share. This yields bounds on the effect of motherhood relative to childlessness, uncontaminated by timing effects. Finally, a symmetric argument applied to women who conceive naturally after their IUI attempts fail delivers bounds on the effect of conceiving after the failed sequence rather than at the first attempt—the effect of later relative to earlier motherhood.

The approach requires only that success at each attempt be as good as random, conditional on observables. Crucially, it imposes no restrictions on heterogeneity across women or over time, selection into additional attempts, or natural conception. I show that the resulting bounds are sharp and that intuitive additional assumptions—with testable implications—can progressively narrow them, up to point identification.

Inference for the bounds is complicated by the fact that their construction selects observations from the tails of the outcome distribution, which requires nonparametric estimation of a quantile function. To address this, I propose orthogonal moment functions that eliminate first-order sensitivity to estimation error in this function, justifying standard asymptotic inference.

My analysis focuses on opposite-sex couples who undergo IUI for their first child. The procedure is typically used as a first-line treatment for couples experiencing difficulties conceiving naturally, particularly when fertility issues are male-factor-related or unexplained. By placing sperm directly into the uterus using a flexible catheter, it closely mimics natural conception while remaining minimally invasive, quick, and generally painless.

While IUI users are a selected clinical population, the effects of parenthood in this group are likely informative about the broader population of parents who intended to have children. Most IUI users are otherwise healthy and receive treatment for unexplained or mild fertility impairments, of which they are typically unaware before attempting to conceive. This means that IUI users effectively come from a larger group of families who would have undergone treatment had they experienced sufficient difficulty conceiving. In Section 4.3, I use fecundity rates from the medical literature to obtain a lower bound on the size of this group and estimate that potential IUI users account for at least one quarter of all parents.

The analysis proceeds in five steps. First, I assess the assumption that IUI success is as good as random. Because success cannot be directly manipulated, the main threat is systematic health differences across women. Because health differences relevant to career outcomes should be reflected in labor-market trajectories, outcomes measured before each attempt provide a test. I find that success at both the first and subsequent IUI attempts is uncorrelated with a rich set of pre-IUI labor-market outcomes, conditional on age. I complement this with a sensitivity analysis in which, among women who conceive at a given attempt, those with better subsequent labor-market outcomes are assumed to have had systematically higher ex ante success probabilities. I find that selection strong enough to generate meaningful imbalance in post-IUI outcomes is inconsistent with the observed balance in pre-IUI outcomes.

Second, I estimate how parenthood affects labor market outcomes and contributes to gender inequality. When motherhood begins at the first IUI attempt, women’s annual work hours decline by 12%–19% and their earnings by 13%–23%, relative to remaining childless; these effects are stable and persistent over at least six years. For men, the bounds are of comparable width but centered near zero, ruling out even small negative effects. Taken together, parenthood accounts for 35%–53% of the gender gap in work hours and 21%–45% of the gap in earnings among parents.

Third, I estimate the effects of parenthood timing. Women whose IUI sequence fails conceive, on average, 2.3 years later than under first-procedure success. Over the following six years, they work 8%–14% more hours and earn 5%–14% more than under first-procedure conception. I decompose this effect into two margins. The first operates while later mothers remain childless and earlier mothers have children, so delay avoids motherhood-related labor-market losses entirely. The second operates once both groups have children and captures the effect of having entered motherhood later. Although the second margin fades over time, both contribute similarly to the overall effect.

Fourth, I investigate heterogeneity and the role of the intensive fertility margin. Delaying motherhood increases work hours most for women with lower pre-IUI earnings, women who earn less than their partner, and younger women. Parenthood effects are also somewhat larger for these groups, though not significantly so. While I also find that women who conceive later have fewer children, fertility responses vary little along dimensions where labor-market responses differ. To assess the intensive margin further, I use couples undergoing treatment for a second child and exploit multiple births at first parity in both the IUI and representative samples. Overall, the evidence suggests that the intensive fertility margin is of limited importance in shaping mothers’ labor-market outcomes. Taken together, these patterns are consistent with the gains from postponement arising primarily from additional time to establish a career before motherhood.

Lastly, I assess the sensitivity of my estimates to potential selection on unobservables. First, I focus on differences in treatment-related behavior that may be associated with both IUI success and labor-market outcomes. I add detailed procedure-level protocol controls that capture clinical differences and may proxy for other conception-related health behaviors. The estimates are virtually unchanged. Second, I use my estimates of the extent of selection on unobservables consistent with the observed balance to adjust realized outcomes so as to restore balance between success and

failure groups, and re-estimate the effects. The estimates remain similar. Finally, I emphasize that if lower success probabilities reflect worse health that also worsens labor-market outcomes, any remaining selection makes the estimated benefits of postponement conservative, because later-conceiving women are negatively selected relative to those who conceive earlier.

I conclude by discussing the broader implications of my results for the role of parenthood in the labor market, especially for women deciding early in their careers whether and when to have children. A central consideration is that, for women whose IUI attempts fail, both remaining childless and becoming mothers later are unplanned outcomes. My estimates may therefore further represent a lower bound on the gains from planned childlessness or delay. This may occur both because women cannot fully optimize their careers in anticipation of these outcomes (see [Bensnes et al., 2025](#), for a discussion) and because unmet childbearing goals may impose a psychological toll (see [Bögl et al., 2024](#), for a discussion). To assess severe manifestations of the latter channel, I examine antidepressant use and find that it can explain little of the estimated labor-market effects. I end with a suggestive calculation of the extent to which postponing childbearing can mitigate the career costs of motherhood. I find that a 2.3-year delay not only postpones the losses but also reduces them over the first four years after childbirth by 12% for earnings and 23% for work hours.

My work builds on studies that estimate the effects of parenthood using quasi-experimental variation from infertility, miscarriages, success in in vitro fertilization, or contraceptive failure ([Hotz et al., 2005](#); [Agüero & Marks, 2008](#); [Cristia, 2008](#); [Miller, 2011](#); [Lundborg et al., 2017, 2024](#); [Gallen et al., 2024](#); [Bensnes et al., 2025](#)). These studies compare women who give birth at a particular time for plausibly random reasons with women who do not, many of whom become mothers later. They report reduced-form effects that capture a weighted average of parenthood and timing effects, or isolate parenthood effects under restrictions on the role of timing. My approach extends this work by identifying parenthood effects without restricting how they vary with timing, and by quantifying timing effects directly.

There is little causal evidence on the effects of parenthood timing. [Bíró et al. \(2019\)](#) compare women who give birth with women who miscarry and subsequently give birth, addressing selection into later childbirth by balancing observables. [Lundborg et al. \(2017\)](#) and [Gallen et al. \(2024\)](#) compare childbirth effects across groups that differ in the age at which they undergo in vitro fertilization or begin using contraception. These comparisons, however, cannot separate timing effects from heterogeneity in fertility intentions across groups. I provide the first causal estimates of the effect of parenthood timing that address selection on unobservables.

I also contribute evidence from a new treatment population and institutional setting. The closest studies exploit IVF success in Scandinavia ([Lundborg et al., 2017, 2024](#); [Gallen et al., 2024](#); [Bensnes et al., 2025](#)). I focus on IUI, a minimally invasive treatment used in cases of mild infertility, and use data from the Netherlands, where parental leave and childcare use are closer to OECD averages. Whereas Scandinavian IVF studies generally find modest earnings losses, I estimate larger and more persistent declines, similar in magnitude to those found by [Gallen et al. \(2024\)](#) following unplanned births after contraceptive failure. This difference is consistent with

family-friendly policies mitigating the career costs of motherhood.

My work also contributes to the literature on dynamic treatments. In biostatistics, numerous methods have been developed to evaluate sequential experiments under full compliance (see [Hernán & Robins, 2020](#), for an overview). In economics, [Van den Berg & Vikström \(2022\)](#) study settings where treatment is assigned dynamically among eligible individuals who selectively exit eligibility, while [Heckman et al. \(2016\)](#) and [Han \(2021\)](#) allow treatment to be obtained outside assignment, provided instruments exist for each treatment margin. I contribute a method for settings where individuals selectively undergo additional assignments and may obtain treatment outside assignment without valid instruments—a structure shared by education admission lotteries, job training programs with partially randomized access, and clinical trials with imperfect compliance.

My work also relates to the literature on treatment-effect bounds, beginning with [Manski \(1989, 1990\)](#); [Horowitz & Manski \(1995\)](#), and in particular to the methods of [Zhang & Rubin \(2003\)](#) and [Lee \(2009\)](#). Bounding methods can be applied to virtually any treatment-effect setting, but a central drawback is that the resulting bounds may be economically uninformative. I contribute by showing how bounding techniques can be combined with sequential quasi-experimental variation to obtain bounds that are strictly narrower and valid for a broader population, without additional assumptions. This improvement is crucial in application: existing methods adapted to separate treatment and timing effects yield bounds for my main outcomes that are several times wider and fail to rule out large positive or negative effects.

My work further relates to the literature on dynamic non-compliance with one-time instruments ([Cellini et al., 2010](#); [Ferman & Tecchio, 2023](#); [Angrist et al., 2024](#); [Gallen et al., 2024](#); [Bensnes et al., 2025](#)), which exploits initial assignment and repeated outcome measurements to identify treatment effects when treatment duration is central. These approaches require restrictions linking effects for initially- and later-treated individuals, typically equal average effects conditional on treatment duration. Such restrictions may be particularly strong for parenthood, where later mothers may differ systematically from earlier mothers and births at different ages may have different career and fertility consequences (see [Gallen et al., 2024](#); [Bensnes et al., 2025](#), for detailed discussions). My approach instead delivers sharp bounds on parenthood effects under unrestricted heterogeneity across women and conception timings, while separately identifying the timing effect itself.

Finally, I contribute to the broader literature on the consequences of fertility. This includes studies exploiting quasi-experimental variation in the intensive margin, ([Rosenzweig & Wolpin, 1980](#); [Angrist & Evans, 1998](#); [Bronars & Grogger, 1994](#); [Black et al., 2005](#)); studies comparing mothers with fathers or with women who have not yet had children ([Fitzenberger et al., 2013](#); [Angelov et al., 2016](#); [Chung et al., 2017](#); [Bütikofer et al., 2018](#); [Kleven, Landais, & Søgaaard, 2019](#); [Kleven, Landais, Posch, et al., 2019](#); [Eichmeyer & Kent, 2022](#); [Kleven et al., 2024](#); [Melentyeva & Riedel, 2023](#); [Adams et al., 2024](#)); and structural analyses of fertility and career choices ([Adda et al., 2017](#); [Jakobsen et al., 2022](#)). The causal evidence I provide on how motherhood timing affects fertility and labor market outcomes helps explain differences among women who enter motherhood at different life-cycle stages.

The remainder of the paper is structured as follows. Section 2 introduces a framework for sequential quasi-experiments. Section 3 discusses existing methods, develops the intuition for my econometric approach, states the formal results, outlines estimation, and introduces a procedure for assessing selection on unobservables. Section 4 describes the institutional setting, intrauterine insemination, and the data, and presents evidence supporting the identifying assumptions. Section 5 presents the results. Section 6 discusses generalizability and life-cycle implications. Section 7 concludes.

2 A Framework for Sequential Quasi-experiments

My framework generalizes the local average treatment effect (LATE) framework (Angrist & Imbens, 1995) by making explicit how a treatment—parenthood—depends not only on an initial treatment assignment, such as the success of the first IUI procedure, but also on subsequent decisions to pursue additional assignments and on their outcomes.

Since the analysis involves timing effects and sequential decisions, the framework could be formulated dynamically. For clarity, I instead use a cross-sectional representation that summarizes the relevant events and decisions within a fixed observation window. Appendix SA1 formally maps the dynamic framework into this representation and shows that there is no loss of generality. Section 2.1 presents the framework, and Section 2.2 cover interpretation details.

2.1 Setup

I observe each woman over a fixed window beginning at her first IUI attempt for a first child; in the main analysis, the window spans seven years. All events considered below—IUI attempts, their outcomes, and the onset of motherhood—occur within this period, and all latent variables are defined with respect to it. All variables are woman-specific, and I suppress individual subscripts throughout. Expectations and probabilities are taken over the population of women.

Each woman is described by two latent variables. First, $W \in \{1, \dots, \bar{w}\}$ is the number of IUIs she would have undergone for her first child within the window had every attempt failed, with upper bound $\bar{w}$; I refer to W as *willingness* to undergo IUI. Second, $R \in \{0, 1\}$ indicates whether she would have remained childless throughout the window had all W IUIs failed; I refer to R as *reliance* on IUI. Women with $R = 1$ (*reliers*) rely on IUI to become mothers, while women with $R = 0$ (*non-reliers*) would have become mothers even if all their IUIs had failed. The timing of IUI attempts and non-IUI motherhood may vary across women. Non-IUI motherhood may include adoption, but it is almost nonexistent in the data.

Potential outcomes summarize a woman’s earnings or work hours over the window. I define three: Y^E is her outcome if the first IUI succeeds (the *early motherhood* outcome), Y^L is her outcome if the first IUI fails but she becomes a mother within the window (the *later motherhood* outcome), and Y^N is her outcome if she remains childless throughout the window (the *childless* outcome). Depending on the analysis, I consider either outcomes cumulated over the entire window, such as total earnings over the period, or contemporaneous outcomes measured at a given horizon relative to the first IUI attempt, such as earnings in the third year after the attempt.

Y^L is not explicitly indexed by the timing of motherhood, although different conception times may correspond to different values of Y^L ; the framework places no restrictions on how Y^L varies with timing. Throughout the analysis, I focus on the case in which motherhood occurs through non-IUI conception after the full IUI sequence has failed. When Y^L refers to a different conception time, I make this explicit. I discuss the timing interpretation in more detail in the next section.

The first parameter of interest is the *effect of parenthood*, defined as the difference between conceiving earlier and remaining childless: $\tau = Y^E - Y^N$. The second is the *effect of parenthood timing*, the effect of later relative to earlier motherhood: $\delta = Y^L - Y^E$. Both τ and δ are defined at the individual level and may vary arbitrarily across women. The two main effects I focus on are the average treatment effect among reliers, $\tau_{ATR} = \mathbb{E}[\tau \mid R = 1]$, and the average timing effect among non-reliers, $\delta_{ANR} = \mathbb{E}[\delta \mid R = 0]$. The two estimands reflect the two margins along which IUI success induces variation. For reliers, who remain childless after the IUI sequence fails, success affects the incidence of motherhood. For non-reliers, who conceive even when the IUI sequence fails, success shifts the timing of motherhood.

The observed indicator for the success of the j th IUI for the first child is Z_j , which equals 1 if the procedure succeeded and 0 if it failed or was not undertaken. The first key variable linking observed and latent variables is the realized number of IUIs, $A = \min(\{j : Z_j = 1\} \cup \{W\})$: a woman undergoes IUIs until one succeeds or until she reaches her willingness. The outcome of the last attempt is Z_A . Willingness is thus observed exactly when the sequence ends in failure, in which case $A = W$; when an attempt succeeds, W is only known to satisfy $W \geq A$. This censoring structure is central to the identification argument in Section 3.

The second key variable is the parenthood (*treatment*) indicator $D = Z_A + (1 - Z_A)(1 - R)$: a woman becomes a mother if an IUI succeeds or if she is a non-relier, who conceives independently of IUI success. Similar to willingness, reliance is only observed when the sequence ends in failure, in which case $R = 1 - D$. A woman's realized labor market outcome is $Y = Z_1 Y^E + (1 - Z_1)[D Y^L + (1 - D) Y^N]$, and it depends on whether she conceived at the first IUI, later, or not at all.

After introducing the core idea, I also leverage information on non-IUI births among women who conceive their first child via IUI. For these women, $R^+ \in \{0, 1\}$ is a latent indicator of whether a woman relies on IUI for all additional children after conceiving her first child via IUI. Women with $R^+ = 1$ (*subsequent reliers*) would have only IUI-conceived children after conceiving their first via IUI, whereas those with $R^+ = 0$ would also have one or more non-IUI children.

The indicator for having at least one non-IUI child is $D^+ = Z_A(1 - R^+) + (1 - Z_A)D$: a woman has at least one such child if an IUI succeeded and she is not a subsequent relier, or if all IUIs failed and she became a mother regardless.

2.2 Framework Interpretation and Discussion

In this subsection, I discuss the interpretation of reliance and willingness, as well as the treatment and timing effects, and connect my framework to the LATE framework.

2.2.1 Interpreting Willingness and Reliance

Willingness and reliance are counterfactual summary statistics that capture all endogeneity in the realized number of IUIs, and in fertility outside observed IUI, that is relevant for identification. Willingness reflects how many IUIs a woman would undergo if all attempts failed, while reliance reflects whether she would remain childless under that same counterfactual, regardless of how these outcomes arise. Because this counterfactual path may end through non-IUI conception, willingness counts the IUIs completed along the path rather than a fixed maximum number of procedures a woman is prepared to undergo.

Willingness and reliance do not describe every dimension of fertility behavior. For example, two women with the same willingness and reliance may start the procedure sequence at different ages or progress through subsequent procedures at different paces. The framework leaves these dimensions implicit and unrestricted, but they matter for identification because IUI success probabilities depend on age. Section 3.1 introduces the identifying assumption and explains how the analysis accounts for age throughout the procedure sequence. Women may likewise conceive outside IUI at different ages if the sequence fails, which affects the interpretation of timing effects; I discuss this in Section 2.2.2.

Willingness and reliance can be given an economic interpretation, but they may also be treated as statistical constructs. Reliance, for instance, may reflect a woman’s decision to attempt natural conception alongside IUI and hence her effort to conceive naturally. Alternatively, all women undergoing IUI may attempt natural conception simultaneously, with success largely random. The two variables also impose no particular information structure. Willingness and reliance may reflect plans made under perfect foresight, or emerge from a dynamic process in which women update their behavior in response to earlier procedures, medical advice, career changes, or other information. The identification results accommodate all of these possibilities.

The interpretation of reliance matters for the parenthood and timing effects. If reliance mainly reflects effort to conceive naturally, the estimated parenthood effect pertains to women less likely to conceive outside IUI, whereas the timing effect pertains to women more likely to do so. If reliance is instead largely driven by random success in non-IUI conception, reliers and non-reliers need not differ systematically, making the two sets of estimates more directly comparable. Willingness, by contrast, is averaged over in the main estimands, so heterogeneity in willingness is less consequential for their interpretation.

Most analyses in this paper estimate effects separately for reliers and non-reliers and therefore do not rely on comparisons between the two groups; where such comparisons are made, I state the assumptions explicitly. This matters because the groups do differ on observables: Section 4.4 characterizes their pre-IUI characteristics and shows that non-reliers are younger, more educated, and have higher employment and earnings.

Two measurement details are important for interpreting willingness and reliance. The first concerns the measurement of IUI procedures. The design exploits variation in IUI success that is plausibly exogenous only once sperm has been injected into the uterus. Willingness is defined accordingly: it counts only the counterfactual injections that would have occurred had all previous

attempts failed, rather than procedures that would have been intended or scheduled but never carried out. The data therefore measure completed procedures.

The second concerns the definition of IUI births. Some IUIs are unobserved, for example when performed abroad. Births that cannot be linked to an observed IUI are therefore treated as non-IUI births, and the corresponding women as non-reliers. This does not undermine identification, since the method relies only on randomness in observed IUI success and allows arbitrary selection into all other conceptions; it only widens the treatment effect bounds relative to a setting with complete information on IUI use.

2.2.2 Interpreting Treatment and Timing Effects

In line with existing work, motherhood is an absorbing state, so subsequent births are included in both the early- and later-motherhood states. Their number may differ across states, making the intensive fertility margin one potential mechanism behind timing effects. Section 5.1 presents effects on total fertility, while Section 5.5 assesses the role of subsequent births. These analyses indicate that the intensive margin contributes little to the estimated labor market effects.

The timing effect is the total effect of postponing entry into motherhood, not the partial effect of maternal age holding the rest of the life course fixed. When childbirth occurs later, a woman enters motherhood at a later career stage and, at any subsequent age, has a younger first child. These changes cannot be separated: age at first birth, child age, and the mother’s current age are linked by an identity. There is therefore no feasible intervention that varies birth timing while holding fixed both career stage and child age. The estimand captures this joint change, which is also the object relevant for women deciding whether to postpone motherhood and for policies that shift the age at which women have children.

The conception dates associated with a woman’s potential outcomes are shaped partly by endogenous decisions. Whether and when she becomes a mother depends on IUI success, her decisions about treatment continuation, and conception outside the IUI process. The framework exploits plausibly exogenous variation in IUI success while placing no restrictions on the selectivity of the other two. For example, if a woman undergoes her first IUI at age 25, her early motherhood outcome corresponds to conception at age 25. If that attempt fails, she decides whether to continue with IUI and whether to attempt conception outside IUI. If subsequent IUIs also fail, she may remain childless or eventually conceive without IUI, say at age 30. That potentially endogenous conception date defines her later motherhood outcome. For this woman, the variation induced by IUI failure is therefore the shift from conception at age 25 to conception at age 30.

The delays induced by IUI failure may also be heterogeneous across women. For example, two women may both undergo their first IUI at age 25 and a series of unsuccessful IUIs at identical ages, yet one may conceive naturally at age 27 and the other at age 30. The timing effect therefore averages the effects of these heterogeneous delays, with each delay weighted by the share of non-reliers who experience it. These weights are identified: among women whose IUI sequence fails, the realized delay is the difference between the first IUI date and the subsequent non-IUI conception date, which recovers the full distribution. What is not observed is the counterfactual delay when

the first IUI succeeds, so effects cannot be estimated separately by delay length.¹

I consider three dimensions of timing effects. The first is cumulative: how does postponement affect total earnings or hours worked over the observation window, or over the life cycle? The second asks whether the effects arise only while motherhood is delayed or persist after the delayed birth. For example, when comparing motherhood at ages 25 and 30, one can compare outcomes between ages 25 and 30, when only the early-birth woman is a mother, and after age 30, when both are mothers. The third compares effects over a fixed number of years after each woman’s own first birth. I focus on the first two in the main analysis; the third requires comparisons between reliers and non-reliers, and Section 6 discusses the required assumptions and presents the results.

The heterogeneity in delays complicates the second comparison. In the baseline analysis, I use the average delay of 2.3 years to define the periods before and after the later birth. This creates some contamination: some later-birth women have already become mothers during the first period, while others remain childless during part of the second. If parenthood weakly reduces women’s labor market outcomes, this biases the first-period effect downward and the second-period effect upward. To assess the importance of this issue, I complement the baseline decomposition with an adaptation of the identification approach that aligns fertility states across groups by defining reliance at a particular horizon rather than over the full window. Section 5.3 and Appendix SA2.1 provide details and show that this adjustment leaves the results largely unchanged.

2.2.3 Relation to the Local Average Treatment Effects Framework

The willingness–reliance framework is closely related to the local average treatment effects framework, which separates treatment take-up into selective and quasi-random components. In standard potential-outcome notation, let $Y_z(d)$ denote the outcome under first-IUI success status z and parenthood status d , so that $Y_1(1) = Y^E$, $Y_0(1) = Y^L$, and $Y_0(0) = Y^N$. The remaining cell $Y_1(0)$ is empty, since a woman whose first IUI succeeds necessarily becomes a mother; the IUI setting therefore contains only compliers—who would remain childless if the first IUI failed—and always-takers—who would become mothers even if the first IUI failed.

Connecting the two frameworks requires specifying not only how many subsequent attempts a woman would undergo (willingness) and whether she would conceive outside IUI (reliance), but also the counterfactual outcomes of those subsequent attempts. Let $Z_j(0)$ denote the success of the j th IUI in the counterfactual in which all previous attempts fail. Compliance with respect to the first IUI then satisfies $C = R \prod_{j=2}^W (1 - Z_j(0))$. A woman is therefore a complier if she is a relier and every subsequent IUI she would undertake after failure of the first also fails. Compliers are thus a subset of reliers; reliers additionally include women who would remain childless if all IUIs failed but would conceive through a subsequent IUI. The parenthood effects identified by the

¹This is also why the framework does not index potential outcomes by conception age; Appendix SA1 shows that doing so yields no additional identifiable parameters. Identifying effects of specific conception timings would require exogenous variation between precisely defined conception ages, which quasi-experimental fertility designs generally do not provide because subsequent conception remains partly endogenous.

willingness–reliance framework therefore cover a broader population than the standard LATE.

A simple example clarifies the distinction. Suppose all women would conceive only through IUI and would undergo exactly one additional procedure if the first failed. All women are then reliers, but only those whose second IUI would fail are compliers with respect to first-IUI success; those whose second IUI would succeed are always-takers. Compliance status therefore combines potentially selective components—willingness and conception outside IUI—with later IUI outcomes that may themselves be quasi-random. The willingness–reliance framework keeps these components separate, allowing variation in later IUI success to contribute to identification.

There are two natural ways to define compliance in a dynamic setting, corresponding to different treatment definitions and identifying assumptions. The first defines compliance over the full data window and is suited to cumulative outcomes when the effect of motherhood does not depend on when within the window it begins. The second defines compliance at a particular horizon and is suited to contemporaneous outcomes under the assumptions that earlier births have the same effect regardless of their exact timing and later births have no anticipatory effects; I discuss these assumptions in Section 3.2. Reliance can be defined analogously at a particular horizon, which I use to separate timing effects before and after the later birth.

3 Econometric Approach

Section 3.1 introduces the central identification assumption. Section 3.2 describes the limitations of conventional methods. Section 3.3 presents the intuition behind my approach. Section 3.4 formalizes the procedure. Section 3.5 outlines estimation. Section 3.6 introduces procedure to assess selection on unobservables.

3.1 Local sequential unconfoundedness

To formalize the intuition underlying the approach, I introduce the following assumption.

Assumption 1 (Local sequential unconfoundedness). $(Y^m, R, W) \perp\!\!\!\perp Z_j \mid A \geq j$, for all m, j .

This assumption states that, among women who undergo IUI attempt j —that is, who actually have sperm inserted into the uterus—the success of that attempt is effectively random: independent of potential outcomes and individual type. This aligns with the standard unconfoundedness assumption used in studies exploiting IVF, where pregnancy conditional on embryo transfer is assumed to be random. Unlike in those settings, however, the assumption here applies not only to the first procedure but also to all subsequent attempts.

To simplify exposition, I abstract from covariates. The main method in Section 3.4 allows IUI success rates to vary flexibly with observables, most importantly age at each procedure. It can also accommodate additional covariates, such as calendar time, time since the previous attempt, education, and current labor market outcomes, either at the time of the particular procedure or over their full histories across previous attempts. These covariates may themselves be endogenous. For instance, women may choose the age at which they initiate each IUI. Success rates are also allowed to differ across attempts for both medical and behavioral reasons.

Crucially, each quasi-experiment is defined among realized inseminations. Assumption 1 concerns only the success of procedure j (Z_j), conditional on that procedure—i.e., insertion of sperm into the uterus—occurring ($A \geq j$). Everything leading up to it—whether a woman seeks subsequent procedures, how long she waits after a failure, and what she does to conceive in between—may depend on observed and unobserved determinants of outcomes. Consequently, women who reach later attempts may form a selected sample; for example, career potential may be systematically related to treatment continuation. The assumption requires only that, among women who undergo attempt j , whether that insemination succeeds is unrelated to their potential outcomes.

Decisions to continue IUI or attempt natural conception may also respond to observable and unobservable shocks realized between attempts that affect labor market outcomes. Women may postpone or cancel planned treatment following a health shock, a career change, adverse experiences with earlier procedures, or planned or unplanned natural conception. Importantly, the assumption requires only that, among women who selectively undergo a given procedure at a given age, IUI success at that attempt is independent of potential outcomes and shocks to future success rates.

In line with the literature, Assumption 1 is stated as joint independence for expositional simplicity, but identification allows correlation between type and the factors determining IUI success, provided those factors are unrelated to potential labor market outcomes. Sterility is a leading example. It mechanically affects conception probabilities and is therefore correlated with reliance, while learning about it may also affect treatment continuation. Yet sterility need not otherwise affect women’s health or labor market outcomes, and women are often unaware of it before attempting to conceive. In that case, these correlations do not threaten identification. It is worth emphasizing that this feature is inherent in quasi-experimental fertility designs, where women with underlying sterility are disproportionately represented among those who remain childless.

In contrast, many factors may jointly shape continuation decisions, conception outside IUI, and labor market outcomes without affecting the success of individual IUI attempts. Work-family preferences provide a clear example. They may influence how many attempts a woman is prepared to undergo, how intensively she tries to conceive outside IUI, and her labor market outcomes, while remaining unrelated to whether a given insemination succeeds. The willingness–reliance framework accommodates precisely this form of selection, allowing fertility choices to be correlated with potential labor market outcomes.

Identification is thus threatened primarily by factors related to both per-attempt success probabilities and potential labor market outcomes. Although the assumption is not directly testable, medical evidence suggests that many natural candidates are only weakly associated with IUI success. For instance, studies of psychological factors, such as stress and anxiety (Syam et al., 2017; Guan et al., 2021), and lifestyle factors, such as body mass index and obesity (Kan et al., 2020; Kim et al., 2023), generally find little association with IUI success. Even for smoking, a well-established predictor of fertility, the evidence is mixed, with studies often finding only small negative associations or no effect on IUI outcomes (see ASRM Practice Committee, 2024, for an overview). The main remaining concern is therefore that success may be correlated with latent health differences

that also shape labor market outcomes.

I provide three pieces of evidence in support of the assumption. First, if IUI success probabilities were systematically related to labor market potential, these differences might also appear in pre-IUI outcomes. In Section 4.6, I find little imbalance between successes and failures at either the first or later procedures, suggesting that success remains unrelated to labor market potential throughout the sequence. This also speaks directly to sterility: women with lower success probabilities are mechanically overrepresented among failures, yet their pre-IUI labor market outcomes are similar, suggesting that underlying sterility is not systematically associated with labor market performance.

Second, Section 3.6 develops an approach to quantify the extent of selection on unobservables that may remain despite these balance results. Applying it in Section 4.6, I find little scope for such selection. Section 5.6 then uses these results to assess the sensitivity of the main estimates to violations of the assumption and finds that the estimates change little.

Third, I investigate whether women who continue to subsequent procedures differ systematically in their likelihood of success. If women who are generally healthier are both more likely to succeed and more likely to continue treatment, success rates may change across attempts as the sample becomes increasingly selected. Instead, Section 4.6 shows that age-conditional per-attempt success rates remain stable across the sequence, providing little evidence of such sorting.

In sensitivity analysis in Section 5.6, I further relax the assumption by conditioning on clinical protocol details at each procedure, including semen testing and preparation, ovulation monitoring, and hormone monitoring and use. Success then need only be as good as random among women receiving the same protocol at the same attempt. The results are virtually unchanged.

Finally, even if some unobserved selection due to latent health differences remain, the direction of the resulting bias is predictable. If women whose IUIs succeed are positively selected on latent health, those whose IUIs fail are negatively selected, and if latent health is positively correlated with labor market outcomes, estimates will understate both the cost of parenthood and the benefit of delaying motherhood. This is particularly relevant for the timing effects: under selection on unobserved health, the estimated gains from delay would provide a lower bound on the true effects.

3.2 Limitations of Standard Methods

The standard IV approach uses the success of a woman’s first IUI as an instrument for parenthood. First, consider a setup in which parenthood status is measured at the end of the observation window, as in my analysis, and outcomes capture cumulative earnings over the full window. The reduced form is then the difference in average outcomes between women whose first IUI succeeds and those whose first IUI fails:

$$\lambda_{RF} = \mathbb{E}[Y \mid Z_1 = 1] - \mathbb{E}[Y \mid Z_1 = 0].$$

The first group conceives at the first IUI, whereas the second contains both women who remain childless (compliers) and women who become mothers later (always-takers). Following Angrist & Imbens (1995), under unconfoundedness the reduced form identifies a weighted combination of the

average parenthood effect for compliers and the average timing effect for always-takers:

$$\lambda_{RF} = \mathbb{E}[Y^E - Y^N \mid C = 1] \Pr(C = 1) + \mathbb{E}[Y^E - Y^L \mid C = 0] \Pr(C = 0) \quad (1)$$

$$= \mathbb{E}[\tau \mid C = 1] \Pr(C = 1) - \mathbb{E}[\delta \mid C = 0] \Pr(C = 0). \quad (2)$$

Scaling by the first stage—the difference in the share of mothers between the two groups, which identifies the complier share—gives

$$\frac{\mathbb{E}[Y \mid Z_1 = 1] - \mathbb{E}[Y \mid Z_1 = 0]}{\mathbb{E}[D \mid Z_1 = 1] - \mathbb{E}[D \mid Z_1 = 0]} = \mathbb{E}[\tau \mid C = 1] - \mathbb{E}[\delta \mid C = 0] \frac{\Pr(C = 0)}{\Pr(C = 1)}. \quad (3)$$

The standard IV exclusion restriction requires that the instrument have no effect among always-takers. In this setting, this means that outcomes do not depend on the timing of motherhood, so that $\delta = 0$. Otherwise, the second term in (3) contaminates the IV estimand.

The direction of this bias is ambiguous. Career costs may be underestimated when younger children require more care and delayed motherhood coincides with women’s peak career years ($Y^L < Y^E$), and overestimated if early motherhood generates lasting career setbacks ($Y^L > Y^E$). Its magnitude also depends on the scaling factor $\Pr(C = 0)/\Pr(C = 1)$. In the context of IUI (and IVF; [Lundborg et al., 2017](#)), around 75% of women whose first procedure fails eventually have children, implying $\Pr(C = 0) = 0.75$ and $\Pr(C = 0)/\Pr(C = 1) = 3$. Even relatively small timing effects can therefore generate substantial bias.²

An alternative is to define parenthood status and compliance at a particular horizon within the observation window. Suppose the econometrician observes two periods, indexed by j , with Y_j and D_j the outcome and motherhood status in period j . Assume that no woman whose first IUI fails becomes a mother in the first period, so that $D_1 = Z_1$, and let C_2 indicate whether a woman would remain childless through the second period if her first IUI failed. The approach also imposes no anticipation: before becoming a mother, later-motherhood and childless potential outcomes are equal, $Y_1^L = Y_1^N$.

In the first period every woman with $Z_1 = 0$ is childless, so the reduced form identifies the effect of one period of motherhood. In the second period women with $Z_1 = 0$ are a mixture—those with $C_2 = 1$ remain childless, those with $C_2 = 0$ have become mothers—so the reduced form again combines the parenthood effect for women who would otherwise remain childless with the timing effect for those who become mothers later. The exclusion restriction now requires $Y_2^E = Y_2^L$, so that once motherhood begins, outcomes do not depend on when it began.

One way to address this contamination is to account for parenthood duration. [Gallen et al. \(2024\)](#) and [Bensnes et al. \(2025\)](#) use a recursive IV approach that first identifies short-run effects and then uses them to adjust longer-run estimates. In the two-period example, identification requires

$$\mathbb{E}[Y_1^E - Y_1^N] = \mathbb{E}[Y_2^L - Y_2^N \mid C_2 = 0], \quad (4)$$

²The same bias can be expressed in terms of negative weights (see [Bensnes et al., 2025](#)). In this example, weight 4 is assigned to the average parenthood effect and weight -3 to the average later-motherhood effect among always-takers, $\mathbb{E}[Y^L - Y^N \mid C = 0]$.

equating the mean effect of one period of motherhood beginning at the first IUI with the effect of one period of motherhood in the second period for always-takers who become mothers in that period. The former is identified by the first-period reduced form and $\Pr(C_2 = 0) = \mathbb{E}[D_2 \mid Z_1 = 0]$ is observed, so the second-period parenthood effect follows by substitution.

The two approaches therefore impose different restrictions, but both ultimately restrict the relationship between early- and later-motherhood outcomes, Y^E and Y^L . Conventional IV allows effects to vary over the life cycle, but requires that, holding life-cycle stage fixed, they do not depend on when motherhood began. Recursive IV allows effects to vary with the duration of motherhood, but requires that, holding duration fixed, they do not depend on the current life-cycle stage. Crucially, both approaches rule out effects that depend on the age or career stage at which motherhood begins, as well as differences arising from subsequent fertility paths induced by earlier or later motherhood.³

Bensnes et al. (2025) show that recursive IV estimates can differ markedly from conventional IV estimates, illustrating the importance of assumptions about the role of parenthood timing. In Section 5.7, I replicate this divergence in my setting and compare both approaches with my estimates.

Before turning to the method, I discuss two intuitive ways to use the IUI sequence for identification and why neither resolves the timing challenge.

The first is to treat each attempt as a separate observation. One can construct a stacked dataset in which each block contains women who reached a given attempt and estimate the blocks separately or pool them. This does not solve the identification problem: within each block, women whose attempt fails include both those who remain childless and those who conceive later than women whose attempt succeeds. Comparisons between successful and failed attempts therefore still combine parenthood and timing effects, and averaging across attempts does not separate them.

The second is to redefine the instrument as success of the last attempt a woman undergoes, Z_A . This may seem appealing because compliers are the compliers with respect to Z_A , since $D = Z_A + (1 - Z_A)(1 - R)$. However, Z_A need not be as good as random because the number of attempts is selected. Suppose maximum willingness is two and attempt j succeeds with probability p_j . For women with $W = 1$, $\Pr(Z_A = 1 \mid W = 1) = p_1$. For women with $W = 2$, $\Pr(Z_A = 1 \mid W = 2) = p_1 + (1 - p_1)p_2$. Thus, the probability of $Z_A = 1$ varies with willingness, generating bias whenever willingness is correlated with potential outcomes. This concern is empirically relevant: Section 4.5 shows that willingness is systematically related to pre-IUI labor market outcomes.

Lastly, while it may seem helpful to control for the number of procedures undergone, this would introduce a bad-control problem because the realized number of attempts is affected by earlier

³There is also no guaranteed ordering between these estimates and the true effects. For example, suppose motherhood before age 30 permanently reduces annual work hours by 50, whereas motherhood after age 30 reduces them by 100. Both approaches would then imply reductions of less than 50 hours. In other settings, the two estimates may lie on opposite sides of the true effect. Ultimately, both can be expressed as weighted combinations of parenthood and timing effects, including negative weights on later-motherhood effects.

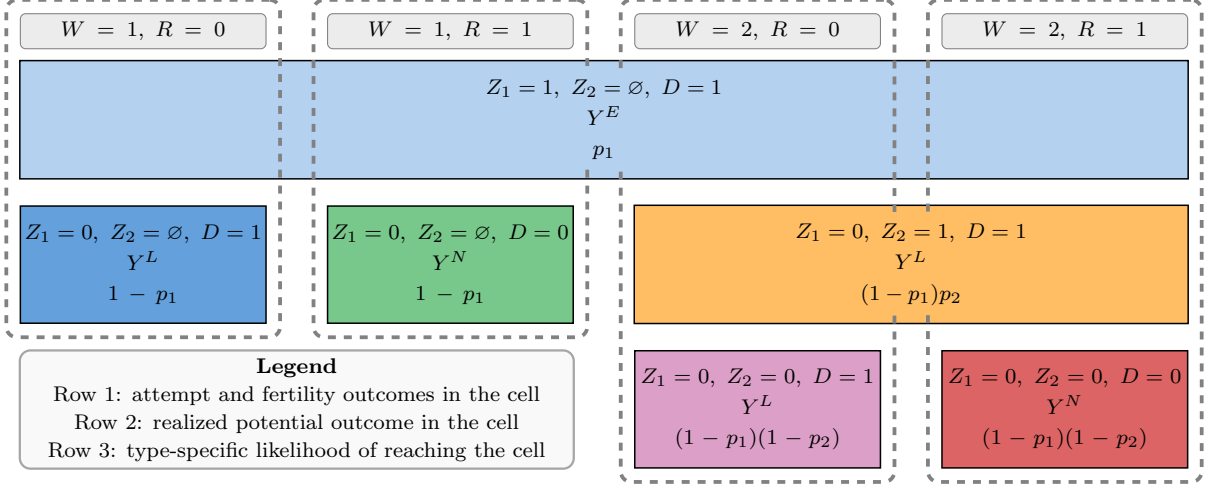


Figure 1: Illustration of the Identification Argument

Note: Column headers denote latent types. Cells show the possible states in which each type may be observed, as determined by Z_1 , Z_2 , and D . The notation $Z_2 = \emptyset$ indicates that no second attempt took place. The likelihoods use p_1 and p_2 for the success probabilities at the first and second procedures. Dashed outlines group states belonging to the same latent type and indicate that, conditional on type, the realized cell is determined by as-good-as-random variation in attempt success. Researchers observe Z_1 , Z_2 , D , and Y , but not the latent type. Horizontally merged cells represent observationally identical states in which researchers cannot distinguish among the latent types.

successes. Conditioning on willingness would avoid this selection, but willingness is unobserved when the sequence ends in success.

3.3 Intuition for the Econometric Approach

I first illustrate the method in a simplified case, then introduce the regularity conditions and covariate-dependent success rates needed for the formal results.

The identification argument has two main steps. The first uses women who experience no IUI success within the window. For these women, willingness and reliance are revealed by definition from the realized number of completed procedures and fertility status. Conditional on willingness, random IUI success implies that they form an as-good-as-random sample of the corresponding type. Outcomes after complete IUI failure therefore identify the relevant average counterfactual for each type—childlessness for reliers and later motherhood for non-reliers—by willingness. The frequencies of failed sequences identify the corresponding type shares, allowing these counterfactual outcomes to be aggregated across willingness types.

The second step bounds the average early-motherhood outcome for reliers and non-reliers using women whose first IUI succeeds. For these women, the early-motherhood outcome is observed and, under random first-attempt success, they form an as-good-as-random sample of the overall population. While their latent types are unknown, I can use the type shares identified in the first step to bound each type's average early-motherhood outcome by assuming that the type fully occupies either the lower or upper tail of the distribution. Combining these bounds with the identified childless and later-motherhood outcomes yields bounds on the average parenthood and timing effects for reliers and non-reliers, respectively.

To illustrate the idea, I focus on the case in which willingness is capped at two attempts, $W \leq 2$; the argument extends straightforwardly to an arbitrary cap. Throughout this section I write $Z_2 = \emptyset$ when no second attempt takes place and $Z_2 = 0$ when it takes place but fails. I write p_1 and p_2 for the success rates at the first and second procedures, the latter among women who actually undergo a second attempt.

Figure 1 separates each latent type into a column, indexed by willingness and reliance, while each row within a column describes a state that the type can reach. The resulting rectangles group individuals by observed fertility histories, summarized by IUI outcomes and parenthood status. When only one type can reach a given state, the corresponding rectangle is contained within that type's column, so the observed group is known to contain only that type; when several types can reach it, the rectangle spans the relevant columns, so the type of individuals in the observed group is not known. Within each rectangle, the first row shows the realized and observed fertility path, the second shows the realized potential outcome corresponding to that path—early motherhood, later motherhood, or childlessness—and the third shows the probability that an individual of that type reaches the cell.

Consider non-reliers willing to undergo one attempt, $(W = 1, R = 0)$. Women who belong to this type may end up in one of two fertility paths. If the first IUI succeeds, they reach the state $(Z_1 = 1, Z_2 = \emptyset, D = 1)$, where the early-motherhood outcome Y^E is realized; the probability of reaching that state is p_1 . Women observed in this state cannot be distinguished from other types, since every type reaches it under first-attempt success, which is why the rectangle in the first row spans all four columns. Alternatively, if the first IUI fails, no second attempt takes place, and the woman nonetheless has a child, she reaches the state $(Z_1 = 0, Z_2 = \emptyset, D = 1)$, where the later-motherhood outcome Y^L is realized. Only the $(W = 1, R = 0)$ type can reach this state: reliers would remain childless, while women willing to undergo two attempts would proceed to a second procedure. Women observed in this state are therefore known to be of this type.

The dashed outlines around each column represent the key identifying assumption. Because success rates are independent of potential outcomes and type, conditional on type, the state, or row, in which a woman ends up within her column is as good as random. For example, the $(W = 1, R = 0)$ type can end up only in the first row or in the first column of the second row. Since the latter cell contains only this type, and assignment between these two cells is as good as random, women observed in the state $(Z_1 = 0, Z_2 = \emptyset, D = 1)$ are a random sample of the $(W = 1, R = 0)$ type. Since their later-motherhood outcome is realized, the type-specific average is identified:

$$\mathbb{E}[Y \mid Z_1 = 0, Z_2 = \emptyset, D = 1] = \mathbb{E}[Y^L \mid W = 1, R = 0].$$

The same argument applied to the second column identifies the average childless outcome among reliers willing to undergo one attempt:

$$\mathbb{E}[Y \mid Z_1 = 0, Z_2 = \emptyset, D = 0] = \mathbb{E}[Y^N \mid W = 1, R = 1].$$

The argument extends directly to types willing to undergo two attempts. Only $(W = 2, R = 0)$ can reach the state in which both attempts fail and they remain childless, so

$$\mathbb{E}[Y \mid Z_1 = 0, Z_2 = 0, D = 1] = \mathbb{E}[Y^L \mid W = 2, R = 0].$$

Symmetrically, for $(W = 2, R = 1)$ the childless outcome is identified from the state with $D = 0$. Extending the argument to an arbitrary number of willingness types simply requires adding the corresponding columns and proceeding to the terminal failure state within each column.

A similar argument identifies the share of each type. The probability that a $(W = 2, R = 1)$ woman reaches the state $(Z_1 = 0, Z_2 = 0, D = 0)$ is the probability that both attempts fail, $(1 - p_1)(1 - p_2)$, so her type's share solves:

$$\Pr(Z_1 = 0, Z_2 = 0, D = 0) = \Pr(W = 2, R = 1)(1 - p_1)(1 - p_2).$$

Because the probability of reaching a fully failed sequence declines with willingness, women willing to undergo more attempts are underrepresented among those observed there; dividing by $(1 - p_1)(1 - p_2)$ is what corrects for this.

Together, these arguments identify the relier and non-relier shares, $\Pr(R = 1)$ and $\Pr(R = 0)$, the average later motherhood outcome for non-reliers, $\mathbb{E}[Y^L \mid R = 0]$, and the average childless outcome for reliers, $\mathbb{E}[Y^N \mid R = 1]$, by averaging the type-specific quantities over willingness.

The figure also shows a state the analysis does not use. Women willing to undergo at least two attempts who conceive at the second, $(Z_1 = 0, Z_2 = 1, D = 1)$, occupy a cell spanning the third and fourth columns, so the two types cannot be separated there. Their realized outcome could in principle be used to bound timing effects between the first and second attempts for women willing to undergo at least two procedures. However, these attempts typically occur only a month apart, so such comparisons may reveal little about timing effects. Focusing on non-reliers instead isolates the group for whom IUI failure generates the largest timing shift: conception at the first attempt versus only after the entire IUI sequence fails.

When outcomes have bounded support, having identified the average childless outcome for reliers and the average later-motherhood outcome for non-reliers, one could bound the remaining unobserved outcomes using the support of the outcome distribution.⁴ These bounds may, however, be very wide. I instead bound the average early-motherhood outcome separately for reliers and non-reliers, which yields tighter bounds on the corresponding average parenthood and timing effects by exploiting observed outcome variation rather than only the limits of outcome support.

I start from the observation that, since IUI outcomes are as good as random, the outcome distribution among women whose first IUI succeeded represents the full distribution of early motherhood

⁴The childless outcome for non-reliers and the later-motherhood outcome for reliers can be assigned their lowest or highest feasible values. Combined with the identified type-specific outcomes and the overall early-motherhood outcome, which is identified from first-IUI successes, this yields bounds on the average parenthood and timing effects.

outcomes in the IUI sample. Using the relier share identified in the previous step, I construct worst-case bounds for average relier early motherhood outcome by assuming that reliers are those with the highest or lowest treated outcomes. For instance, suppose 100 women had a successful first IUI and the relier share is 80%. It is not known which 80 are reliers, but I can obtain the upper (lower) bound on their average outcome by taking the average of the top (bottom) 80 outcomes.⁵ a symmetric argument applies to bounding the non-relier early motherhood outcome.

I then refine the bounds by incorporating pre-IUI covariates. Suppose that after splitting the sample by pre-IUI earnings, each group is estimated to have an 80% relier share. Selecting the bottom 80% of outcomes among women whose first IUI succeeded, without accounting for pre-IUI earnings, may yield a set of potential reliers whose pre-IUI earnings are inconsistent with the group-specific shares. Because this selection produces the most conservative lower bound, any alternative selection can only raise it. To construct the refined bounds, I identify the relier share within each pre-IUI earnings group and select the corresponding share of the lowest and highest treated outcomes in that group. The argument for non-reliers is symmetric.

I can further narrow the bounds only by imposing assumptions on which women are (not) reliers. For example, it may be reasonable to assume that women who have a second or third child without IUI after having their first through IUI would have had at least one child even if all IUIs had failed—or, equivalently, that women who are reliant on IUI for their first child are also reliant for subsequent children. To see how this helps narrow the bounds, consider the previous example. If 10 of the 100 women whose first IUI succeeded subsequently have a non-IUI child, they can be excluded from the pool of potential reliers, as they are certainly not reliant on IUI. Selecting the 80 lowest (highest) outcomes from the remaining 90 can only yield a higher (lower) average than selecting from the full set of 100.

Assumption 2 (Subsequent reliance monotonicity). $R^+ \geq R$.

The monotonicity condition ensures that women whose first IUI succeeded and who have any non-IUI children, and are therefore not subsequent reliers, are also not reliers. This allows them to be excluded from the pool of potential reliers used to construct the bounds.

One way to interpret the monotonicity condition is that families are more determined to have a first child than additional ones. From a fertility-choice perspective, it rules out couples who prefer multiple children but would rather have none than only one. This restriction seems reasonable, as couples who initiate IUI intend to have at least one child and are likely aware that they may not be able to have multiple.

However, fertility may not be entirely determined by choice. For example, having a first child may improve relationship stability or mental health relative to failing to conceive, leading to more natural conception attempts. This may lead to non-IUI births that would not have occurred had the

⁵Trimming outcome distribution tails to bound subgroup averages follows the logic of [Zhang & Rubin \(2003\)](#) and [Lee \(2009\)](#) for addressing sample selection. The innovation lies in identifying and overcoming the challenges involved in appropriately leveraging sequential assignment with bounding, yielding narrower bounds that apply to a broader subpopulation (see Appendix [SA2.2](#)).

first IUI failed, thereby violating monotonicity. I relax the assumption and address such concerns in Section SA2.3.

The monotonicity assumption also yields testable implications, namely that the subsequent relier share at each covariate value exceeds the relier share. I present the testing procedure and results in Appendix SA2.4. In Appendix SA2.5, I also report estimates based on a relaxed specification that allows the direction of monotonicity to vary with covariates. Monotonicity is not rejected, and the estimates under the alternative assumption remain similar.⁶

Finally, I discuss what the remaining uncertainty in the bounds reflects and conditions under which the bounds collapse to a point. Under monotonicity, all reliers are subsequent reliers, but not all subsequent reliers need be reliers. The remaining uncertainty therefore concerns which women among the subsequent reliers would have remained childless had all IUIs failed. Since all women in this group are actively trying to conceive, it may be reasonable to assume that this selection contains a random component. In the extreme case, reliers are a random subset of subsequent reliers with respect to early-motherhood potential outcomes:

Assumption 3 (Random reliance). $\mathbb{E}[Y^E \mid R = 1] = \mathbb{E}[Y^E \mid R^+ = 1]$.

This assumption is a genuine restriction on selection into non-IUI fertility. The formal identification approach weakens it by imposing the restriction only conditional on pre-IUI covariates, including prior labor market histories. Crucially, the bounds remain valid even if Assumption 3 fails.

Under Assumption 3, the average early-motherhood outcome for reliers is point identified by the observed average among women whose first IUI succeeds and who subsequently have no non-IUI children—the subsequent reliers. The corresponding non-reliar outcome can then be recovered from the overall average early-motherhood outcome and the identified type shares. Since the average childless outcome for reliers and the average later-motherhood outcome for non-reliars are already point identified, the parenthood and timing effects are consequently point identified.

Both the monotonicity and random reliance assumptions imply testable restrictions when the procedure is applied to predetermined outcomes. Because outcomes realized before the first IUI cannot depend on subsequent fertility, all three potential outcomes coincide for every woman, so the parenthood and timing effects are identically zero. The procedure itself does not impose this. If monotonicity fails, excluding women with non-IUI children from the pool of potential reliers may exclude some true reliers, so the resulting bounds on average relier pre-IUI outcomes need not contain the relier mean identified using women who actually remain childless after IUI failure. Similarly, if subsequent reliers differ systematically from reliers, the point estimate of the relier pre-IUI outcome obtained from subsequent reliers may differ from the mean identified using women who actually remain childless after IUI failure. Section 5.2 shows that the placebo bounds contain zero and that the point estimates are close to zero, providing support for both assumptions.

⁶The remaining results assume monotonicity. In settings without variables that can credibly help identify reliers, sharp bounds without monotonicity (i.e., under trivial monotonicity) can be obtained by redefining $R^+ = 1$ and $D^+ = (1 - Z_A)D$, thus treating everyone as subsequent reliers.

3.3.1 Treatment Effect Identification in the Absence of Timing Effects

Before turning to the formal results, I consider an alternative way to identify the average treatment effect for reliers under an exclusion restriction analogous to standard IV. This approach targets the same population as my main estimand—reliers rather than compliers—while imposing assumptions similar to existing IV approaches. It therefore helps distinguish differences arising from identifying assumptions from those arising from the populations covered by the estimands.

The previous section identifies the overall average early-motherhood outcome, the average childless outcome for reliers, the average later-motherhood outcome for non-reliers, and the corresponding type shares. These quantities can be combined into a *sequential reduced form* comparing average outcomes when the first IUI succeeds with those when all IUIs fail. Let $Y_0^* = Y^N R + Y^L(1 - R)$ denote the outcome when all IUIs fail. For reliers, it is the childless outcome, for non-relier it is the later-motherhood outcome. Then $\mathbb{E}[Y_0^*] = \mathbb{E}[Y^N | R = 1] \Pr(R = 1) + \mathbb{E}[Y^L | R = 0] \Pr(R = 0)$, where all terms are identified in the previous section. It follows that $\mathbb{E}[Y^E - Y_0^*] = \tau_{ATR} \Pr(R = 1) - \delta_{ANR} \Pr(R = 0)$. Scaling by the relier share yields

$$\frac{\mathbb{E}[Y^E - Y_0^*]}{\Pr(R = 1)} = \tau_{ATR} - \delta_{ANR} \frac{\Pr(R = 0)}{\Pr(R = 1)}. \quad (5)$$

I refer to this as the *sequential IV* estimand because it uses the sequence of IUI attempts to construct an estimand analogous to conventional IV in Equation (3), but targets reliers rather than compliers. Under the standard exclusion restriction that the timing of motherhood does not affect outcomes, $\delta_{ANR} = 0$, and sequential IV identifies τ_{ATR} . When timing matters, the estimand is contaminated by δ_{ANR} , but the timing term receives less weight than under conventional IV because non-reliers are a subset of always-takers: $\Pr(R = 0)/\Pr(R = 1) \leq \Pr(C = 0)/\Pr(C = 1)$.

Importantly, the sequential IV estimand need not lie within the bounds on τ_{ATR} . If it falls outside the bounds, the exclusion restriction can therefore be rejected; which serves as a direct test for the presence of timing effects.

3.4 Formal Identification Statement

This section formally establishes the identification results introduced intuitively in the previous section. Whereas the previous section established identification using conditional expectations, this section establishes it using inverse probability weighting. The two approaches are combined in the estimation section, analogously to how doubly robust estimators combine outcome regression and inverse probability weighting under a single treatment assignment (e.g., [Abadie, 2003](#); [Chernozhukov et al., 2018](#)).

As in previous sections, the outcome (Y) is measured either over the entire data window or at a particular horizon from the first IUI, whereas parenthood status (D), the IUI attempt history (A), and attempt outcomes (Z_j) summarize events occurring between the first IUI and the end of the data window. I begin by introducing the conditional local sequential unconfoundedness assumption:

Assumption 4 (Conditional local sequential unconfoundedness).

$(Y^m, R^+, R, W) \perp\!\!\!\perp Z_j \mid X_j, A \geq j \text{ for all } m, j, \text{ and } X_j \in \mathcal{X}_j.$

where $X_j \in \mathcal{X}_j$ are covariates measured at assignment j , such as age at the time of the procedure, time since the last procedure, and other observable characteristics. In words, the success of IUI j is independent of potential outcomes and type, conditional on undergoing at least j IUIs and on these covariates. Note that the measurement timing of X_j may vary across women and be correlated with potential outcomes: it is determined by when they choose to undergo each procedure. The next assumption provides regularity conditions. Let $e_j(x) = \Pr(Z_j = 1 \mid X_j = x, A \geq j)$.

Assumption 5 (Regularity).

1. $0 < \underline{e} < e_j(x) < \bar{e} < 1$ for all j and $x \in \mathcal{X}_j$, for some fixed $\underline{e}$ and $\bar{e}$.
2. Y has a probability density function for $Z_1 = 1$, $D^+ = 0$, and all $x \in \mathcal{X}_1$.

It contains two parts. First, the probability of IUI success conditional on undergoing the procedure and covariates at the time differs from 0 and 1. Second, Y is a continuous random variable conditional on the first IUI succeeding, having only IUI children, and any value of X_1 . In practice, adding a negligible amount of continuously distributed noise to Y is sufficient to avoid ties in trimming without meaningful bias.

The bounding procedure begins with identifying several nuisance functions involved in the trimming step. First, the covariate-conditional relier share is identified using the share of women without children among those whose IUIs failed:

$$r(x) = \mathbb{E} \left[(1 - D^+) \prod_{j=1}^A \frac{(1 - Z_j)}{(1 - e_j(X_j))} \mid X_1 = x \right]. \quad (6)$$

Because the denominator decreases in A , women who undergo more IUIs receive larger weights, correcting for their lower likelihood of never experiencing IUI success and thus their underrepresentation in this group. Next, the covariate-conditional share of subsequent reliers is identified from the share of women having only IUI children among those whose first IUI succeeded $r^+(x) = \mathbb{E}[1 - D^+ \mid Z_1 = 1, X_1 = x]$. Under monotonicity, the covariate-conditional share of reliers among subsequent reliers is then $p(x) = r(x)/r^+(x)$.

The covariate-conditional quantile function of the treated outcome distribution among subsequent reliers is identified from the outcome distribution of women whose first IUI succeeded and who have only IUI children:

$$q(u, x) = \inf \{ q : u \leq \Pr(Y \leq q \mid X_1 = x, Z_1 = 1, D^+ = 0) \}. \quad (7)$$

Finally, $q(p(x), x)$ and $q(1 - p(x), x)$ identify the covariate-conditional $p(x)$ -th and $1 - p(x)$ -th quantiles of the treated outcome distribution among subsequent reliers. These quantiles are used to trim the outcome distribution and select reliers in scenarios where they have either the lowest or highest treated outcomes.

The nuisance functions are combined with the data to construct the moments:

$$m^L(G, \eta^0) = Y(1 - D^+)1_{\{Y < q(p(X_1), X_1)\}} \frac{Z_1}{e_1(X_1)} - Y(1 - D^+)\prod_{j=1}^A \frac{(1 - Z_j)}{(1 - e_j(X_j))} \quad (8)$$

$$m^U(G, \eta^0) = Y(1 - D^+)1_{\{Y > q(1-p(X_1), X_1)\}} \frac{Z_1}{e_1(X_1)} - Y(1 - D^+)\prod_{j=1}^A \frac{(1 - Z_j)}{(1 - e_j(X_j))}, \quad (9)$$

where vector G contains observed variables and η^0 contains the nuisance functions:

$$\eta^0(x_1, \dots, x_A) = \{r^+(x_1), r(x_1), q(p(x_1), x_1), q(1 - p(x_1), x_1), e_1(x_1), \dots, e_{\bar{w}}(x_{\bar{w}})\}. \quad (10)$$

The first term in $m^L(G, \eta^0)$ is used to bound the relier average treated outcome. It assigns positive weights to women whose first IUI succeeded, who have only IUI children, and whose outcomes fall below the covariate-conditional trimming threshold $q(p(x), x)$. The second term, used to identify the relier average control outcome, assigns positive weights to outcomes of childless women. Larger weights are given to women who underwent more IUIs to account for the fact that reliers willing to undergo more IUIs are less likely to not experience IUI success, making them underrepresented in this group. $m^U(G, \eta^0)$ mirrors this for the scenario where reliers have the highest treated outcomes. The moments are then scaled by the relier share.

Theorem 1 (Bounds on the Treatment Effect). *Under Assumptions 2, 4, and 5, sharp lower and upper bounds on τ_{ATR} are given by $\theta_L = \mathbb{E}[m^L(G, \eta^0)]/\mathbb{E}[r(X_1)]$ and $\theta_U = \mathbb{E}[m^U(G, \eta^0)]/\mathbb{E}[r(X_1)]$.*

Proof. See Appendix 7.

The next moment is used to identify the sequential reduced-form estimand, which captures the average effect of the first IUI succeeding relative to the entire sequence of IUIs failing. This estimand is then used both to back out timing effects from the treatment-effect bounds and to identify the sequential IV estimand, which identifies the relier average treatment effect under the assumption of no timing effects.

$$m^{RF}(G, \eta^0) = Y \frac{Z_1}{e_1(X_1)} - Y \prod_{j=1}^A \frac{(1 - Z_j)}{(1 - e_j(X_j))} \quad (11)$$

Corollary 1 (Identification in the Absence of Timing Effect). *Under Assumptions 4 and 5, and if $\delta = 0$ a.s., then $\tau_{ATR} = \mathbb{E}[m^{RF}(G, \eta^0)]/\mathbb{E}[r(X_1)]$.*

Proof. See Appendix 7.

Since the sequential reduced-form estimand is a weighted average of the parenthood effect for reliers and the timing effect for non-reliers, the moments used to bound the average timing effect for non-reliers combine the moment identifying the sequential reduced form with those used to bound the relier average treatment effect:

Theorem 2 (Bounds on the Timing Effect). *Under Assumptions 2, 4, and 5, sharp lower and upper bounds on δ_{ANR} are given by $\gamma_L = \mathbb{E}[m^L(G, \eta^0) - m^{RF}(G, \eta^0)]/\mathbb{E}[1 - r(X_1)]$ and $\gamma_U = \mathbb{E}[m^U(G, \eta^0) - m^{RF}(G, \eta^0)]/\mathbb{E}[1 - r(X_1)]$.*

Proof. See Appendix 7.

The next assumption yields point identification when non-IUI conception selection is limited:

Assumption 6 (Conditionally random reliance). $\mathbb{E}[Y^E|R = 1, X_1] = \mathbb{E}[Y^E|R^+ = 1, X_1]$ *a.s.*

The assumption states that, among subsequent reliers with the same covariates, reliance carries no information about average outcomes under early motherhood. In the IUI context, it formalizes the idea that, conditional on pre-IUI labor market outcomes and other observables, whether a couple would remain reliant on IUI after a failed sequence may be unrelated to career potential.

$$m^{RND}(G, \eta^0) = Y(1 - D^+)p(X_1) \frac{Z_1}{e_1(X_1)} - Y(1 - D^+) \Pi_{j=1}^A \frac{(1 - Z_j)}{(1 - e_j(X_j))}. \quad (12)$$

This moment replaces the trimming indicator in m^L and m^U with the covariate-conditional relier share $p(X_1)$. Rather than assigning reliers the lowest or highest early-motherhood outcomes among subsequent reliers, it assigns them the average.

The following theorem establishes identification and shows that the identified parameters are guaranteed to lie within the bounds.

Theorem 3 (Identification under Random Reliance). *Under Assumptions 4, 5, and 6, $\tau_{ATR} = \mathbb{E}[m^{RND}(G, \eta^0)]/\mathbb{E}[r(X_1)]$ and $\delta_{ANR} = \mathbb{E}[m^{RND}(G, \eta^0) - m^{RF}(G, \eta^0)]/\mathbb{E}[1 - r(X_1)]$. Moreover, $\theta_L \leq \tau_{ATR} \leq \theta_U$ and $\gamma_L \leq \delta_{ANR} \leq \gamma_U$.*

Proof. See Appendix 7.

3.5 Estimation

The bounds can be estimated by replacing the expectations and nuisance functions in the theorem with their empirical counterparts. However, inference for such estimators is complicated by their sensitivity to first-stage estimation error in the conditional quantile function, which itself must be estimated nonparametrically. If overly conservative—i.e., non-sharp—bounds are acceptable, one can overcome this challenge by restricting attention to discrete covariates or by partitioning continuous covariates into bins. In that case, covariate-conditional bounds can be estimated separately within each covariate cell and then aggregated. Asymptotic normality and a consistent estimator of the asymptotic variance then follow from conventional GMM results.

Ensuring sharp bounds, however, requires using all available covariates, including continuous ones, making such an approach infeasible. To address this challenge, I draw on insights from [Seменова \(2025\)](#), who develops an estimator for the [Lee \(2009\)](#) bounds, which also rely on conditional quantiles. The method combines orthogonalization with sample splitting. I adapt this approach by constructing new orthogonal moment functions for my bounds.

The moment functions given in Table 1 identify the same parameters as the baseline moments:

$$\mathbb{E}[\psi^L(G, \xi^0)] = \mathbb{E}[m^L(G, \eta^0)], \quad \mathbb{E}[\psi^U(G, \xi^0)] = \mathbb{E}[m^U(G, \eta^0)], \quad (13)$$

$$\mathbb{E}[\psi^{RF}(G, \xi^0)] = \mathbb{E}[m^{RF}(G, \eta^0)], \quad \mathbb{E}[\psi^{RND}(G, \xi^0)] = \mathbb{E}[m^{RND}(G, \eta^0)]. \quad (14)$$

However, the original moments are sensitive to small errors in the nuisance parameters, whereas the new moments are not. I use this property together with sample splitting to justify asymptotic inference as if the nuisance parameters were known, following the argument in [Semenova \(2025\)](#).⁷

The moments in this paper generalize those in [Semenova \(2025\)](#) by incorporating the sequential structure of my procedure. This shifts the focus from compliers to reliers and requires additional correction terms for sequential propensity scores. When only the first IUI attempt is used, rather than the full sequence, variables can be mapped directly between the two frameworks, and the moments ψ^L and ψ^U exactly match their counterparts in [Semenova \(2025\)](#).

The estimator for θ_b , for $b \in \{L, U\}$, is given by:

$$\hat{\theta}_b = \left(\sum_i \left(\psi^b(G_i, \hat{\xi}_i) 1_{\{p(X_1) \leq 1\}} + \psi^{RND}(G_i, \hat{\xi}_i) 1_{\{p(X_1) > 1\}} \right) \right) / \left(\sum_i \psi^R(G_i, \hat{\xi}_i) \right) \quad (15)$$

where G_i is the data for observation i , $\hat{\xi}_i$ is the nuisance parameter for observation i , estimated on a subsample that excludes observation i , and ψ^R is the orthogonal estimator of the relier share. ψ^b is the orthogonal counterpart of m^b , and ψ^R is the orthogonal counterpart of r . ψ^{RND} is used as a fallback when the estimated relier share exceeds the subsequent relier share; I discuss monotonicity further in Appendix [SA2.4](#). The moment for the sequential reduced form is ψ^{RF} , and the estimator for γ_b , for $b \in \{L, U\}$, follows Theorem [2](#).

I use 4-fold cross-fitting, estimating the nuisance functions for each fourth of the sample using the remaining three-fourths.⁸ Propensity scores are estimated using logistic regressions. The key covariates are women’s and their partners’ ages at the time of the procedure, each included as a second-order polynomial. Because success rates vary smoothly with age, I model age parametrically to avoid sparse cells at later attempts; other nuisance functions control for age nonparametrically.

Following [Heiler \(2024\)](#), I estimate the remaining nuisance functions using Generalized Random Forests ([Athey et al., 2019](#)).⁹ For attempt k , covariates include all propensity-score covariates from attempts 1 through k and both partners’ work hours and earnings in the year before the first IUI. The pre-IUI outcomes are highly predictive of post-IUI outcomes and thus key for obtaining narrow bounds. Including age at the current and previous attempts controls for the time between attempts. Confidence intervals for the partially identified parameters follow [Heiler \(2024\)](#), based on [Stoye \(2020\)](#), and account for uncertainty in the bounds when covering the underlying parameter.

I use data on the first ten procedures, which makes the bounds only negligibly wider than if

⁷For example, consider a perturbation of the propensity score at attempt j , $e_j^*(X_j^*, r) = e_j^*(X_j^*) + r(\hat{e}_j^*(X_j^*) - e_j^*(X_j^*))$. For the orthogonal upper-bound moment, $\partial_r \mathbb{E}[\psi^U(G, \xi_r) | X_j^*]_{r=0} = 0$ almost surely, whereas the corresponding derivative for the original moment is generally nonzero. Orthogonality is established by repeated application of Lemma [1](#).

⁸The main results are insensitive to this choice (see Appendix [SA2.6](#)).

⁹The baseline specification uses 250 trees; results are insensitive to this choice (Appendix [SA2.6](#)). I add $u \sim U(0, 0.001)$ noise to outcomes to avoid ties in the trimming step. I estimate z_t^{U+} and z_t^{L+} by trimming above or below the estimated quantiles and estimating the resulting conditional expectations. Theoretical results for nonparametric estimation of truncated expectations exist ([Olma, 2021](#)), but no implementation is currently available.

Table 1: Orthogonal Moments

Moment functions	
$\psi^L(G, \xi^0)$	$Y(1 - D^+)1_{\{Y < q(p(X_1^*), X_1^*)\}} \frac{Z_1}{e_1^*(X_1^*)} - Y(1 - D^+)\Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $+ q(p(X_1^*), X_1^*) \left[\Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} (1 - D^+ - r_1(X_1^*)) \right.$ $\left. - \frac{Z_1}{e_1^*(X_1^*)} p(X_1^*)(1 - D^+ - r^+(X_1^*)) - \frac{Z_1}{e_1^*(X_1^*)} (1 - D^+) (1_{\{Y < q(p(X_1^*), X_1^*)\}} - p(X_1^*)) \right]$ $- \frac{Z_1 - e_1^*(X_1^*)}{e_1^*(X_1^*)} z^{L+}(1, X_1^*) r_1(X_1^*) + \sum_{k=1}^{\bar{w}} 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} \frac{e_k^*(X_k^*) - Z_k}{1 - e_k^*(X_k^*)} [r_k(X_k^*) \beta_k(X_k^*)$ $+ q(p(X_1^*), X_1^*) (r_1(X_1^*) - r_k(X_k^*))]$
$\psi^U(G, \xi^0)$	$Y(1 - D^+)1_{\{Y > q(1-p(X_1^*), X_1^*)\}} \frac{Z_1}{e_1^*(X_1^*)} - Y(1 - D^+)\Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $+ q(1 - p(X_1^*), X_1^*) \left[\Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} (1 - D^+ - r_1(X_1^*)) \right.$ $\left. - \frac{Z_1}{e_1^*(X_1^*)} p(X_1^*)(1 - D^+ - r^+(X_1^*)) - \frac{Z_1}{e_1^*(X_1^*)} (1 - D^+) (1_{\{Y > q(1-p(X_1^*), X_1^*)\}} - p(X_1^*)) \right]$ $- \frac{Z_1 - e_1^*(X_1^*)}{e_1^*(X_1^*)} z^{U+}(1, X_1^*) r_1(X_1^*) + \sum_{k=1}^{\bar{w}} 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} \frac{e_k^*(X_k^*) - Z_k}{1 - e_k^*(X_k^*)} [r_k(X_k^*) \beta_k(X_k^*)$ $+ q(1 - p(X_1^*), X_1^*) (r_1(X_1^*) - r_k(X_k^*))]$
$\psi^{RND}(G, \xi^0)$	$Y(1 - D^+) \frac{Z_1}{e_1^*(X_1^*)} p(X_1^*) - Y(1 - D^+)\Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $- \beta^+(X_1^*) \left[\frac{Z_1}{e_1^*(X_1^*)} \frac{(1-D^+ - r^+(X_1^*))}{r^+(X_1^*)} r_1(X_1^*) - \Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} (1 - D^+ - r_1(X_1^*)) \right]$ $- \frac{Z_1 - e_1^*(X_1^*)}{e_1^*(X_1^*)} \beta^+(X_1^*) r_1(X_1^*) + \sum_{k=1}^{\bar{w}} 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} \frac{e_k^*(X_k^*) - Z_k}{1 - e_k^*(X_k^*)} [r_k(X_k^*) \beta_k(X_k^*)$ $+ \beta^+(X_1^*) (r_1(X_1^*) - r_k(X_k^*))]$
$\psi^{RF}(G, \xi^0)$	$Y \frac{Z_1}{e_1^*(X_1^*)} - Y \Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $- \gamma^1(X_1^*) \frac{Z_1 - e_1^*(X_1^*)}{e_1^*(X_1^*)} + \sum_{k=1}^{\bar{w}} \gamma_k^0(X_k^*) 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} \frac{e_k^*(X_k^*) - Z_k}{1 - e_k^*(X_k^*)}$
$\psi^R(G, \xi^0)$	$r_1(X_1^*) + (1 - D^+ - r_1(X_1^*)) \Pi_{j=1}^{\bar{w}} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $+ \sum_{k=1}^{\bar{w}} 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{(1-Z_j)}{(1-e_j^*(X_j^*))} \frac{e_k^*(X_k^*) - Z_k}{1 - e_k^*(X_k^*)} [r_1(X_1^*) - r_k(X_k^*)]$
Nuisance functions	
$\xi^0(x_1^*, \dots, x_{\bar{w}}^*)$	$\{e_1^*(x_1^*), \dots, e_{\bar{w}}^*(x_{\bar{w}}^*), r_1(x_1^*), \dots, r_{\bar{w}}(x_{\bar{w}}^*), r^+(x_1^*), q(p(x_1^*), x_1^*), q(1 - p(x_1^*), x_1^*),$ $\beta_1(x_1^*), \dots, \beta_{\bar{w}}(x_{\bar{w}}^*), \beta^+(x_1^*), z^{U+}(x_1^*), z^{L+}(x_1^*), \gamma_1^0(x_1^*), \dots, \gamma_{\bar{w}}^0(x_{\bar{w}}^*), \gamma^1(x_1^*)\}$
$r_k(x)$	$\mathbb{E}[(1 - D^+)/(\Pi_{j=k+1}^A (1 - e_j^*(X_j^*))) \mid X_k^* = x, Z_A = 0]$
$\beta_k(x)$	$\mathbb{E}[\Pi_{j=k+1}^A (1 - e_j^*(X_j^*)) \mid X_k^* = x, Z_A = 0]$
$\beta^+(x)$	$\mathbb{E}[Y/(\Pi_{j=k+1}^A (1 - e_j^*(X_j^*))) \mid X_k^* = x, D = 0]$
$z^{U+}(x)$	$\mathbb{E}[\Pi_{j=k+1}^A (1 - e_j^*(X_j^*)) \mid X_k^* = x, D = 0]$
$z^{L+}(x)$	$\mathbb{E}[Y \mid X_1^* = x, Z_1 = 1, D^+ = 0, Y \geq q(1 - p(x), x)]$
$\gamma_k^0(x)$	$\mathbb{E}[Y \mid X_1^* = x, Z_1 = 1, D^+ = 0, Y \leq q(p(x), x)]$
$\gamma^1(x)$	$\mathbb{E}[Y/(\Pi_{j=k+1}^A (1 - e_j^*(X_j^*))) \mid X_k^* = x]$
	$\mathbb{E}[\Pi_{j=k+1}^A (1 - e_j^*(X_j^*)) \mid X_k^* = x]$
	$\mathbb{E}[Y \mid X_1^* = x, Z_1 = 1]$
Covariate vector	
X_k^*	$(X_k, 1_{\{A \geq k\}})$

additional procedures were included, while avoiding the need to estimate nuisance functions on small subsamples of women who undergo more than ten (fewer than 10% of the sample). This means that reliers are women who would not have a child if their first ten observed procedures failed. Varying the number of procedures used between 9 and 13 has little impact on the estimates (Appendix SA2.7).

3.6 Characterizing Potential Selection on Unobservables

This section develops a procedure for bounding the potential extent of selection on unobservables. The results characterize how such selection may bias the estimates and, in particular, how much the bias may be amplified by using the full sequence of IUI attempts rather than only the first attempt. The procedure also provides an empirical benchmark for the degree of selection consistent with the observed data, which I use in the sensitivity analysis to assess how the estimates change under selection models that cannot be rejected.

The procedure formalizes the idea that women may differ in latent health, unobserved in the data, that is correlated with both potential outcomes and IUI success rates. For simplicity, suppose women belong to one of two latent health groups, $H = 1$ and $H = 0$, although the approach extends straightforwardly to multiple types or a continuous distribution. Women with $H = 1$ succeed at the first attempt with probability p^H and women with $H = 0$ with probability p^L , where $p^H > p^L > 0$.¹⁰

Although the procedure can assess selection across all three potential outcomes and the entire IUI sequence, it is easiest to illustrate using women whose first IUI succeeds, since all of them realize the same potential outcome, Y^E . Let $\pi = \Pr(H = 1)$ and $p = \mathbb{E}[Z_1]$ denote the share of high-health women and the observed first-attempt success rate. Since $p = \pi p^H + (1 - \pi)p^L$, the sample share of high-health women is pinned down by the two success rates: $\pi = (p - p^L)/(p^H - p^L)$. Let $\mu_h = \mathbb{E}[Y^E \mid H = h]$ denote the average early-motherhood potential outcome in group h , and $\Delta = \mu_1 - \mu_0 > 0$ the difference between groups.

Under this model, the outcomes of women whose first IUI succeeds differ from those of the overall sample by

$$\mathbb{E}[Y^E \mid Z_1 = 1] - \mathbb{E}[Y^E] = \frac{\pi(1 - \pi)(p^H - p^L)}{p} \Delta. \quad (16)$$

The difference is increasing in the health gap in outcomes, Δ , in the health gap in success probabilities, $p^H - p^L$, and in the extent to which both health groups are represented in the sample.

The next step is to quantify how large this imbalance could be. The first term in (16), the average outcome of women whose first IUI succeeds, is directly observed because Y^E is realized for all of these women. The second term, the average early-motherhood outcome, is not. If latent health were observed, however, it could be recovered using a standard reweighting argument. Conditional on H , first-attempt success does not select women on their potential outcomes, so

$$\mathbb{E}[Y^E \mid H = 1, Z_1 = 1]\pi + \mathbb{E}[Y^E \mid H = 0, Z_1 = 1](1 - \pi) = \mathbb{E}[Y^E]. \quad (17)$$

¹⁰The exercise therefore abstracts from sterility and instead captures latent health factors or behaviors that may jointly affect IUI success and labor market outcomes.

Thus, the outcomes of successful women can be made representative of the sample by reweighting the two health groups back to their sample shares.

Neither the two type-specific success rates nor H is observed. The procedure therefore fixes a candidate pair (p^H, p^L) , which determines π , and assigns H among women whose first IUI succeeds so as to generate the strongest possible relationship between latent health and the outcome. By Bayes' rule, the share of successful women who must belong to the high-health group is $\pi p^H/p$. I therefore rank women whose first IUI succeeds by their realized outcome and assign the fraction $\pi p^H/p$ with the highest outcomes to $H = 1$, with the remainder assigned to $H = 0$. For a given pair (p^H, p^L) , this is the most adverse assignment because it maximizes the relationship between latent health and Y^E . Evaluating (17) under this assignment yields the bias implied by this selection model and, equivalently, the largest proportional gap $\mathbb{E}[Y^E \mid Z_1 = 1]/\mathbb{E}[Y^E]$ between the outcomes of first-attempt successes and the mean.

The second part of the procedure asks whether a candidate selection model is consistent with the observed data. The key observation is that the same reweighting argument can be applied to characteristics measured before IUI. Let X denote a pre-IUI characteristic, or a vector of characteristics. Under the candidate model,

$$\mathbb{E}[X \mid H = 1, Z_1 = 1]\pi + \mathbb{E}[X \mid H = 0, Z_1 = 1](1 - \pi) = \mathbb{E}[X]. \quad (18)$$

Unlike (17), the right hand side of (18) is directly observed. The candidate pair (p^H, p^L) and the associated worst-case assignment of H therefore imply a testable restriction on pre-IUI characteristics. If the reweighted characteristics of women whose first IUI succeeds differ significantly from those of the full sample, the corresponding selection model is inconsistent with the observed data.

Notably, however, failure to reject (18) does not provide support for the selection model. Even when there is no actual selection on Y^E , falsely assuming such selection may generate little or no imbalance in a particular X if X is only weakly related to Y^E . I therefore use the largest degree of selection among the models that cannot be rejected as a conservative sensitivity benchmark.

The framework also clarifies how much selection can be amplified by focusing on women who experience a sequence of unsuccessful IUIs. A potential concern is that even modest selection at each attempt could accumulate across repeated failures and generate substantial imbalance among women who remain in the failure group.

By an argument symmetric to (16), the potential outcomes of women whose first IUI fails differ from the average by

$$\mathbb{E}[Y^E] - \mathbb{E}[Y^E \mid Z_1 = 0] = \frac{\pi(1 - \pi)(p^H - p^L)}{1 - p} \Delta. \quad (19)$$

Taking the ratio of imbalance in the success group (19) and in the failure group (16) gives

$$\frac{\mathbb{E}[Y^E] - \mathbb{E}[Y^E \mid Z_1 = 0]}{\mathbb{E}[Y^E \mid Z_1 = 1] - \mathbb{E}[Y^E]} = \frac{p}{1 - p}. \quad (20)$$

The ratio in (20) has two important implications. First, because it depends only on the ob-

served success probability p , and not on π , Δ , p^H , or p^L , selection among failures follows directly from selection among successes and the observed success rate. This makes it possible to quantify imbalance in their potential early-motherhood outcomes without observing those outcomes directly.

Second, because the ratio is increasing in p , it shows that when the success rate is low, even a strong relationship between success probabilities and potential outcomes generates only limited selection among women whose IUI fails. The intuition is straightforward: when success is uncommon, most women remain in the failure group, so its average potential outcome cannot move far from the population mean.

Because the per-procedure IUI success rate remains around 10% throughout the sequence, most of any imbalance is concentrated among women whose attempt succeeds. For example, if the relevant potential outcomes of women whose first IUI succeeds are 1% above the mean, then the corresponding potential outcomes of women whose first IUI fails are only 0.11% below the mean.

This substantially limits the extent to which selection can accumulate across a sequence of unsuccessful attempts. If the same imbalance—women whose IUI succeeds having relevant potential outcomes 1% above the mean—arose at each of ten consecutive attempts, the maximum number considered in the analysis, the cumulative selection factor among women experiencing ten failures would be $(1 - 0.01/9)^{10} \approx 0.989$. Thus, even after ten failures, the group’s average potential outcomes would lie only about 1.1% below the mean, barely more than the 1% imbalance among women whose first attempt succeeded.

These calculations also provide a natural way to correct the main estimates for potential selection. Suppose the sensitivity exercise implies that the ratio of the potential outcomes of women whose first IUI succeeds to the mean is x . Their realized outcomes can then be adjusted by dividing by x . Similarly, if the ratio for women who experience a particular sequence of unsuccessful attempts to the corresponding type’s mean is y , their outcomes can be divided by y . For repeated failures, the relevant ratio can be constructed by multiplying the attempt-specific failure-selection ratios implied by (20). The main analysis can then be repeated using the adjusted outcomes.

The procedure cannot rule out every form of selection on unobservables, since it imposes a particular two-type model of latent heterogeneity. It nevertheless provides a transparent benchmark for how much selection is consistent with the data and how strongly it could accumulate over repeated attempts. In the sensitivity analysis, the main estimates change little even under the most extreme models I cannot reject.

Appendix SA3 provides the derivations and explains how the procedure incorporates covariates. In short, I consider selection on potential outcomes conditional on the baseline propensity scores, thereby netting out selection already captured by the woman’s and her partner’s ages. I then test balance in pre-IUI labor market characteristics and quantify imbalances in post-IUI outcomes after additionally controlling for these characteristics. These two steps mirror the baseline balance tests and the specification used for the main estimates, respectively.

4 Institutions, Intrauterine Insemination, and Data

Section 4.1 describes Dutch family policies and the labor market context. Section 4.2 discusses IUI. Section 4.3 describes the data. Section 4.4 presents sample descriptives, compares the IUI and representative samples, and characterizes pre-IUI differences between reliers and non-reliers. Section 4.5 documents procedure profiles and pre-IUI differences across women with different levels of willingness. Section 4.6 provides empirical support for the unconfoundedness assumption.

4.1 Family Policies in the Netherlands

Dutch women are entitled to 4 to 6 weeks of pregnancy leave before the due date and at least 10 weeks of maternity leave following childbirth, totaling a minimum of 16 weeks. In the case of multiple births, the total entitlement increases to 20 weeks. During this period, mothers receive full wage replacement from the unemployment insurance agency, subject to a daily maximum. In the sample period, fathers are entitled to one week of fully paid leave within the first four weeks after childbirth, financed by the employer.

Children can enroll in private daycare at three months old. In 2022, 72% of children under two attended formal child care, averaging 20 hours per week (OECD, 2023a). After turning four and starting elementary school, they become eligible for out-of-school care. In 2023, families using child care paid an average of 8,950 euros, of which 64% was reimbursed by the government, resulting in a net cost equivalent to 10% of median disposable household income.¹¹

The Netherlands has average family policies compared to other OECD countries. Paternity and maternity leave durations are slightly below the OECD averages of 2.5 and 21 weeks, respectively (OECD, 2023c). While formal child care enrollment for children under two is the highest among OECD countries, average time spent in care is the lowest (OECD, 2023a). After age four, enrollment rates and out-of-school care hours align with OECD averages (OECD, 2022).

While employment rates for mothers, fathers, and non-parents in the Netherlands exceed their respective OECD averages, part-time work is far more common, making average hours worked comparable to the OECD average (OECD, 2023b). In 2021, the maternal employment rate was 80%, compared to the OECD average of 71%. However, in 2023, 52% of women and 18% of men worked part-time (less than 30 hours per week), more than twice the respective OECD averages (OECD, 2023d). Among two-parent families, only 14% had both parents working full-time, 52% had a full-time working father and a part-time working mother, and 12% had both parents working part-time.¹²

¹¹www.cbs.nl/nl-nl/nieuws/2024/30/ouders-betaalden-gemiddeld-3-210-euro-aan-kinderopvang-in-2023, longreads.cbs.nl/materiele-welvaart-in-nederland-2024/inkomen-van-huishoudens/.

¹²www.cbs.nl/en-gb/news/2024/10/fewer-and-fewer-families-in-which-only-the-father-works

4.2 Assisted Conception Procedures

My analysis focuses on Dutch couples who undergo IUI for their first child. As in most countries, the procedure is usually a first-line treatment for couples who are unable to conceive naturally within a year, especially in cases of male-factor or unexplained infertility. Prior to the procedure, women typically undergo cycle tracking and may receive hormonal stimulation to enhance egg production. During IUI, sperm is injected directly into the uterus via a catheter, mimicking natural conception by facilitating fertilization within the body. With a success rate of about 10% per attempt, the procedure is minimally invasive, typically lasting five minutes and causing little or no discomfort.

Some couples who do not succeed with IUI eventually proceed to IVF, which has previously been used to study the career impact of parenthood in Denmark, Norway, and Sweden ([Lundborg et al., 2017, 2024](#); [Gallen et al., 2024](#); [Bensnes et al., 2025](#)). IVF is a surgical procedure in which eggs are retrieved through the vaginal wall, fertilized in the laboratory, and transferred as embryos into the uterus. It is more invasive than IUI, performed under sedation or anesthesia, and in most countries it is used for women with severe fertility problems, such as tubal damage or advanced endometriosis. It also has a higher success rate than IUI, of about 25% per embryo transfer. Because IVF also generates as-good-as-random fertility variation, I include eventual IVF attempts in the analysis. To account for differences in success rates between IUI and IVF, all terms in each propensity score are interacted with a procedure-type indicator.

In the Netherlands, couples without a specific infertility diagnosis are typically required to undergo six IUI cycles before becoming eligible for IVF. Compulsory health insurance covers an unlimited number of IUI cycles and up to three IVF procedures. In 2022, each additional IVF cycle cost approximately 4,000 euros, but because multiple embryos can be frozen per cycle, subsequent transfers could cost 1,000 euros or less.

4.3 Data

I use administrative data from Statistics Netherlands, covering all residents. Medical records from 2012–2017 are drawn from the Diagnosis-Treatment Combination system, which Dutch hospitals are required to maintain. The main variables are the procedure type—IUI or IVF—and the date of sperm or embryo insertion.

Importantly, these records indicate procedures that were actually carried out: a cycle that a woman cancels or chooses not to undergo does not appear. An attempt is therefore defined as an actual insertion, corresponds directly to the conceptual experiment, and the key variation is whether a completed insertion results in a birth. Procedure success is defined as having a child born within ten months of insertion with no subsequent procedures, a definition validated against medical records by [Lundborg et al. \(2017\)](#).

The medical registry also contains information on the treatment protocol surrounding each attempt, including semen testing and preparation, ovulation monitoring, and hormone monitoring. I use these variables as additional controls in sensitivity analyses.

Labor market data span 2011–2024 and include annual work hours and gross earnings derived

from tax records. Gross earnings are expressed in 2024 euros using the consumer price index. Annual work hours and gross earnings are censored at zero and the 99th percentile of their respective population distributions. Reported work hours include maternity leave, and earnings include maternity pay. In Appendix [SA2.8](#), I replicate the main analyses using adjusted hours that account for the maximum potential duration of unobserved leave. This adjustment affects estimates only in the first year after childbirth. I also use several demographic variables, including an indicator for higher education attainment, number of children, year and month of birth, marital status, and cohabitation status. Fertility is measured based on legal parenthood and is independent of coresidence. Adoptions in the Netherlands are extremely rare, with fewer than 40 occurring annually.

My main sample includes 12,892 Dutch-born women who underwent IUI to conceive their first child and were cohabiting with or in a registered partnership with a Dutch-born man at the time of the procedure. Throughout the analysis, I refer to these men as the partners. I address potential complications related to separation in Appendix [SA2.3](#). Following [Lundborg et al. \(2017\)](#), I exclude women whose first observed procedure occurred in the first data year to reduce the risk of including women with unobserved prior procedures, and those whose first observed procedure occurred in the last data year to avoid misclassifying failed procedures as successful because subsequent procedures are unobserved. Thus, the sample consists of women whose first observed fertility procedure is an IUI between 2013 and 2016.

For comparison, I use 84,337 Dutch-born women who conceived their first child between 2013 and 2016 without prior assisted conception procedures and were cohabiting with or in a registered partnership with the child’s Dutch-born father. To assess the role of the intensive fertility margin, I also use a sample of 11,426 women who underwent IUI or IVF to conceive a second child, imposing the same restrictions as in the main sample. All analyses use the full samples, with hours and earnings set to zero for individuals not in paid employment.

4.4 Sample Descriptives and Representativeness

Table [2](#) compares pre-IUI characteristics of couples whose first procedure succeeded (column 1) and failed (column 2). Because age strongly predicts IUI success, these comparisons are descriptive and not intended to test unconfoundedness. Despite this, in the year before the procedure, the groups are similar in earnings, employment rates, and education. The only notable difference is age: both women and their partners whose first IUI succeeded were around nine months younger, consistent with age being the key determinant of procedure success.

Table [2](#) also compares pre-IUI characteristics of reliers (column 4) and non-reliers (column 5), estimated using the baseline specification. Non-reliar women have 5–6% higher employment, earnings, and hours worked and are more likely to have a bachelor’s degree, indicating positive selection on labor market outcomes among women who conceive after IUI failure. Reliers are also about two years older on average, consistent with age being an important determinant of natural conception success. Given these baseline differences, comparisons between relier and non-reliar effects should be interpreted cautiously; when making such comparisons in the discussion, I therefore compare effects in relative terms.

Table 2: Pre-Parenthood Descriptives by First IUI Outcome, Latent Type, and Sample

	First IUI outcome			Latent type			Representative sample		
	Success (1)	Fail (2)	Dif. (1)-(2)	Relier (4)	Non-rel. (5)	Dif. (4)-(5)	Rep. (7)	vs relier (7)-(4)	vs non-rel. (7)-(5)
<i>Panel A: Woman</i>									
Work	0.910 [0.287]	0.916 [0.278]	-0.006 (0.008)	0.891 (0.008)	0.928 (0.005)	-0.037 (0.010)	0.935 [0.248]	0.043 (0.004)	0.007 (0.003)
Hours	1299.447 [548.456]	1296.947 [556.609]	2.500 (15.872)	1227.727 (15.532)	1314.270 (11.627)	-86.543 (18.933)	1307.439 [543.748]	79.712 (7.758)	-6.831 (6.632)
Earn. 1000s EUR	52.651 [30.414]	52.111 [30.804]	0.539 (0.879)	49.257 (0.771)	52.033 (0.559)	-2.776 (0.985)	46.264 [27.116]	-2.993 (0.388)	-5.768 (0.333)
Bachelor deg.	0.517 [0.500]	0.502 [0.500]	0.016 (0.014)	0.470 (0.011)	0.526 (0.009)	-0.056 (0.017)	0.533 [0.499]	0.063 (0.007)	0.007 (0.006)
Age	31.351 [3.883]	32.061 [4.264]	-0.710 (0.121)	33.078 (0.099)	31.096 (0.059)	1.982 (0.135)	28.871 [3.908]	-4.207 (0.056)	-2.225 (0.048)
<i>Panel B: Partner</i>									
Work	0.891 [0.311]	0.884 [0.320]	0.007 (0.009)	0.866 (0.008)	0.889 (0.007)	-0.023 (0.011)	0.894 [0.308]	0.028 (0.004)	0.004 (0.004)
Hours	1501.993 [631.601]	1485.385 [643.639]	16.608 (18.346)	1444.074 (15.694)	1502.663 (13.059)	-58.588 (22.055)	1484.782 [639.634]	40.707 (9.096)	-17.881 (7.799)
Earn. 1000s EUR	68.391 [35.450]	69.063 [36.557]	-0.672 (1.041)	66.739 (0.891)	69.290 (0.701)	-2.551 (1.174)	61.744 [32.606]	-4.994 (0.466)	-7.545 (0.401)
Bachelor deg.	0.416 [0.493]	0.406 [0.491]	0.010 (0.014)	0.351 (0.011)	0.439 (0.009)	-0.088 (0.017)	0.431 [0.495]	0.079 (0.007)	-0.009 (0.006)
Age	34.059 [4.988]	34.855 [5.493]	-0.796 (0.155)	36.056 (0.128)	33.868 (0.086)	2.188 (0.178)	31.429 [4.809]	-4.628 (0.069)	-2.439 (0.059)
Observations	1,372	11,520		2,348	3,141		84,337		

Note: *Success* – average among women whose first IUI succeeded; *Fail* – average among women whose first IUI failed; *Relier* – average among women who would remain childless if all IUIs failed; *Non-rel.* – average among women who would have a child even if all IUIs failed; *Rep.* – average in the representative sample of women who conceived their first child without assisted conception procedures. Relier and non-relier averages are estimated using the baseline specification. Earnings, hours, and employment are measured in the year preceding the first IUI for the IUI sample and in the year preceding conception for the representative sample; age and education are measured in the reference year. *Bachelor deg.* – indicator for completing a bachelor's degree. Standard deviations are in brackets. Standard errors are in parentheses; relier and non-relier statistics use bootstrap standard errors based on 300 replications.

Column (7) reports pre-pregnancy characteristics for the representative sample, while the final two columns report differences relative to reliers and non-reliers. The pattern is mixed: reliers have higher employment and hours worked, while these outcomes are similar for non-reliers; both groups have higher earnings, with a larger difference for non-reliers. Women in the IUI sample are also about three years older, which can largely be explained by eligibility requiring at least a year of unsuccessful natural conception attempts and treatment typically beginning about six months later, as well as by the representative sample additionally including unplanned births, which tend to occur at younger ages.

Six years after the first birth, women whose first IUI succeeded have about as many children as mothers in the representative sample: 1.88 compared with 1.96. They are, however, considerably more likely to have a multiple birth at their first delivery (7.1% vs. 1.4%), which could limit the external validity of the estimated effects. In Section 5.5, I estimate the direct effect of a multiple birth in both the IUI and representative samples and re-estimate the main specification excluding

couples with a multiple birth; both exercises are justified if multiple births are as good as random. A multiple birth raises the number of children six years later by about 0.45 in both samples, yet has little effect on women’s earnings or work hours, and the main estimates are largely unchanged when multiple births are excluded. Since a multiple birth represents a large shift in the intensive fertility margin, these results also suggest that this margin plays a limited role in the effects estimated here.

Beyond observable characteristics, IUI mothers may differ systematically in unobservables, limiting the generalizability of the estimates. For example, they may have a stronger desire for children and therefore respond differently to motherhood. Since only about 5% of Dutch mothers undergo IUI before their first child, such selection could in principle be substantial.

However, conditional on age, fertility problems are largely unpredictable. IUI mothers can therefore be viewed as drawn from a broader group of women with similar preferences who were trying to conceive and would have initiated treatment had they not conceived earlier. Within this group, reaching IUI depends on whether natural conception occurs first, which may itself be largely a matter of chance, making the IUI sample representative of this broader group.

The size of this broader group can be approximated using medical evidence on fecundity and the institutional setting. Medical guidelines typically require couples to try to conceive naturally for at least a year before becoming eligible for treatment, and IUI usually begins about six months later. With a mean IUI initiation age of 32, most women therefore began trying by about age 30.5. At this age, about 80% conceive naturally within 1.5 years (Leridon, 2004). If natural conception within this group is largely a matter of chance, each woman who reaches IUI therefore corresponds to about four otherwise similar women who conceived naturally beforehand. Combined with the 5% of mothers who undergo IUI, this implies that the broader group accounts for roughly a quarter of all mothers.

This approximation is likely conservative because many women who would eventually use IUI may conceive well before the age at which they would otherwise begin treatment, making their cumulative probability of natural conception higher than 80%. The magnitude is also plausible given that most couples who want children and fail to conceive naturally would eventually seek medical assistance, with IUI among the first treatments they receive. The estimates may therefore be relevant for a substantially broader population than IUI mothers alone.

4.5 Procedure Profiles

Figure 2 describes women’s procedure histories. Panel A shows the distribution of the number of procedures women undergo after the first failure and the number they would undergo if the first ten failed, estimated using the baseline specification. Women undergo on average 4.1 additional procedures, while average willingness is 6.8. Among women whose first IUI fails, 43% eventually undergo IVF, which accounts for 20% of all procedures. Panel B reports the relier share six years after the first IUI by willingness and shows little relationship between the two, although this relationship is irrelevant for identification.

Panels C and D report average work hours and earnings in the year before the first IUI by willingness, relative to the sample mean, estimated using the baseline specification. Both increase

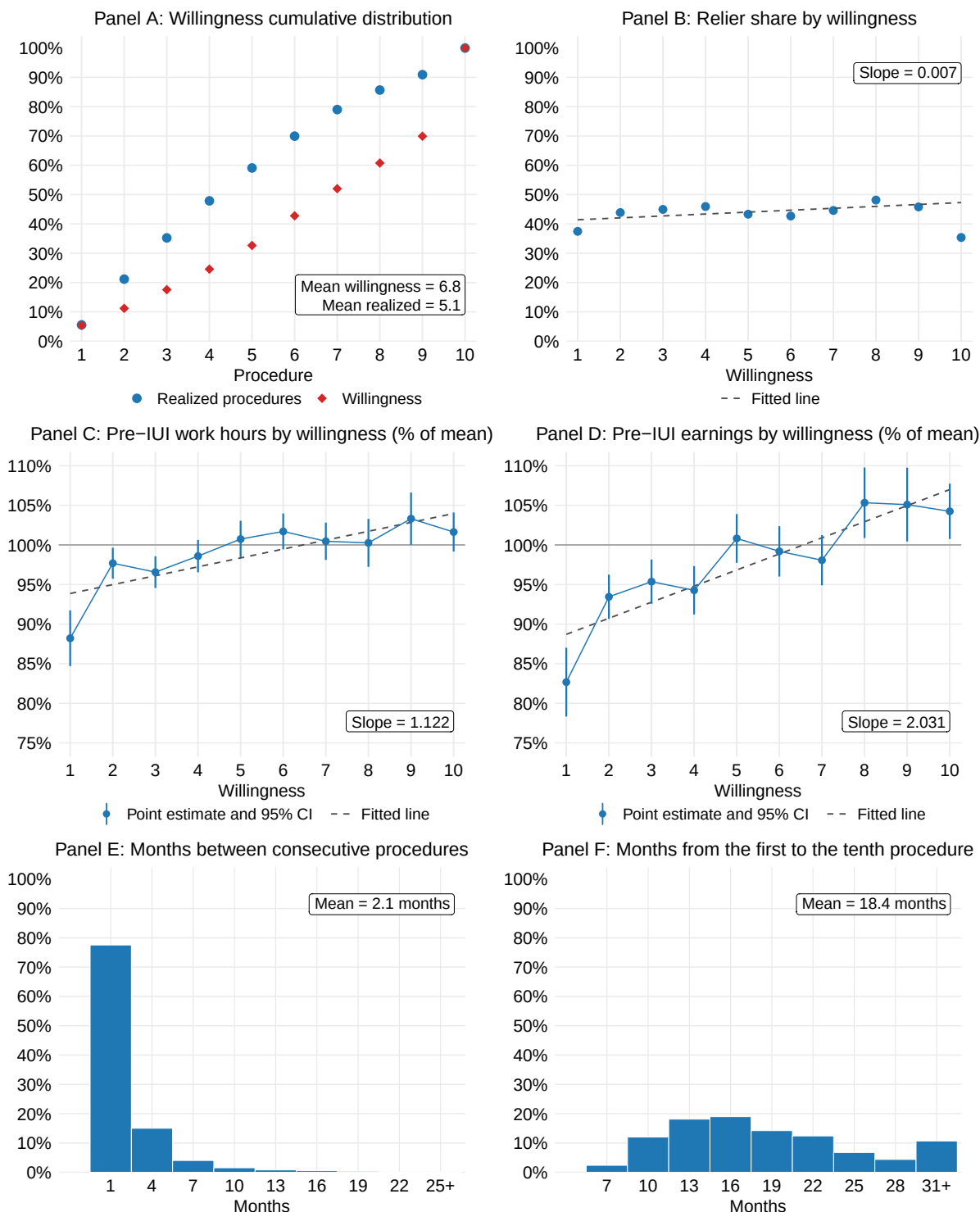


Figure 2: Willingness, Timing and Pre-IUI Characteristics

Note: Panel A: cumulative distribution of the number of assisted conception procedures pursued after the first failure, and of the number women would pursue if the first ten procedures failed. Panel B: relier shares by willingness. Panels C and D: women's work hours and earnings in the year before the first procedure, by willingness, expressed as a percentage of the overall mean; bootstrap standard errors are based on 300 replications. Panel E: distribution of the number of months between consecutive procedures. Panel F: number of months between the first and tenth procedures. Gaps are rounded down to whole months and grouped into three-month bins labeled by their midpoint. Dashed lines in Panels B, C, and D are least-squares fits. Panels A–D use the baseline specification.

systematically with willingness: women who would undergo more procedures in case of failure have higher pre-IUI labor market outcomes. If this pattern extends to post-IUI outcomes, women who eventually succeed through IUI will be positively selected, because those with greater willingness undergo more procedures and are therefore more likely to experience at least one success. This means that, in the absence of timing effects, using the outcome of the last IUI a woman undergoes as an instrument for parenthood would unambiguously understate its career cost.

Panels E and F report the distribution of gaps between consecutive procedures and the elapsed time between the first and tenth procedures among women who reach the tenth, using centered three-month bins. The modal gap between consecutive procedures is one month throughout the sequence, while the average is around two months. The average time between the first and tenth procedures is 18.4 months. Thus, women who continue treatment generally undergo subsequent procedures in relatively quick succession.

4.6 Unconfoundedness

Since women have limited direct control over IUI outcomes (see Section 3.1 for discussion), the main threat to unconfoundedness is that success depends on latent health factors that also affect labor market outcomes. If such factors also affect pre-IUI outcomes, this can be assessed by comparing pre-IUI characteristics between women whose procedures succeed and fail.

I make these comparisons separately for each procedure in the sequence, using only women who reach that procedure. This holds continuation decisions—which may themselves be endogenous—fixed and directly matches the identifying assumption, which conditions on reaching a given attempt. The comparisons use inverse-probability weights from the baseline specification that account for age and education, following [Lundborg et al. \(2017\)](#). Since subsequent procedures also include IVF, I additionally control for each partner’s age interacted with treatment type. This requires procedure success to be as good as random only among couples of the same ages undergoing the same treatment at the same point in the sequence, allowing selection into IUI or IVF to depend on women’s types and potential outcomes.

Table 3 reports the results. Across the sequence, the remaining differences in pre-IUI employment, work hours, and earnings are negligible. The table also reports differences in outcomes measured in the year before each procedure. These measures are often identical for the first few procedures, which typically occur only a few months apart, but increasingly differ later in the sequence. These differences are also small. Notably, because women for whom conception is impossible are necessarily in the failure group, these comparisons also suggest that such underlying fertility conditions are not systematically related to labor-market outcomes. Overall, these results support conditional local sequential unconfoundedness.

The remaining concern is that latent health may affect post-IUI labor market outcomes without leaving a trace in pre-IUI employment, hours, or earnings. I use the approach introduced in Section 3.6 to assess how large such selection might be. Panel A in Figure 3 applies the exercise to the first procedure, using cumulative earnings over the full observation window under first-attempt success—the early-motherhood potential outcome. The procedure assumes two unobserved types

Table 3: Balance Across Procedures

	Procedure number									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Panel A: Woman</i>										
<i>First procedure</i>										
Work	-0.011 (0.009)	0.008 (0.009)	-0.013 (0.010)	0.008 (0.010)	0.001 (0.011)	0.013 (0.012)	-0.003 (0.017)	0.006 (0.014)	0.011 (0.016)	0.019 (0.020)
Hours	0.923 (16.286)	19.299 (17.802)	-17.682 (19.795)	27.530 (20.340)	24.955 (22.272)	50.190 (25.782)	11.717 (30.737)	-4.531 (30.776)	41.572 (36.374)	76.042 (42.895)
Earn. 1000s EUR	1.850 (0.956)	1.234 (1.082)	0.613 (1.105)	0.287 (1.257)	2.796 (1.384)	2.071 (1.516)	-0.067 (1.561)	0.411 (1.843)	0.280 (2.057)	3.203 (2.869)
<i>Current procedure</i>										
Work		0.009 (0.009)	-0.011 (0.010)	0.011 (0.010)	-0.000 (0.012)	0.021 (0.013)	0.005 (0.017)	-0.006 (0.017)	-0.011 (0.022)	0.040 (0.022)
Hours		16.725 (17.960)	-19.258 (19.924)	14.255 (21.111)	18.232 (23.176)	58.119 (25.979)	14.366 (30.531)	-42.638 (31.920)	-10.830 (41.028)	97.769 (47.537)
Earn. 1000s EUR		1.238 (1.092)	0.430 (1.113)	0.040 (1.304)	2.923 (1.447)	2.121 (1.540)	0.758 (1.637)	-0.667 (1.906)	-1.490 (2.183)	3.739 (3.025)
<i>Panel B: Partner</i>										
<i>First procedure</i>										
Work	0.001 (0.010)	0.009 (0.010)	0.020 (0.010)	0.007 (0.012)	0.018 (0.012)	-0.002 (0.016)	-0.006 (0.016)	-0.023 (0.022)	0.002 (0.022)	0.034 (0.024)
Hours	2.719 (19.105)	21.260 (21.103)	25.860 (21.312)	23.631 (24.084)	34.325 (26.310)	-2.022 (31.967)	-12.127 (32.792)	-45.307 (41.345)	-10.891 (46.126)	21.889 (51.120)
Earn. 1000s EUR	0.028 (1.105)	-0.764 (1.221)	0.024 (1.262)	1.808 (1.470)	2.148 (1.598)	-1.705 (1.820)	0.553 (1.877)	1.665 (2.401)	-1.988 (2.630)	1.128 (3.574)
<i>Current procedure</i>										
Work		0.008 (0.010)	0.015 (0.011)	0.013 (0.012)	0.022 (0.013)	0.011 (0.016)	-0.016 (0.017)	-0.030 (0.022)	-0.015 (0.024)	0.061 (0.022)
Hours		20.906 (21.151)	17.521 (21.644)	16.404 (24.375)	50.694 (25.900)	31.633 (31.820)	-24.683 (33.496)	-25.072 (41.788)	-51.934 (48.426)	44.335 (51.225)
Earn. 1000s EUR		-0.568 (1.222)	0.235 (1.268)	1.110 (1.490)	1.856 (1.577)	-1.954 (1.783)	1.612 (2.003)	2.524 (2.534)	-1.506 (2.737)	1.761 (3.707)
Observations	12,892	10,880	9,082	7,462	6,006	4,713	3,463	2,418	1,652	1,048
Joint <i>p</i> -val.	0.415	0.932	0.508	0.297	0.287	0.088	0.747	0.335	0.344	0.546

Note: Column (*k*) reports the difference in average characteristics between women whose *k*-th procedure succeeded and those for whom it failed, among women who underwent a *k*-th procedure, adjusted for age and education using inverse probability weights from the baseline specification. *First procedure* measures characteristics in the year preceding the first procedure and is therefore the same variable in every column; *Current procedure* measures them in the year preceding the procedure of that column. Earn. – earnings. Standard errors in parentheses.

with high and low IUI success propensities. It assigns types among women whose first IUI succeeds by sorting on the potential outcome so as to maximize differences between types. The type-specific probabilities are then used to reweight women whose first procedure succeeded, recovering both the mean potential outcome and the mean pre-IUI outcomes of all who entered the procedure. The first is compared with the average outcome observed among successes, which gives the selection the model implies; the second with average the pre-IUI outcomes observed for everyone, which tests whether the model is consistent with the data.

The horizontal axis fixes the log odds ratio in success propensity between the two types beyond

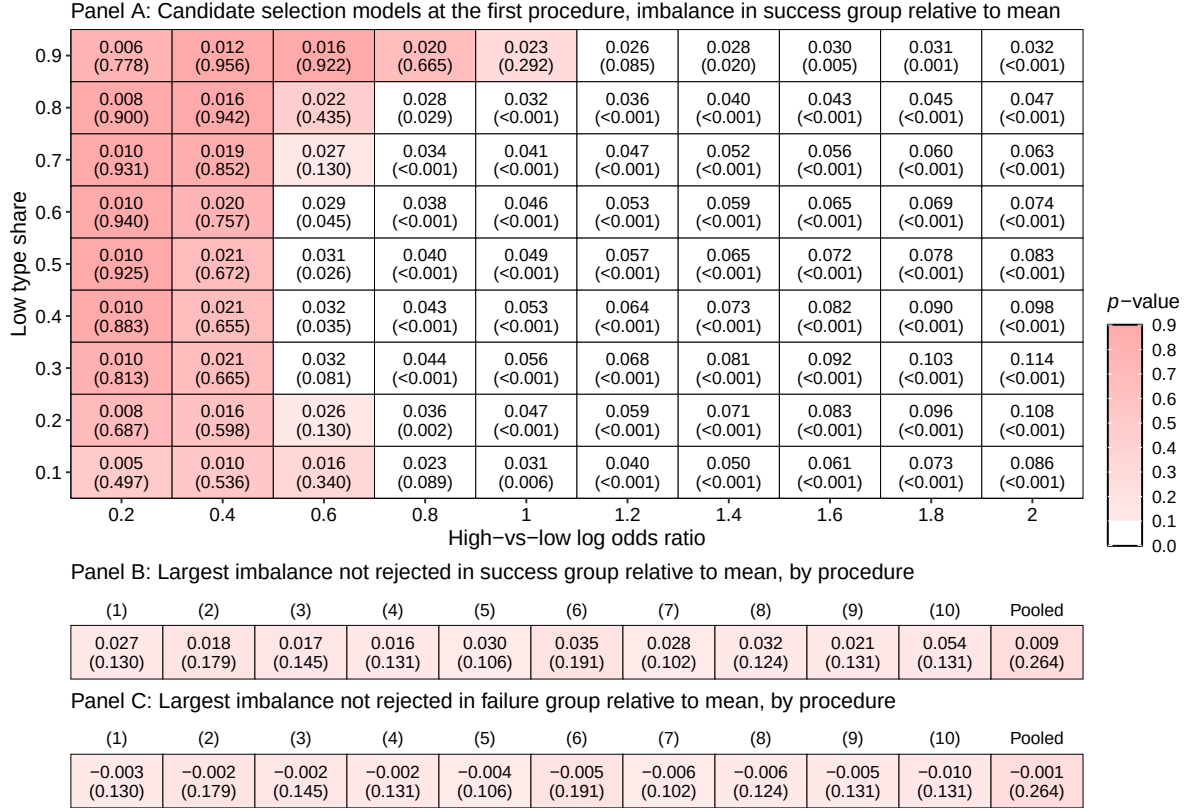


Figure 3: Selection on Unobservables in Potential Earnings

Note: *Panel A:* each cell represents a candidate model of selection on unobservables at the first procedure; Section 3.6 describes the test. The horizontal axis reports the log odds ratio in success propensity between the two unobserved types, beyond what is captured by the baseline specification; the vertical axis reports the share of the low type. The top number gives the implied difference in potential outcomes under IUI success between women whose procedure succeeds and the sample mean, relative to that mean. The outcome is cumulative earnings over the first six years following the first IUI. The number in parentheses is the p -value from a joint test of balance in mean pre-IUI employment, earnings, and hours for the woman and her partner. Cells not rejected at the 10% level are shaded. *Panel B:* largest non-rejected imbalance at each procedure, with the corresponding p -value below. *Panel C:* corresponding imbalance among women whose procedure fails. In Panels B and C, the final column pools procedures 1–10; the pooled p -value accounts for within-woman correlation across attempts.

what is captured by the baseline specification.¹³ The vertical axis fixes the share of the low type among women whose first IUI succeeds. The top number in each cell reports the largest implied difference in potential outcomes between women whose first IUI succeeds and the sample mean, relative to that mean; the number below reports the p -value from a joint test of whether the selection model is consistent with mean pre-IUI employment, earnings, and hours for the woman and her partner.

A high top number indicates sizable selection, while a low p -value indicates that the model can

¹³I use log odds because baseline propensities are heterogeneous and level shifts can imply probabilities outside the unit interval. For intuition, consider a baseline propensity of 10% and suppose the two types are equally common. A log odds ratio of 0.2 then corresponds to success rates of about 11% for the high type and 9% for the low type, while a log odds ratio of one corresponds to about 15% and 6%, respectively.

be rejected. The relevant benchmark is therefore the largest imbalance among models that cannot be rejected. Extreme models are rejected: log odds ratios above one are rejected at every type share. Models with less selection generally survive but imply little imbalance; at a log odds ratio of 0.2, the implied difference in potential outcomes is at most 1.0%. At the 10% level, the largest non-rejected imbalance is 2.7%. This means that women whose first IUI succeeds have potential outcomes 2.7% above the mean. Since about 10% of first procedures succeed, women whose first IUI fails then have potential outcomes at most 0.3% below the mean. As emphasized earlier, failure to reject does not imply such selection; it provides a conservative benchmark for the largest selection consistent with the observable balance tests.

Panel B reports, for each procedure, the largest non-rejected imbalance among women whose procedure succeeds relative to the mean among all women who reach that procedure; Appendix SA3 reports the full grids by type share and log odds ratio. The largest imbalance is similar across the first nine procedures and increases at the tenth as the sample shrinks. Because the analysis of subsequent procedures uses outcomes only among women whose procedures fail, the more relevant quantity is their imbalance relative to the corresponding sample mean, reported in Panel C. These imbalances are substantially smaller because most procedures fail, so the failure-group mean remains close to the overall mean. Across procedures, their potential outcomes lie at most 1% below the corresponding sample mean. The pooled specification, which tests balance jointly across all procedures, implies an imbalance below 0.1%. Applied across ten failures, the first-procedure and pooled benchmarks accumulate to roughly 3% and 1%, respectively. Section 5.6 uses these benchmarks to assess the sensitivity of the main estimates to non-rejected selection models.

Lastly, I investigate whether women who proceed to later procedures differ systematically in success propensity. For instance, if women who are generally healthier are both more likely to succeed at each procedure and more likely to continue treatment, success rates might be expected to change across the sequence as the sample becomes increasingly selected. Because success rates decline with age, I estimate adjusted rates holding age and other covariates fixed at their first-procedure means. Figure 4 shows little variation across procedures, providing little evidence of such sorting. Any remaining sorting would threaten identification only if this variation in success propensity were also related to potential labor market outcomes, for which the balance and selection-test results above leave little scope. Taken together, the evidence supports conditional local sequential unconfoundedness.

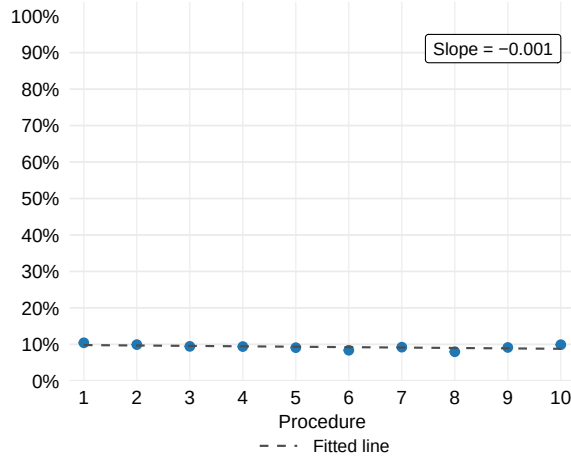


Figure 4: Success Rate by Procedure

Note: IUI success rates by procedure among women reaching each procedure, estimated using the baseline specification and holding covariates at their average values at the first procedure.

5 Results

Section 5.1 presents the effects on fertility. Section 5.2 quantifies the effects of parenthood and their contribution to within-couple gender gaps, and Section 5.3 the effects of parenthood timing. Section 5.4 examines how both sets of effects vary across women, and Section 5.5 isolates the role of the intensive fertility margin. Section 5.6 reports sensitivity analyses, and Section 5.7 compares the estimates with those obtained from conventional methods.

5.1 Fertility

Panel A in Figure 5 reports the share of mothers under first-IUI success and failure, by years since the first IUI, where year 0 denotes the first IUI year. Under success, only around one third have given birth by the end of year 0, since not all successful conceptions result in a birth within the same calendar year. From year 1 onward, the share is mechanically 100%. Under failure, the share rises gradually from zero, begins to plateau around year 4, and reaches 59% by year 6. This 59% is the non-reliar share: the remaining 41%, who are still childless by year 6, are reliers over the observation window.

Panel B reports the timing of motherhood among non-reliers. Under first-IUI failure, around 12% become mothers in the following year, 29% in year 2, 28% in year 3, 16% in year 4, and the remaining 15% in years 5 and 6. IUI failure therefore delays motherhood by 2.3 years on average among non-reliers. The estimated timing effects on labor market outcomes thus capture the consequences of an average delay of 2.3 years.

Panels C and D turn to the intensive fertility margin. Panel C shows that, among reliers, average fertility under first-IUI success is slightly above one child in year 1 and rises gradually to around 1.4 by year 6. Panel D reports the corresponding outcomes for non-reliers. Under success, average fertility rises from slightly above one child in year 1 to around 2.15 by year 6; under failure, it reaches around 1.5. The resulting gap of about 0.64 children in year 6 implies that the estimated timing effects reflect not only later motherhood but also differences in completed fertility over the

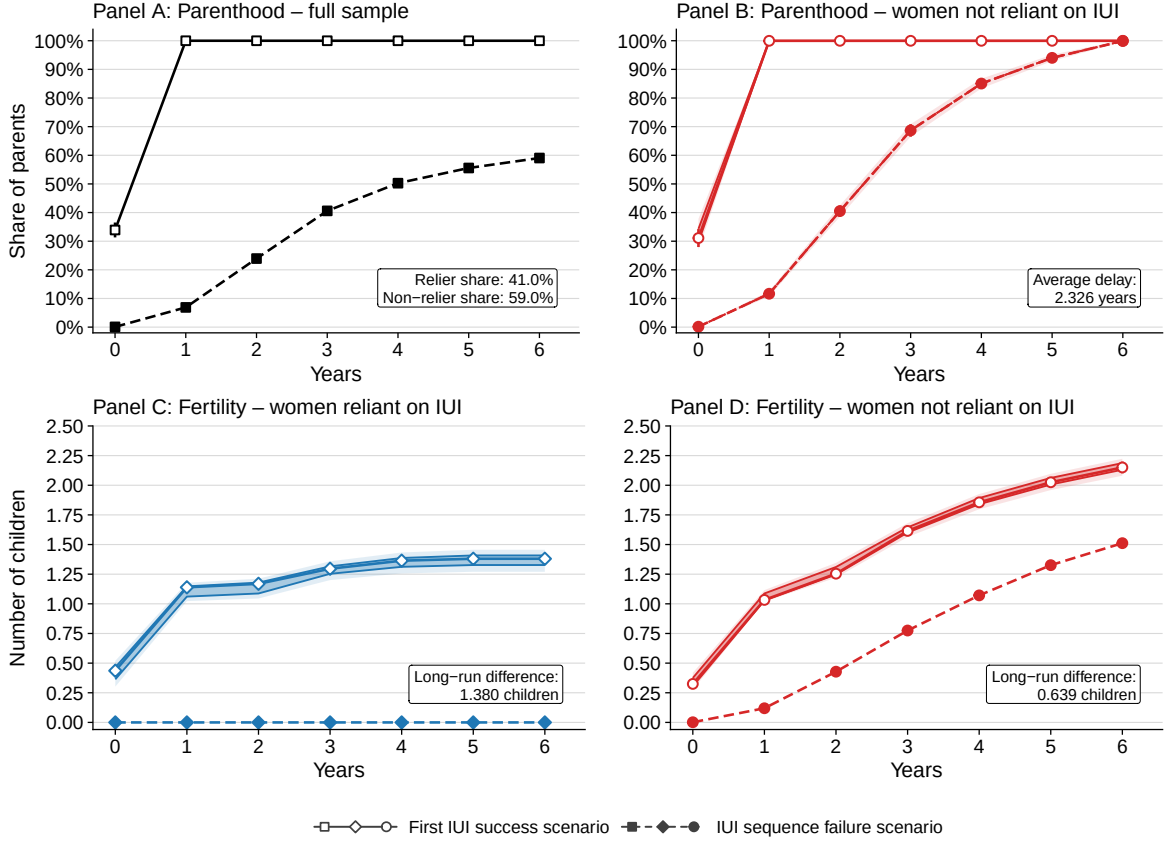


Figure 5: Parenthood and Fertility under Alternative IUI Sequence Scenarios

Note: Parenthood and fertility outcomes, by year relative to the first intrauterine insemination. Panels A and B report the share of women who are parents under each scenario, with Panel B restricted to women who would become parents within the window irrespective of IUI success. Panels C and D report the corresponding number of children born among women who rely on IUI to become parents and women who do not, respectively. The translucent band gives the estimated identified set for the partially identified parameters, and the fainter outer band the corresponding 95% confidence interval for the parameters; markers give point estimates under conditionally random reliance with their 95% confidence intervals. All estimates use the baseline specification.

observation window. I investigate the role of this intensive fertility margin in Section 5.5.

5.2 Effects of Parenthood

Panels A and B in Figure 6 present the effects of parenthood on women's annual work hours and earnings. The estimates apply to reliers and are reported in percentage terms relative to their average childless potential outcome in each year. Years -5 through -1 provide placebo tests of sequential unconfoundedness, monotonicity, and conditionally random reliance (see Sections 4.6 and SA2.4). The bounds include zero throughout the pre-IUI period, and the point estimates are close to zero, supporting these assumptions.

In the conception year, the bounds indicate negligible effects on either outcome. Effects become only slightly more negative in the following year, in part because maternity leave is included in

measured hours and earnings¹⁴. Between the second and sixth years of motherhood, the average bounds imply reductions of 12%–20% in annual work hours and 13%–25% in annual earnings. The point estimates lie approximately midway between the bounds. Averaged over the first six years of parenthood, the bounds imply reductions of 12%–19% in work hours and 13%–23% in earnings, with corresponding point estimates of 16% and 18%.

Figure 7 relates these effects to gender gaps in labor market outcomes, measured as the percentage by which women’s outcomes are lower than men’s. In the year before the first IUI, women work 14.5% fewer hours and earn 25.2% less than their partners. If the couple remains childless, over the six years following the first IUI these gaps increase modestly to 18.5% and 27.6%, respectively. If the first IUI succeeds and the couple becomes parents, the bounds imply gaps of 27.5%–39.3% for work hours and 32.7%–51.1% for earnings, with point estimates of 33.3% and 41.5%. The final block of the figure translates these estimates into the share of the post-parenthood gender gap caused by parenthood. The bounds imply shares of 34.9%–52.8% for work hours and 20.5%–44.5% for earnings, with point estimates of 44.9% and 34.3%, respectively. Thus, parenthood can account for up to half of the gender gaps in work hours and earnings.

One complication is that male partners are, on average, 2.5 years older, so outcomes measured at the same time relative to the first IUI correspond to systematically different ages. There is no unique way to adjust for this difference. Motivated by the absence of meaningful parenthood effects on men, I opt for a simple adjustment that keeps women’s outcomes indexed by years since the first IUI but shifts the partner’s outcomes so that the two are compared at the same age. For example, if a woman is 30 at the first IUI and her partner is 33, her outcome three years after the IUI, when she is 33, is compared with her partner’s outcome at age 33, which in this example is observed in the IUI year. This slightly reduces the sample because outcomes at the corresponding age are not observed for some couples with large age differences.¹⁵ Figure 7 also reports these age-adjusted estimates. The adjustment has little effect: the parenthood gap in work hours moves from 33.3% to 34.8% and in earnings from 41.5% to 40.4%, and the implied share of the post-parenthood gap caused by parenthood from 44.9% to 46.7% and from 34.3% to 33.3%.

5.3 Effects of Parenthood Timing

Figure 8 presents the effects of parenthood timing. The estimates are reported in percentage terms and compare outcomes when the first IUI fails and motherhood occurs later with outcomes when the first IUI succeeds and motherhood begins earlier. Panels A and B report effects on women’s work hours and earnings, respectively. As before, the pre-IUI years provide placebo tests of the identifying assumptions: the bounds contain zero throughout and the point estimates are close to zero. After the first IUI, the benefits of delay increase through year 2, when the bounds imply gains of 15%–21% in work hours and 11%–22% in earnings, with point estimates of around 17% for both outcomes. The effects then decline gradually, with point estimates of around 3% by year 6. Averaged over the full post-IUI window, both work hours and earnings are around 10%

¹⁴See Appendix SA2.8 for leave-adjusted estimates; only first-year estimates change meaningfully.

¹⁵The sample falls from 12,892 to 8,798 post-IUI and 8,762 pre-IUI.

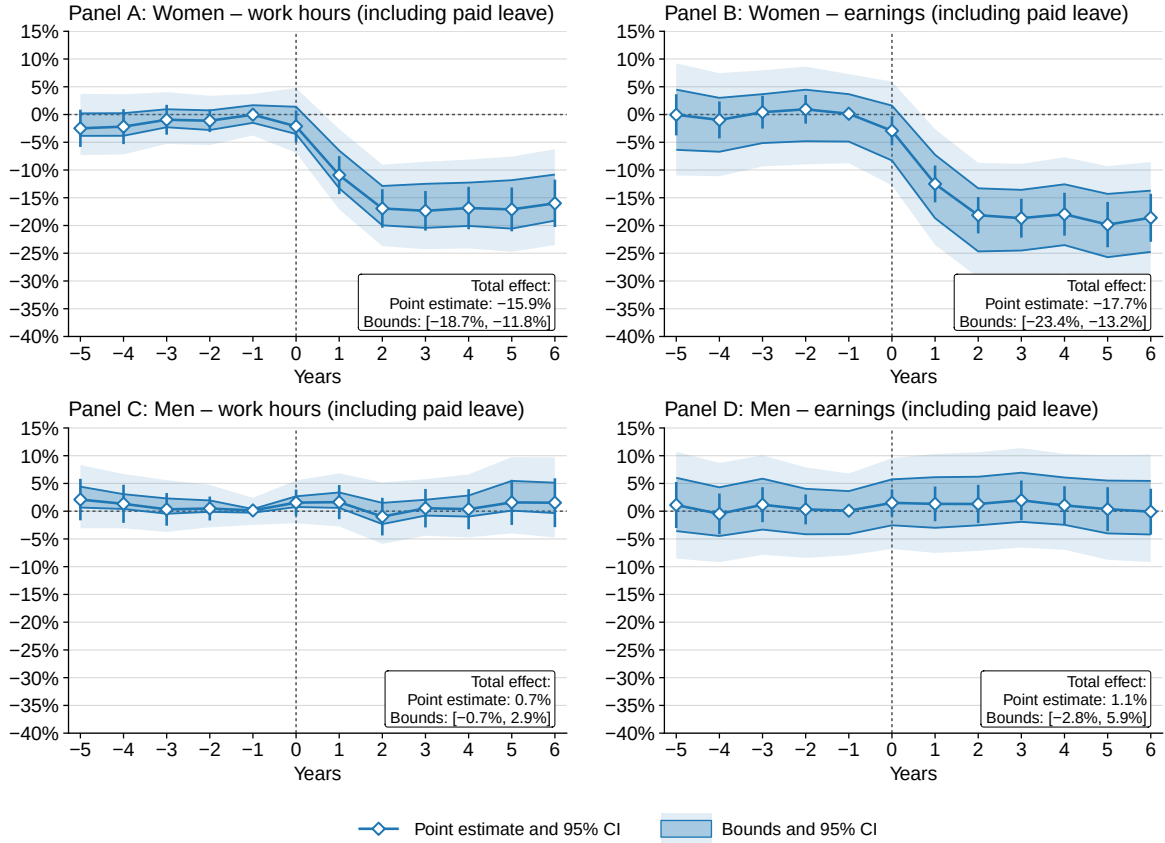


Figure 6: Effects of Parenthood on Work Hours and Earnings

Note: Effects of parenthood on annual work hours and earnings, by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level without children. Panels A and B report estimates for women, and Panels C and D for their partners. The translucent band gives the estimated identified set for the partially identified parameters, and the fainter outer band the corresponding 95% confidence interval for the parameters; markers give point estimates under conditionally random reliance with their 95% confidence intervals. The boxes report effects totalled over years 1–6, expressed as a percentage of the totalled counterfactual level without children.

higher. Thus, the estimates suggest substantial labor market benefits from delaying parenthood.

The vertical dashed line approximately marks the average delay in motherhood among non-reliers, 2.3 years. If all non-reliers became mothers exactly 2.3 years later, estimates before this point would capture the benefit of remaining childless relative to having already become a mother, while estimates afterward would compare later with earlier motherhood. In practice, the timing of later births varies. Before the average later birth, some non-reliers have already become mothers, which tends to understate the benefit of remaining childless. Afterward, some have not yet become mothers, which tends to overstate the benefit of having become a mother later.

To assess the importance of this timing heterogeneity, I construct alternative estimates for the two periods. For the first three years, I use the negative of the parenthood effect estimated in Section 5.2, scaled by the corresponding average outcome under first-IUI success. This directly measures the effect of remaining childless relative to early motherhood. From year 3 onward, I instead re-estimate the timing effect using parenthood status in each period rather than status

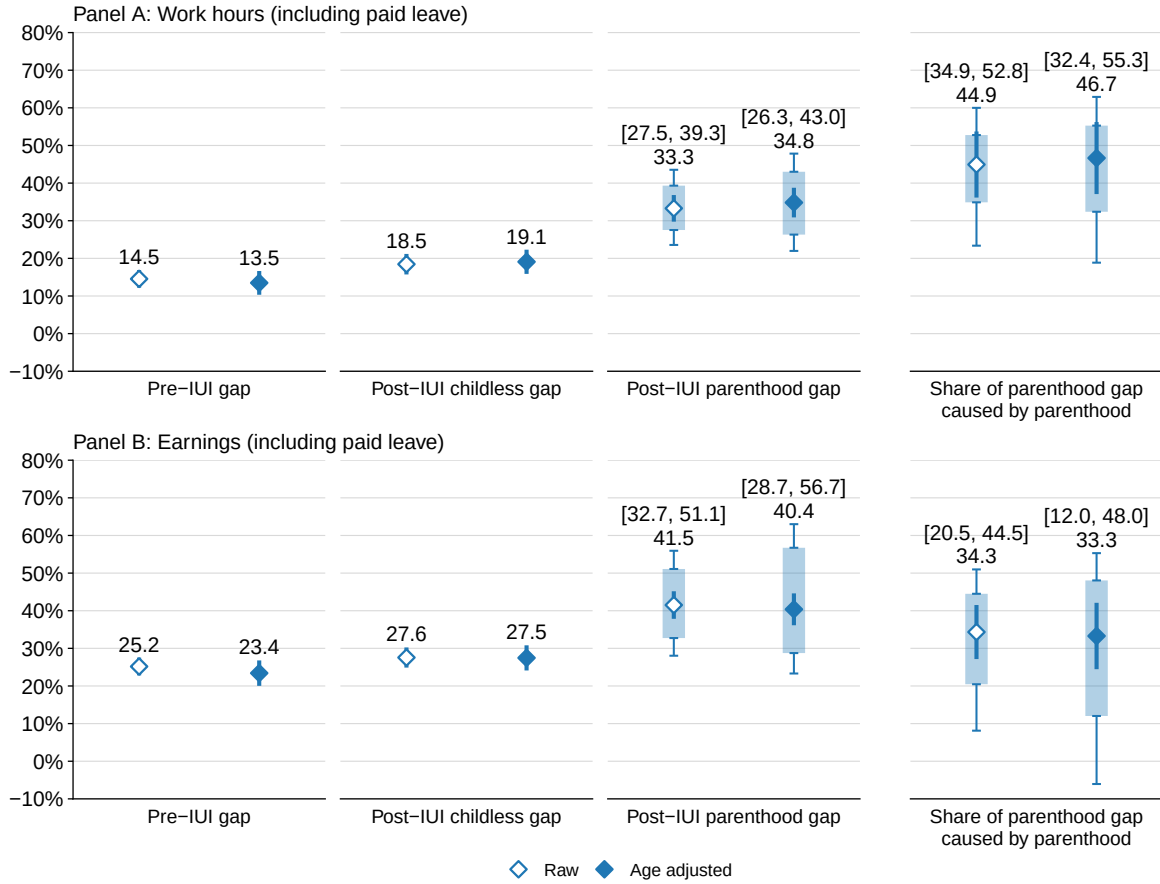


Figure 7: Parenthood and Within-Couple Gender Gaps

Note: Average within-couple gender gaps in work hours and earnings, measured as the percentage by which the woman's outcome falls short of her partner's. *Pre-IUI* is the gap in the year before the first procedure; *childlessness* and *parenthood* are the gaps over the six years following it under each scenario; the final block gives the share of the post-parenthood gap attributable to parenthood. Bars give the estimated identified set for the partially identified parameters, with whiskers showing the corresponding 95% confidence interval for the parameters; markers give point estimates under conditionally random reliance with their 95% confidence intervals. *Age adjusted* shifts the partner's outcomes so that the two are compared at the same age, as described in the text.

at the end of the observation window, effectively treating each period as the end of the window. This ensures that each estimate compares early mothers with non-reliers who have already become mothers by that period, so the effect reflects later rather than earlier motherhood. Appendix SA1 presents the formal identification results.

Figure 9 reports the resulting point estimates, while Figure SA1 in Appendix SA2.1 reports the bounds. The estimates change little overall and move in the expected direction. Before later motherhood, the average estimated gain increases from 9.4% to 11.0% for work hours and from 10.1% to 12.7% for earnings. After later motherhood, it falls from 7.7% to 5.7% for work hours and from 7.1% to 5.4% for earnings. Nearly all of the difference is concentrated in years 2 and 3, when most later births occur. The estimates are therefore very similar in years 0 and 1, when few later births have occurred, and from year 4 onward, when most have already occurred.

Taken together, the results suggest that delaying parenthood substantially increases women's

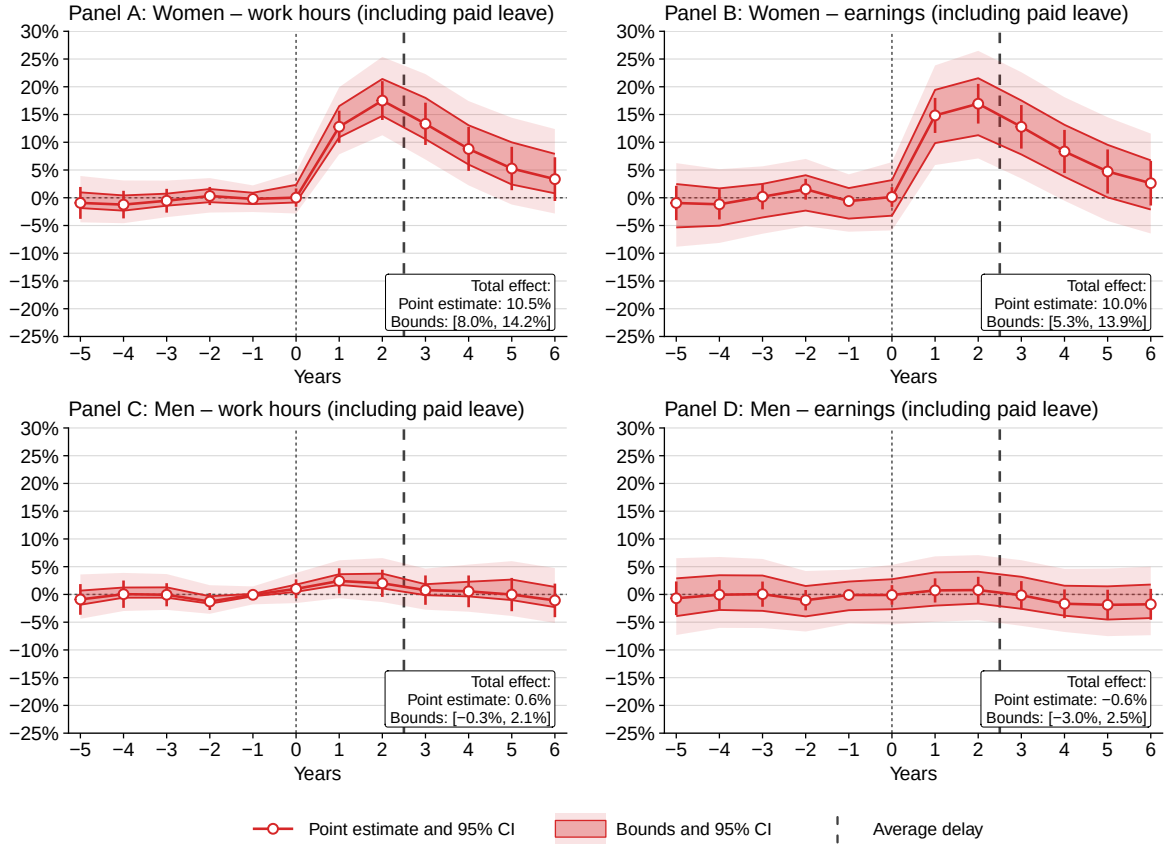


Figure 8: Effects of Later Parenthood on Work Hours and Earnings

Note: Effects of later parenthood on annual work hours and earnings, by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level under early parenthood. Panels A and B report estimates for women, and Panels C and D for their partners. The translucent band gives the estimated identified set for the partially identified parameters, and the fainter outer band the corresponding 95% confidence interval for the parameters; markers give point estimates under conditionally random reliance with their 95% confidence intervals. The boxes report effects totalled over years 1–6, expressed as a percentage of the totalled counterfactual level under early parenthood. The dashed line marks the average later conception moment, 2.3 years after the first procedure.

work hours and earnings. Importantly, these gains are not limited to eliminating the losses before delayed birth: women who become mothers later continue to have better labor market outcomes even after motherhood begins. This occurs even though they have become mothers more recently and impacts are often thought to be largest at the onset of motherhood (see [Lundborg et al., 2017](#), for discussion). It also occurs even though these delays are unintended, arising from IUI failure, which should lead to larger losses if fertility is timed to minimize career costs (see [Bensnes et al., 2025](#), for discussion). Overall, the evidence points to early childbearing as an important driver of career setbacks. It suggests that postponement helps mitigate the losses not only by eliminating them before the later birth but also by reducing them afterward. I present indicative calculations of how much postponement may offset the career cost of motherhood in Section 6.

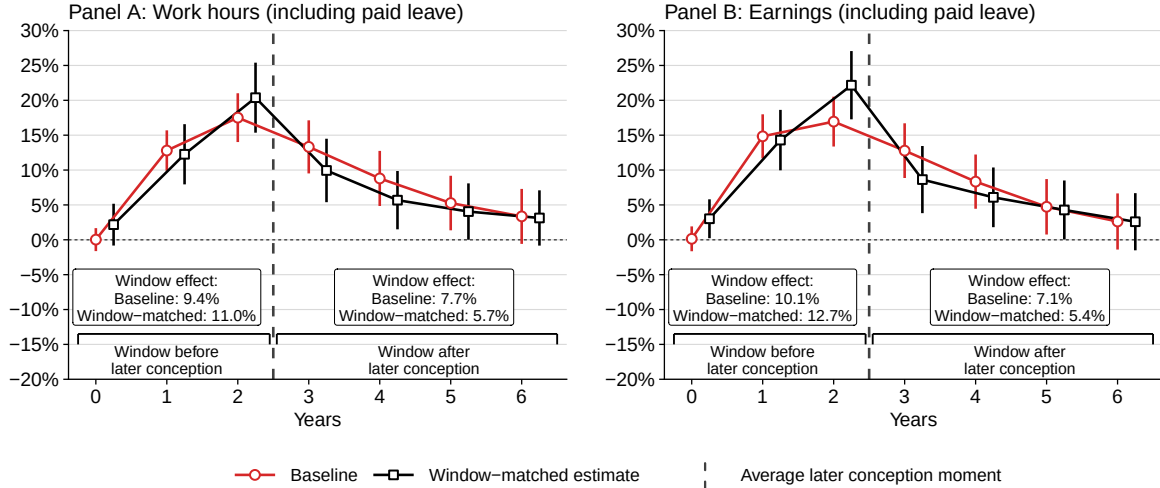


Figure 9: Window-Matched Estimates of the Effect of Later Parenthood on Women's Outcomes

Note: Effects of later parenthood on women's annual work hours and earnings, by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level under early parenthood. Estimates are point identified under conditionally random reliance. *Baseline* uses parenthood status at the end of the observation window. *Window-matched* uses the negative of the parenthood effect from Section 5.2, scaled by the average outcome under first-IUI success, for years 0–2, and re-estimates the timing effect using parenthood status in each period from year 3 onward. The boxes report totalled effects over each window, expressed as a percentage of the corresponding totalled counterfactual level. The dashed line marks the boundary between the two windows. Appendix SA2.1 reports the corresponding bounds.

5.4 Heterogeneity

Figure 10 splits both estimands at the median of three pre-IUI characteristics: the woman's own earnings, the within-couple earnings difference, and her age at the first procedure. The estimation approach follows Heiler (2024): nuisance functions are estimated jointly in the full sample, and the resulting moments are then split by the relevant pre-IUI characteristic to maximize precision. The dashed line indicates the pooled-sample point estimate.

The point estimates show the same pattern throughout. Losses from parenthood are somewhat larger, and gains from delaying it substantially larger, for women with lower pre-IUI earnings, for women who earn less relative to their partner, and for younger women. Although the parenthood and timing effects are estimated for distinct groups, this pattern is consistent with postponement being most valuable for women who face the greatest costs of early motherhood.

Statistical evidence for this heterogeneity is more limited. Only one set of differences in point estimates is statistically significant: the effect of later parenthood on work hours in Panel C, where all three splits differ at conventional levels. The estimates are 17.2% versus 6.9% by pre-IUI earnings ($p = 0.03$), 14.7% versus 6.5% by the within-couple earnings difference ($p = 0.05$), and 14.8% versus 5.9% by age ($p = 0.03$). Elsewhere, the groups cannot be statistically distinguished and the identified sets overlap substantially. The remaining heterogeneity should therefore be interpreted as suggestive.

Panels E and F examine heterogeneity in the intensive fertility margin. Fertility responses

vary along several dimensions, but do not consistently align with heterogeneity in work hours and earnings. For parenthood, younger women have substantially more children, yet labor market effects differ little by age. For timing, lower-earning women experience both a larger reduction in fertility and larger labor market gains, whereas fertility effects vary little by the within-couple earnings gap or age despite differences in labor market effects along these dimensions. Panel G also shows little heterogeneity in the duration of the delay. Changes in total fertility may therefore explain some of the heterogeneity in labor market effects, but do not appear to account for it systematically; I investigate the intensive fertility margin further in the next section.

Overall, the results in this section are consistent with the benefits of postponement being largest when careers are least established.

5.5 The Intensive Fertility Margin

The timing estimates compare women who become mothers earlier with those who become mothers later. As shown in Section 5.1, the latter group also has fewer children within the observation window: about 0.64 fewer by year 6. Differences in subsequent fertility may therefore be one mechanism behind the labor market returns to postponement. Because subsequent births may themselves be selective, the intensive margin cannot be fully separated. This section therefore provides indirect evidence on its importance.

Figure 11 reports three sets of results. The first, in Panels A and B, uses couples who already have one child and undergo IUI or IVF for a second, and reports sequential reduced-form estimates comparing success at the first procedure with failure of the whole sequence. Panel A shows that a successful first procedure raises the number of children by 0.76 by the end of the window, from 1.48 to 2.24, a larger shift than the 0.64 difference in the main analysis. Panel B reports the corresponding labor market effects. Cumulated over years 1–3 the point estimates are -3.2% for work hours and -1.0% for earnings; over years 1–6 they are -3.1% and -0.4% . These are small relative to the timing effects.

The second exercise, in Panels C and D, uses a multiple first birth as a shock to subsequent fertility, like [Rosenzweig & Wolpin \(1980\)](#). In both the IUI sample and a representative sample of couples having a first child between 2013 and 2016, I compare couples with multiple and singleton births, controlling for the baseline covariates. Panel D confirms that a multiple birth substantially increases completed fertility over the following six years, by 0.46 children in the IUI sample and 0.43 in the representative sample. Panel C shows that the corresponding labor market effects over the same six-year window are small: 2.0% for work hours and 0.6% for earnings in the IUI sample, and -2.6% and -2.4% in the representative sample.

Finally, I assess the sensitivity of the main estimates to the exclusion of multiple births. Panel E re-estimates the effect of parenthood dropping couples with a multiple birth at the first IUI. The point estimates are essentially unchanged: -15.9% for work hours, against -15.9% in the baseline, and -16.8% for earnings, against -17.7% .

Together with the evidence in the previous section that the benefits of postponement are larger for younger women with less established careers, despite limited heterogeneity in intensive-margin

responses along these dimensions, the results in this section further suggest that the intensive margin is not a major driver of either the parenthood or timing effects.

5.6 Sensitivity

The baseline specification predicts IUI success using both partners' age, education, and procedure type. Age is the single most important clinical determinant of IUI success, while education may absorb behavioral or health differences associated with success probabilities, for instance through smoking. One may nevertheless worry that women who are similar along these observed dimensions differ in their behavior during treatment, and that these differences are related to potential labor-market outcomes. To investigate this further, I additionally control for detailed features of the procedure protocol. These controls capture differences in treatment behavior directly and may also proxy for remaining differences in the behavior or circumstances surrounding conception.

I construct six indicators for whether a given activity occurs within a fixed window around the insemination date: a semen analysis in the preceding year, sperm preparation on the day of insemination, ovulation ultrasound or hormonal monitoring in the preceding fortnight, and progesterone administration in the fortnight before or after. Each is measured at the level of the individual procedure and included among the controls for the corresponding procedure.

Figure 12 re-estimates the per-period effects adding three progressively richer sets of these controls: a single indicator for whether any of the six activities is recorded; three indicators grouping them into male-factor testing and preparation, cycle monitoring, and progesterone; and the six separately. The estimated paths are indistinguishable from the baseline. Appendix SA2.9 reports the corresponding bounds on the window totals, which are likewise nearly identical to the baseline estimates.

I next assess how sensitive the estimated effects are to selection on unobservables. Section 3.6 quantifies how much imbalance in post-IUI potential outcomes the observed balance in pre-IUI outcomes is unable to rule out, under a particular model in which women with higher success probabilities have systematically higher potential outcomes. I use these magnitudes to adjust observed outcomes accordingly. For example, if the test implies that women who succeed at the first procedure may have potential outcomes up to $x\%$ above the mean among women reaching it, I scale down the outcomes of women who actually succeed there by the corresponding amount. Likewise, if women who fail a given procedure may have potential outcomes a given percentage below the corresponding mean, I scale their outcomes up accordingly. These failure adjustments compound along each woman's realized treatment sequence: a woman who fails three procedures has her outcomes scaled up three times in succession.

The key additional assumption is that selection operates proportionally across potential outcomes. For instance, if women with higher success probabilities have early-motherhood outcomes that are 1% higher, their late-motherhood and childlessness outcomes are also assumed to be 1% higher. This seems natural if the main concern is latent health or other persistent characteristics that affect labor-market outcomes similarly across fertility scenarios. Appendix SA2.10 gives the procedure in detail.

Importantly, the exercise does not require committing to the particular latent-type selection model used in Section 3.6. It accommodates any selection process in which women who conceive at a given procedure have proportionally higher potential outcomes than the mean among women reaching it, while women who fail have proportionally lower potential outcomes; the model in Section 3.6 is used only to calibrate the benchmark magnitudes.

I use two adjustment benchmarks. The first—mild selection—uses the largest differences not rejected over the entire procedure sequence, implying that women whose procedure succeeds have potential outcomes 0.9% higher than the mean among women entering that procedure, while those whose procedure fails have potential outcomes 0.1% lower. The second—strong selection—uses the largest difference not rejected at the first procedure, where the corresponding gaps are 2.7% and 0.3% for the success and failure groups, respectively.

Figure 13 reports the resulting estimates. Under the milder scenario, the effect of parenthood on work hours moves from -15.9% to -17.2% , and the effect on earnings from -17.7% to -18.9% ; under the stronger scenario, the corresponding effects move to -19.8% and -21.4% . The timing effects move from 10.5% to 12.2% and then 15.7% for work hours, and from 10.0% to 11.8% and then 15.4% for earnings. Appendix SA2.10 reports the corresponding bounds, which move in similar proportion. The adjustment therefore changes the estimates only modestly and, in both cases, implies larger career costs of motherhood and larger benefits of postponement.

The Supplementary Appendix reports further sensitivity analyses. Appendix SA2.3 bounds the effects for women who, in the event of procedure failure, would remain partnered and not begin antidepressant use, and reports the effect of conceiving at the first IUI on antidepressant uptake and partnership stability. Appendix SA2.7 varies the maximum number of procedures used for estimation, and Appendix SA2.6 the number of cross-fitting folds and trees. Appendix SA2.4 tests monotonicity and Appendix SA2.5 relaxes it. Appendix SA2.8 adjusts work hours for unobserved parental leave. The main results are generally insensitive to these changes.

5.7 Comparison with Conventional Methods

Figure 14 compares my estimates of the effect of parenthood with those from conventional approaches, all summed over the six years following the first IUI and expressed as a percentage of the corresponding counterfactual level. Appendix SA2.2 reports the corresponding per-period estimates and provides estimation details.

Panels A1 and B1 report my baseline estimates, and Panels A2 and B2 place them alongside bounds that use only the first IUI and do not impose monotonicity, which are equivalent to Zhang & Rubin (2003) and Lee (2009). These bounds run from -59.4% to 20.2% for work hours and from -57.3% to 27.2% for earnings, and therefore do not rule out large effects of either sign. My bounds, at $[-18.7\%, -11.8\%]$ and $[-23.4\%, -13.2\%]$, are roughly ten times narrower. This improvement comes from exploiting the sequential structure of assignment and is a theoretical guarantee of the approach. Using only the first procedure identifies the average control outcome for compliers, whereas the full sequence identifies it for the larger group of reliers. Because this group is larger, less trimming of the treated mean is required.

Panels A3 and B3 report conventional instrumental-variable estimates. Scaling the cumulative reduced form by the first stage at the end of the window—the *endline* estimator—gives -33.1% for work hours and -34.9% for earnings, well outside the identified set. Scaling each period’s reduced form by its contemporaneous first stage before cumulating gives -21.5% and -22.8% , much closer to the lower bound.

Panels A4 and B4 report the sequential IV estimator of Section 3.3.1, which targets *reliers* rather than *compliers*. It gives -28.6% and -29.9% at the endline and -20.2% and -21.3% per period. The fact that the sequential estimates remain close to the conventional IV estimates indicates that the difference from my estimates is not primarily due to heterogeneity in treatment effects between *reliers* and *compliers*. More specifically, because my estimates and sequential IV target exactly the same group, the remaining difference reflects contamination by timing effects, which causes both conventional and sequential IV estimators to overstate the career cost of parenthood.¹⁶

Panels A5 and B5 instead define the instrument as success at the last procedure a woman undergoes. The estimates fall to -25.7% and -30.1% at the endline and to -17.2% and -21.1% per period. This is consistent with the evidence in Section 4.5 that women who undergo more procedures are positively selected, which should generate smaller parenthood effect estimates than conventional and sequential IV. In my application, these estimates therefore move closer to the baseline estimates, since this selection works in the opposite direction from the timing bias that causes conventional IV to overstate the cost of parenthood.

Recursive IV estimates, in Panels A6 and B6, adjust for control women who become mothers later, following [Gallen et al. \(2024\)](#) and [Bensnes et al. \(2025\)](#). The conventional recursive estimator gives -14.9% for both outcomes and the sequential recursive estimator -16.7% for work hours and -17.1% for earnings. Both lie in the upper half of the identified set and close to my point estimates. Unlike in [Bensnes et al. \(2025\)](#), however, the recursive adjustment moves the estimates toward smaller losses rather than larger ones. The direction of this adjustment depends on how motherhood effects evolve with duration. In [Bensnes et al. \(2025\)](#), losses are largest shortly after birth and fade over time, so adjusting for later motherhood in the control group moves the estimate toward larger losses. In my setting, losses instead rise over the first few years before flattening ([Appendix SA2.2](#)), so the same adjustment moves the estimate toward smaller losses.

Panels A7 and B7 report event-study estimates following [Kleven, Landais, & Søgaaard \(2019\)](#). These are -25.4% and -28.5% in the IUI sample and -31.2% and -30.0% in a representative sample of couples having a first child over the same period, all substantially larger than my baseline estimates. One reason is selection into parenthood. If women who become mothers later are positively selected, using them to identify the counterfactual, as in an event-study design, tends to overstate the losses from motherhood. A second reason is that the counterfactuals differ: in my design, women are already trying to conceive and fail to do so, whereas in an event study women observed before conception may not yet have started trying. I return to this in the next section.

¹⁶The timing effects are effectively obtained by comparing the sequential IV estimates with the parenthood-effect bounds, so the presence of timing effects directly implies this contamination.

These results have an important implication for how my estimates compare with the existing literature. Conventional estimators applied to my data span -14.9% to -33.1% for work hours and -14.9% to -34.9% for earnings, while my baseline estimates, -15.9% and -17.7% , lie at the small end of both ranges. The relatively large parenthood effects I find compared with IVF designs in Scandinavian settings, and their similarity to estimates from contraceptive-failure designs—which might be expected to be larger because unintended births generate larger losses—therefore do not appear to be driven by methodological differences. This instead suggests that parenthood losses are larger in the Dutch setting.

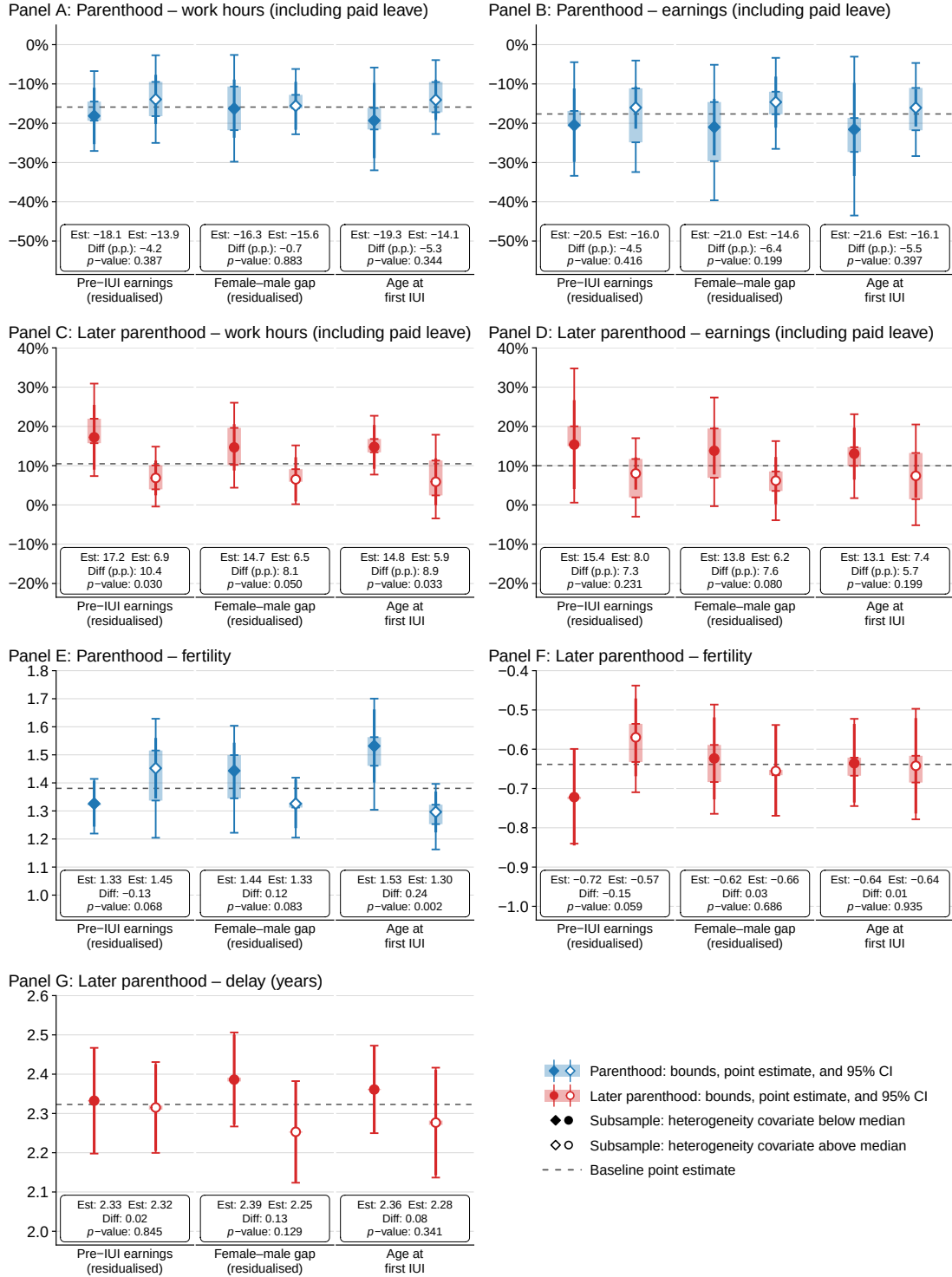


Figure 10: Heterogeneity in the Effects of Parenthood and Its Timing on Women's Outcomes

Note: Heterogeneity in effects by above- versus below-median values of three characteristics measured in the year before the first procedure: (1) pre-IUI earnings residualized on age, (2) the female–male earnings gap residualized on age and female earnings, and (3) age. Panels A–D express effects as percentages of the counterfactual level; Panels E and F report numbers of children, and Panel G years. Bars give the estimated identified set for the partially identified parameters, with whiskers showing the corresponding 95% confidence interval; markers give point estimates under conditionally random reliance with their 95% confidence intervals. The dashed line reports the full-sample estimate. Estimation follows Heiler (2024).

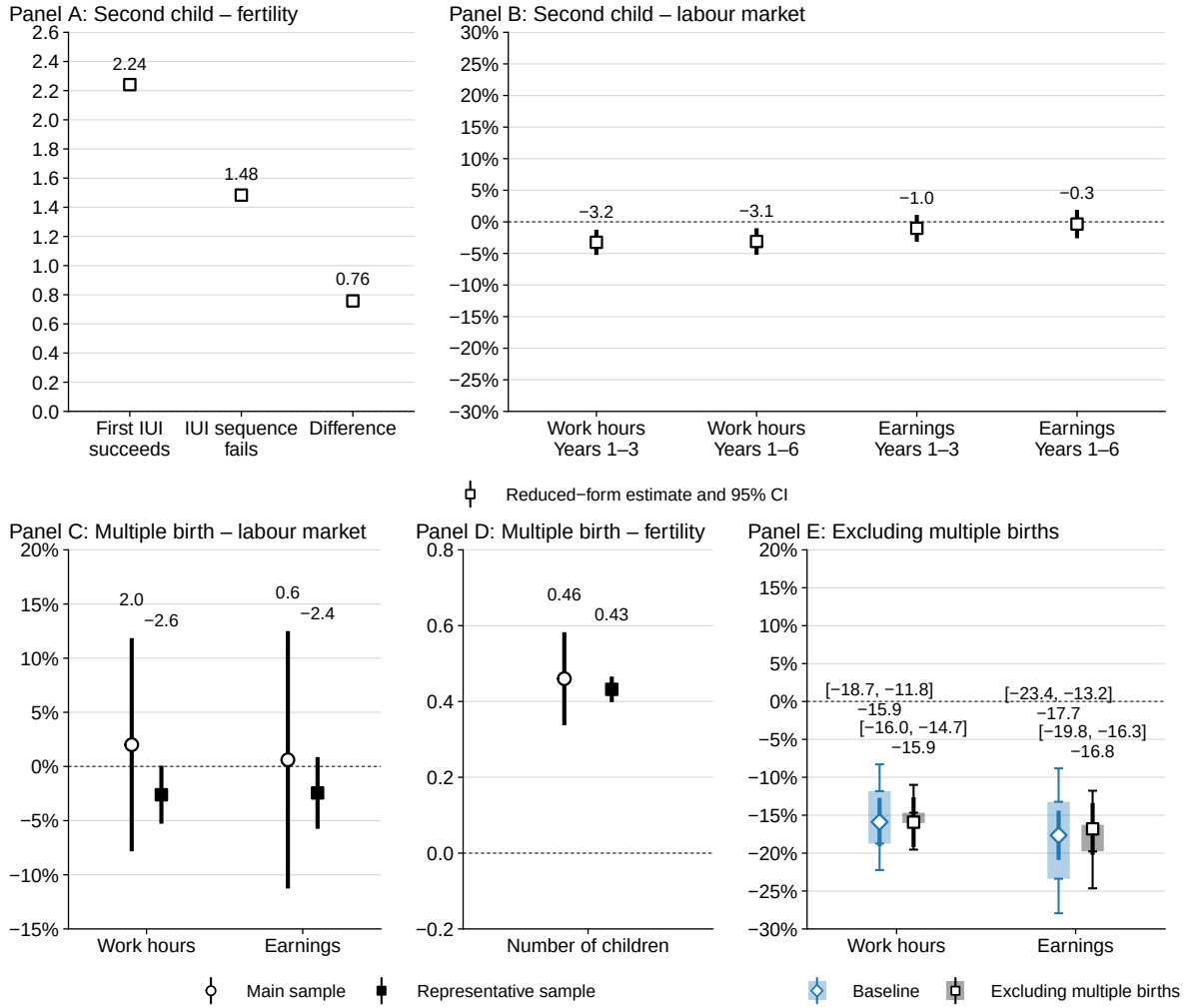


Figure 11: The Intensive Fertility Margin

Note: Panels A and B use couples who already have one child and undergo IUI or IVF for a second, and report point-identified reduced-form estimates. Panel A reports the number of children under first-procedure success and sequence-failure scenarios, six years after the first procedure; Panel B reports the effects of first-procedure success relative to sequence failure on work hours and earnings totalled over years 1–3 and 1–6, expressed as a percentage of the corresponding level under sequence failure. Panels C and D compare couples with multiple and singleton births, controlling for the baseline covariates, separately in the IUI sample and a representative sample of couples having a first child between 2013 and 2016. Work hours and earnings are totalled over six years from first conception, while the number of children is measured six years after first conception. Panel C reports labor market effects as a percentage of the mean among singleton births, and Panel D effects on the number of children. Panel E reports effects of parenthood in the main sample on women’s outcomes totalled over years 1–6, under the baseline specification and the same specification excluding couples with a multiple first birth. Translucent bars give the estimated identified set for the partially identified parameters, with whiskers showing the corresponding 95% confidence interval; markers give point estimates under conditionally random reliance with their 95% confidence intervals. Work hours and earnings include paid leave throughout.

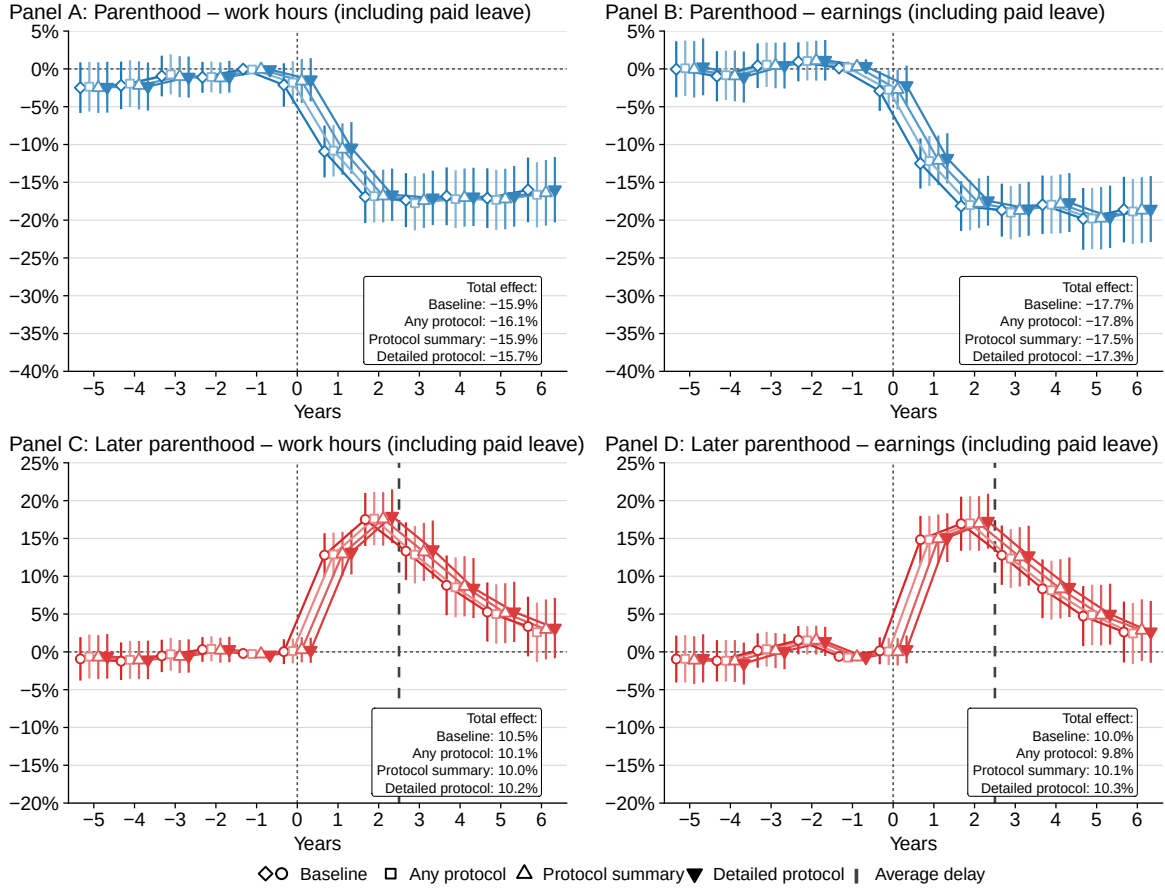


Figure 12: Sensitivity of Effects on Women's Outcomes to Controls for the Clinical Protocol

Note: Point estimates under conditionally random reliance, by year relative to the first intrauterine insemination, for the effects of parenthood (Panels A and B) and later parenthood (Panels C and D), expressed as a percentage of the counterfactual level. *Baseline* is the specification used throughout the paper, which predicts procedure success using both partners' age and education and procedure type. Relative to the baseline, *Any protocol* adds an indicator for whether any of six procedure-level care activities is recorded within its relevant window: semen analysis, sperm preparation, ovulation ultrasound, hormonal monitoring, and progesterone administration before or after insemination. *Protocol summary* adds three indicators grouping these activities into male-factor testing and preparation, cycle monitoring, and progesterone; *Detailed protocol* adds all six separately.

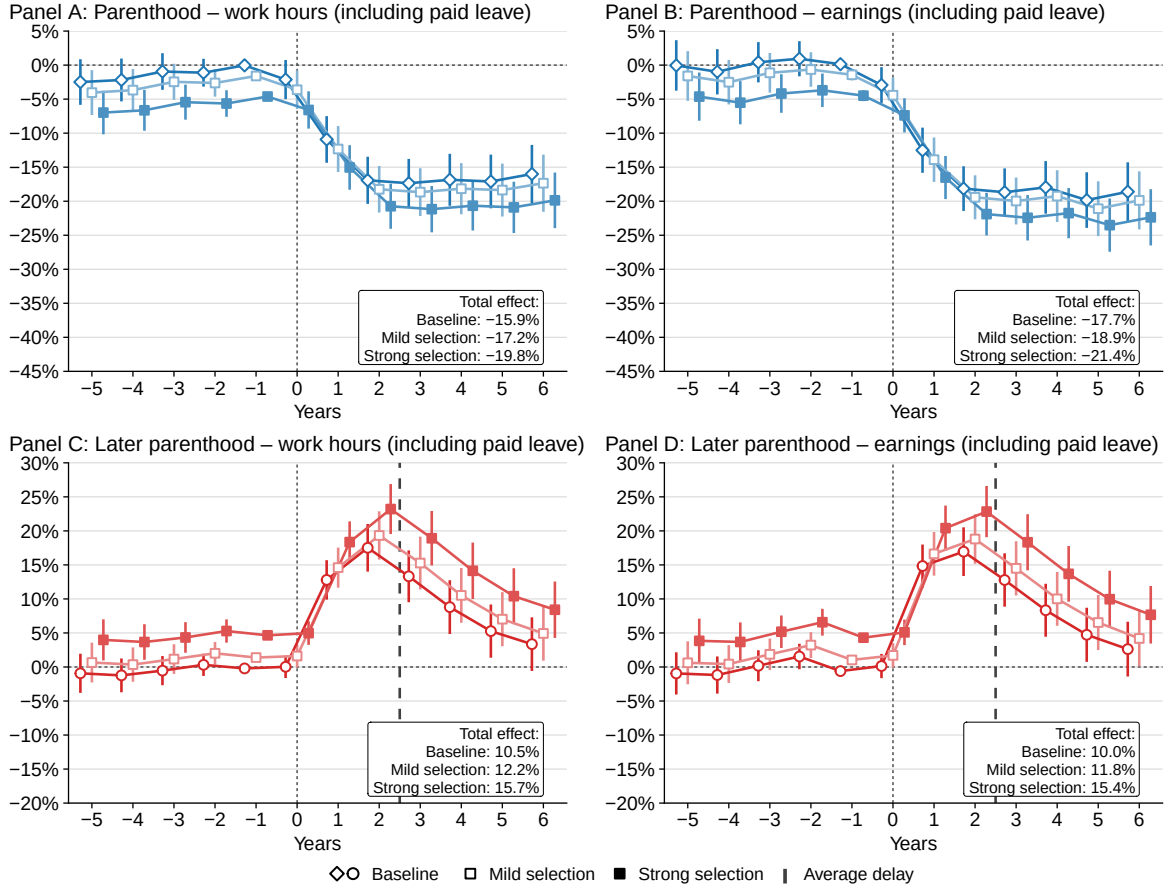


Figure 13: Sensitivity of Effects on Women's Outcomes to Selection on Unobservables

Note: Point estimates under conditionally random reliance, by year relative to the first intrauterine insemination, for the effects of parenthood (Panels A and B) and later parenthood (Panels C and D), expressed as a percentage of the counterfactual level. Realized outcomes are rescaled according to each woman's number of procedures, capped at ten. Under *strong selection*, outcomes are scaled by 1.027^{-1} for women who conceive at the first procedure and by 0.997^{-A} for women with no IUI success; under *mild selection*, the corresponding factors are 1.009^{-A} and 0.999^{-A} , where A denotes the number of procedures. The mild adjustment uses the largest imbalance not rejected by the pooled balance test in Section 3.6; the strong adjustment uses the largest imbalance not rejected at the first procedure. Appendix SA2.10 reports the corresponding bounds.

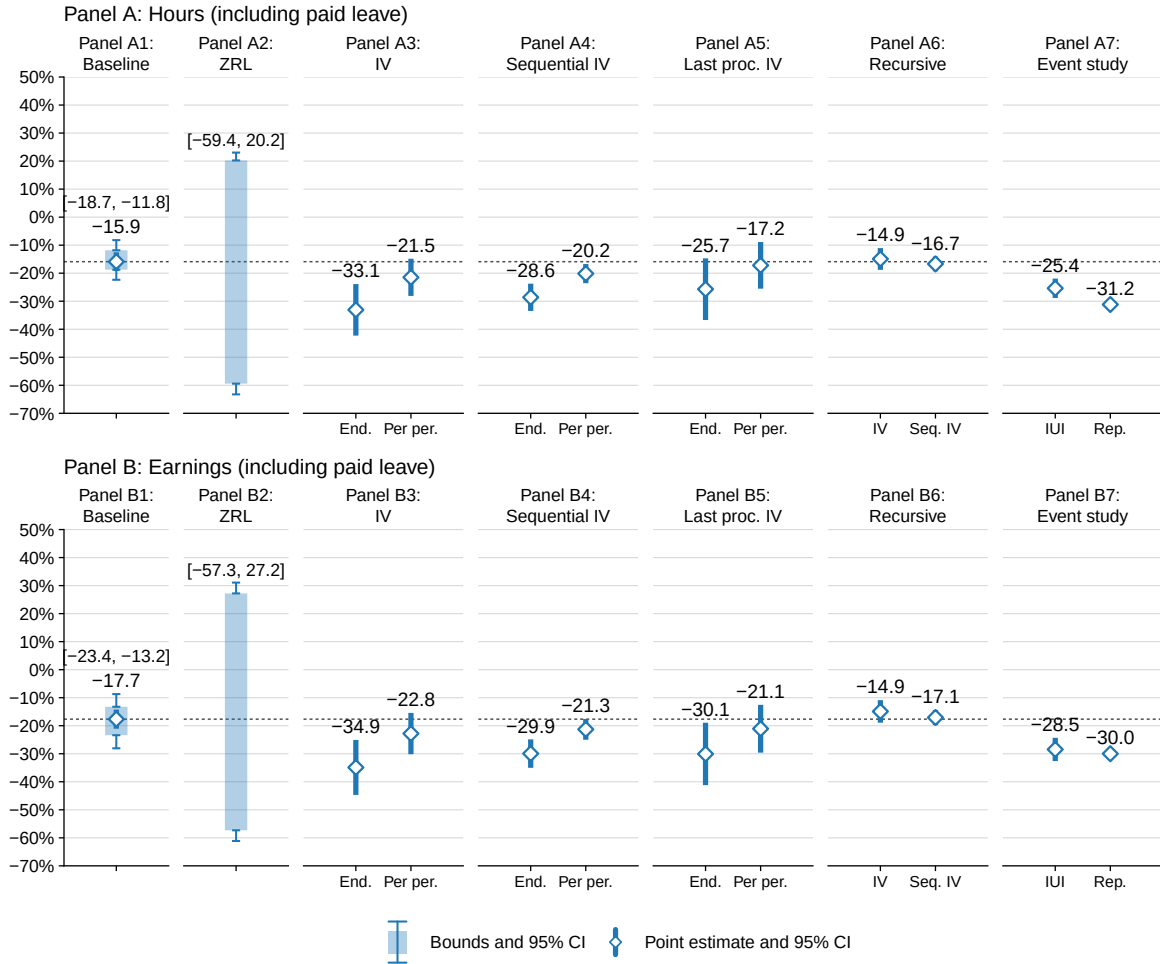


Figure 14: Comparison of Estimators of the Effect of Parenthood on Women's Hours and Earnings

Note: Effects of parenthood on women's work hours and earnings, totalled over years 1–6 relative to the first intrauterine insemination, expressed as a percentage of the totalled counterfactual level without children. Panels A1 and B1 report the baseline estimates: the estimated identified set and the point estimate under conditionally random reliance. Panels A2 and B2 report bounds using only the first procedure and imposing no monotonicity, following Zhang & Rubin (2003) and Lee (2009). Panels A3–A5 and B3–B5 report instrumental-variable estimates: first-procedure success versus failure; first-procedure success versus sequence failure; and success at the last procedure undergone. For each instrument, *endline* scales the cumulative reduced form by the first stage at the end of the window, while *per period* scales each period's reduced form by its contemporaneous first stage before cumulating. Panels A6 and B6 report recursively adjusted estimates, following Gallen et al. (2024) and Bensnes et al. (2025), applied to the first-procedure-success versus first-procedure-failure and first-procedure-success versus sequence-failure reduced forms. Panels A7 and B7 report event-study estimates for the intrauterine-insemination sample and a representative sample of women having a first child between 2013 and 2016, following Kleven, Landais, & Sogaard (2019). The horizontal dotted line gives the baseline point estimate from Panels A1 and B1. See Appendix SA2.2 for further details.

6 Discussion

In this section, I discuss how my estimates may generalize to scenarios in which women make different fertility and career plans, and I present suggestive calculations on how much delaying motherhood may mitigate its career costs.

The focal point of my analysis is the moment of the first IUI procedure, whose timing is chosen by the woman. My estimates for the effect of parenthood therefore reflect the consequences of failing to conceive when intended, while my estimates for the effect of timing reflect the consequences of conceiving later than intended. To understand the broader role of parenthood in the labor market, two other counterfactuals are especially important: planned childlessness and deliberate postponement of childbearing.

How my estimates differ from the impacts of parenthood relative to planned childlessness depends on whether women would invest more or less in their careers under that scenario. On the one hand, women may invest more if they plan to remain childless, since the absence of children may allow them to reap greater returns. In that case, my estimates would serve as a lower bound on the career cost of motherhood. Alternatively, women may invest more if they anticipate motherhood—both to accumulate resources for raising children and to develop careers compatible with family life. In that case, my estimates would instead provide an upper bound on the career cost.

The impacts of planned delayed childbearing are arguably easier to anticipate. One reason why planned and unplanned delays may differ concerns the experience of infertility itself: women’s outcomes may be depressed by the failure to achieve pregnancy. Another reason is that women may plan fertility and their careers, at least in part, to minimize career costs. Both imply that planned delays would result in better labor market outcomes than unplanned delays. This suggests that the results may be seen as a lower bound on the gains from postponement.

I conclude with a suggestive calculation, based on my estimates, of how much postponing motherhood can mitigate its career cost. Figure 15 constructs three career paths: remaining childless, conceiving at the first IUI, and conceiving 2.3 years later—the average delay in the analysis. The childless path is the estimated relier path and serves as the level anchor.¹⁷ The early-motherhood path scales this path by the ratio of reliers’ estimated mean outcome under early motherhood to that under childlessness. From year 3 onward, the later-motherhood path further scales the early-motherhood path by the corresponding ratio of non-reliers’ estimated mean outcomes under later and early motherhood, using the window-matched specification. The exercise is suggestive because the effects of parenthood and its timing are estimated for different groups. The comparison would be exact under the additional assumption that timing effects are proportional across reliers and non-reliers.¹⁸

Relative to the childless path, the early mother works 15.9% fewer hours and earns 17.7%

¹⁷Appendix SA2.11 shows that anchoring the levels on the full-sample conception-at-first-IUI path or on the non-relier early-motherhood path, rather than on reliers, leaves the loss percentages within a tenth of a percentage point and the implied reductions within 1.6 points.

¹⁸ $\mathbb{E}[Y^L|R=0]/\mathbb{E}[Y^E|R=0] = \mathbb{E}[Y^L|R=1]/\mathbb{E}[Y^E|R=1]$

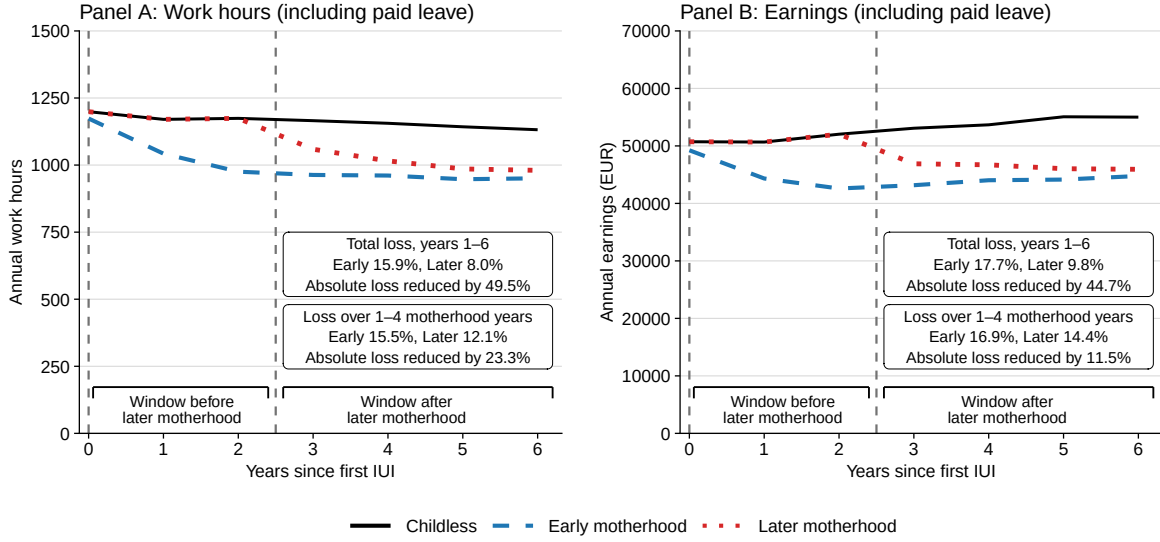


Figure 15: Childless, Early-Motherhood and Later-Motherhood Paths

Note: Work hours and earnings under alternative motherhood scenarios, by year relative to the first intrauterine insemination. The childless path is the estimated relier path; the early-motherhood path scales it by the ratio of reliers' mean outcomes under early motherhood and childlessness; from year 3 onward, the later-motherhood path additionally scales it by the corresponding ratio of non-reliers' mean outcomes under later and early motherhood from the window-matched specification. The upper box reports reductions relative to childlessness over years 1–6; the lower box reports reductions over the first four years of motherhood, years 1–4 for the early mother and years 3–6 for the later mother. The final line in each box gives the implied mitigation from postponement. Appendix SA2.11 reports alternative level anchors.

less over years 1–6. The later mother works 8.0% fewer hours and earns 9.8% less over the same window. Delaying motherhood by 2.3 years therefore reduces the loss in hours by 49.5% and the loss in earnings by about 45%.

The gain does not arise solely from postponing the reduction. Comparing each woman over her own first four years of motherhood—years 1–4 for the early mother and years 3–6 for the later one—the early mother works 15.5% fewer hours than under childlessness, while the later mother works 12.1% fewer relative to her corresponding childless path. In levels, the total reduction in hours over these windows is 23.3% smaller for the later mother. For earnings, the corresponding figures are 16.9% and 14.4%, while the total reduction in earnings is 11.5% smaller. Thus, even at the same stage of motherhood, the effects remain smaller for the later mother.

7 Conclusion

This paper develops a new methodology to estimate treatment and treatment-timing effects in quasi-experiments where individuals not initially assigned to treatment may obtain it later through repeated assignments. I use this method together with Dutch administrative data to assess the career impacts of parenthood. I find that motherhood persistently reduces women's work hours and earnings, whereas fatherhood has little effect. Taken together, parenthood causes up to half of the gender inequality among parents. Finally, my results show that women's losses are substantially mitigated when childbearing occurs later in their careers, with benefits both before and after the

later birth.

My results inform debates about policies affecting the costs and timing of parenthood, including work-family supports and access to contraception and reproductive technologies. They suggest that later childbearing can mitigate the effects of motherhood on earnings and work hours. However, this does not necessarily imply that later childbearing is welfare-improving, as it may entail health costs outside the labor market and reduce completed fertility.

Appendix

To simplify exposition, let $X_j^* \in \mathcal{X}_j^*$ contain X_j and $1_{A \geq j}$, and define $\mathcal{X}_j^{*1} = \{x \in \mathcal{X}_j^* : 1_{A \geq j} = 1\}$. Further, define $e_j^*(x) = \Pr(Z_j = 1 | X_j^*)$ and $Z_l^* = (1 - Z_l)/(1 - e^*(X_l))$. Note that $e_j^*(x) = 0$ and $Z_j = 0$ if $1_{A \geq j} = 0$. Hence, Assumption 5 implies $\mathbb{E}[Z_l^* | X_l^*] = 1$. Moreover, by definition, $\Pi_j^A \frac{(1 - Z_j)}{(1 - e_j(X_j))} = \Pi_j^{\bar{w}} Z_j^*$.

Lemma 1. *Under Assumption 4:*

$$(Y^E, Y^N, R^+, R, W, Z_1, \dots, Z_{j-1}, X_1^*, \dots, X_{j-1}^*) \perp\!\!\!\perp Z_j^* | X_j^* \text{ for all } j > 1. \quad (21)$$

Proof of Lemma 1. X_j^* includes $1_{\{A \geq j\}}$, and when $A < j$, we have $Z_j = 0$, which covers the cases when $X_j^* \in \mathcal{X}_j^* \setminus \mathcal{X}_j^{*1}$. The remainder follows from Assumption 4, since $1_{\{A \geq j\}}$, $Z_1, \dots, Z_{j-1}$, $X_1^*, \dots, X_{j-1}^*$, and $e_j^*(X_j^*)$ are fixed given X_j^* . $\square$

Lemma 2. *For any l s.t. $1 \leq l \leq \bar{w}$ and any measurable function $g(M_l)$, where $M_l = (Y^E, Y^N, R^+, R, W, Z_1, \dots, Z_l, X_1^*, \dots, X_l^*)$, under Assumptions 4 and 5: $\mathbb{E}[g(M_l) \Pi_{j=l+1}^{\bar{w}} Z_j^* | X_l^*] = \mathbb{E}[g(M_l) | X_l^*]$.*

Proof of Lemma 2. For any l s.t. $l < \bar{w}$:

$$\mathbb{E}[g(M_l) \Pi_{j=l+1}^{\bar{w}} Z_j^* | X_l^*] = \mathbb{E}[\mathbb{E}[g(M_l) \Pi_{j=l+1}^{\bar{w}} Z_j^* | X_{\bar{w}}^*] | X_l^*] \quad (22)$$

$$= \mathbb{E}[g(M_l) \Pi_{j=l+1}^{\bar{w}-1} Z_j^* \mathbb{E}[Z_{\bar{w}}^* | X_{\bar{w}}^*] | X_l^*] \quad (23)$$

$$= \mathbb{E}[g(M_l) \Pi_{j=l+1}^{\bar{w}-1} Z_j^* | X_l^*] \quad (24)$$

$$= \mathbb{E}[g(M_l) | X_l^*], \quad (25)$$

where (22) holds by law of iterated expectations and because X_j^* includes X_l^* for $j \geq l$, (23) holds by the Lemma (1), (24) holds because $\mathbb{E}[Z_l^* | X_l^*] = 1$ under Assumption 5, and where (25) follows from steps similar to (22) through (24) for X_j^* for j s.t. $l < j < \bar{w}$. $\square$

Proof of Theorem 1. I demonstrate the result for the upper bound, the result for the lower bound is symmetric. First, I demonstrate that $\mathbb{E}[Y(1 - D^+) \Pi_{j=1}^{\bar{w}} Z_j^*] / \mathbb{E}[r(X_1^*)] = \mathbb{E}[Y^N | R = 1]$. Note that:

$$\mathbb{E}[Y(1 - D^+) \Pi_{j=1}^{\bar{w}} Z_j^*] = \mathbb{E}[Y^N R \Pi_{j=1}^{\bar{w}} Z_j^*] \quad (26)$$

$$= \mathbb{E}[\mathbb{E}[Y^N R \Pi_{j=1}^{\bar{w}} Z_j^* | X_1^*]] \quad (27)$$

$$= \mathbb{E}[\mathbb{E}[Y^N R Z_1^* | X_1^*]] \quad (28)$$

$$= \mathbb{E} [\mathbb{E} [Y^N R | X_1^*] \mathbb{E} [Z_1^* | X_1^*]] \quad (29)$$

$$= \mathbb{E} [Y^N R] \quad (30)$$

$$= \mathbb{E} [Y^N | R = 1] \Pr(R = 1), \quad (31)$$

where (26) follows from the definitions, (27) holds by law of iterated expectations, (28) holds by Lemma (2), (29) holds by Assumption 4, and (30) holds because $\mathbb{E}[Z_l^* | X_l^*] = 1$ under Assumption 5. Next, note that:

$$\mathbb{E} [r(X_1) | X_1] = \mathbb{E} [R \Pi_{j=1}^{\bar{w}} Z_j^* | X_1^*] \quad (32)$$

$$= \mathbb{E} [R Z_1^* | X_1^*] \quad (33)$$

$$= \Pr(R = 1 | X_1^*), \quad (34)$$

where (32) follows from definitions, (33) holds by Lemma (2), and where (34) is obtained using that $\mathbb{E}[Z_l^* | X_l^*] = 1$ under Assumption 5 and applying Assumption 4. Since $\mathbb{E}[\Pr(R = 1 | X_1^* = x)] = \Pr(R = 1)$, the result holds.

It remains to show that $\mathbb{E}[Y(1 - D^+) 1_{\{Y > q(1-p(X_1^*), X_1^*)\}} Z_1 / e_1^*(X_1^*)] / \mathbb{E}[r(X_1^*)]$ is a sharp upper bound for $\mathbb{E}[Y^E | R = 1]$. I first demonstrate that $p(x) = \Pr(R = 1 | D^+ = 0, Z_1 = 1, X_1^* = x)$. Assumption 4 together with $D^+ = 1 - R^+ | Z_1 = 1$ implies that $r^+(x) = \Pr(R^+ = 1 | X_1^* = x)$. Under Assumption 2, $\Pr(R = 1 | X_1^* = x) = \Pr(R = 1, R^+ = 1 | X_1^* = x)$. Applying the definition of conditional probability gives $p(x) = \Pr(R = 1 | R^+ = 1, X_1^* = x)$. Assumption 4 together with $D^+ = 1 - R^+ | Z_1 = 1$ gives $\Pr(R = 1 | D^+ = 0, Z_1 = 1, X_1^* = x) = \Pr(R = 1 | R^+ = 1, X_1^* = x)$, which implies the result.

The remainder of the proof is similar to Lee (2009). Let $\gamma_x = \mathbb{E}[Y | Z_1 = 1, D^+ = 0, Y \geq q(1-p(x), x), X_1^* = x]$. I next demonstrate that γ_x is a sharp upper bound for $\mathbb{E}[Y^E | X_1^* = x, R = 1]$. Using that $p(x) = \Pr(R = 1 | D^+ = 0, Z_1 = 1, X_1^* = x)$, Corollary 4.1 in Horowitz & Manski (1995) gives $\gamma_x \geq \mathbb{E}[Y | Z_1 = 1, D^+ = 0, R = 1, X_1^* = x]$. Using that $D^+ = 0 | R = 1$ and $Y = Y^E | Z_1 = 1$ and by Assumption 4, $\mathbb{E}[Y | Z_1 = 1, D^+ = 0, R = 1, X_1^* = x] = \mathbb{E}[Y^E | X_1^* = x, R = 1]$, meaning that γ_x is an upper bound for $\mathbb{E}[Y^E | X_1^* = x, R = 1]$. Since $p(x)$ is identified and Y^E is observed only among those initially assigned treatment, Corollary 4.1 in Horowitz & Manski (1995) implies sharpness.

Let $f_{x|R=1}(x)$ be the p.d.f. of X_1^* conditional on $R = 1$. Applying Bayes rule for densities to $\Pr(R = 1 | X_1^* = x)$ identified by $r(x)$ and p.d.f. of X_1^* identified directly identifies $f_{x|R=1}(x)$, making $\int_{\mathcal{X}_1^*} \gamma_x f_{x|R=1}(x) dx$ the sharp upper bound for $\mathbb{E}[Y^E | R = 1]$.

The last step is to show that:

$$\int_{\mathcal{X}_1^*} \gamma_x f_{x|R=1}(x) dx = \mathbb{E}[Y(1 - D^+) 1_{\{Y > q(1-p(X_1^*), X_1^*)\}} Z_1 / e_1^*(X_1^*)] / \mathbb{E}[r(X_1^*)]. \quad (35)$$

Using the law of iterated expectations and the definition of conditional probability:

$$\mathbb{E}[Y(1 - D^+) 1_{\{Y > q(1-p(X_1^*), X_1^*)\}} Z_1 / e_1^*(X_1^*)] \quad (36)$$

$$= \mathbb{E}[\mathbb{E}[\gamma_{X_1^*} | X_1^*] \Pr(D^+ = 0, Z_1 = 1, Y > q(1 - p(X_1^*), X_1^*) | X_1^*) / e_1^*(X_1^*)]. \quad (37)$$

Applying the definition of conditional probability twice and the definition of $p(X_1^*)$:

$$\Pr(D^+ = 0, Z_1 = 1, Y > q(1 - p(X_1^*), X_1^*)) = r(X_1^*)e_1^*(X_1^*) \quad (38)$$

Thus:

$$\mathbb{E}[Y(1 - D^+)1_{\{Y > q(1 - p(X_1^*), X_1^*)\}} Z_1 / e_1^*(X_1^*)] = \mathbb{E}[\gamma_{X_1^*} r(X_1^*)]. \quad (39)$$

Applying Bayes rule for densities: $\mathbb{E}[\gamma_{X_1^*} r(X_1^*)] = \int_{\mathcal{X}_1^*} \gamma_x f_{x|R=1}(x) dx \Pr(R = 1)$. Since $\mathbb{E}[r(X_1^*)] = \Pr(R = 1)$, the statement holds. $\square$

Proof of Corollary 1. Following reasoning analogous to Theorem 1 yields:

$$\mathbb{E}[Y(Z_1/e_1(X_1) - \Pi_{j=1}^{\bar{w}} Z_j^*)] = \tau_{ATR} \Pr(R = 1) - \delta_{ANR} \Pr(R = 0), \quad (40)$$

and $\mathbb{E}[r(X_1)] = \Pr(R = 1)$. Since $\delta = 0$ implies $\delta_{ANR} = 0$, the statement holds. $\square$

Proof of Theorem 2. I demonstrate the result for the upper bound, the result for the lower bound is symmetric. Following reasoning analogous to Theorem 1, yields:

$$\mathbb{E}[Y(Z_1/e_1(X_1) - \Pi_{j=1}^{\bar{w}} Z_j^*)] = \tau_{ATR} \Pr(R = 1) - \delta_{ANR} \Pr(R = 0), \quad (41)$$

$\mathbb{E}[r(X_1)] = \Pr(R = 1)$, and $\mathbb{E}[1 - r(X_1)] = \Pr(R = 0)$. Theorem 1 further implies that $\mathbb{E}[m^U(G, \eta^0)]$ is a sharp upper bound on $\tau_{ATR} \Pr(R = 1)$. Together, these results imply that $\mathbb{E}[m^U(G, \eta^0) - m^{RF}(G, \eta^0)]$ is a sharp upper bound on $\delta_{ANR} \Pr(R = 0)$. Thus, the statement holds. $\square$

Proof of Theorem 3. As in Theorem 1, $\mathbb{E}[Y(1 - D^+) \Pi_{j=1}^{\bar{w}} Z_j^*] = \mathbb{E}[Y^N | R = 1] \Pr(R = 1)$, $p(x)r^+(x) = r(x)$, and $\mathbb{E}[r(X_1)] = \Pr(R = 1)$.

I next show that the first term of $m^{RND}(G, \eta^0)$ identifies the relier average treated outcome. By the law of iterated expectations and the definition of conditional expectation:

$$\mathbb{E} \left[Y(1 - D^+) p(X_1^*) \frac{Z_1}{e_1^*(X_1^*)} \right] = \mathbb{E} \left[\frac{p(X_1^*)}{e_1^*(X_1^*)} \mathbb{E}[Y | Z_1 = 1, D^+ = 0, X_1^*] \Pr(D^+ = 0, Z_1 = 1 | X_1^*) \right]. \quad (42)$$

Assumption 4 together with $D^+ = 1 - R^+ | Z_1 = 1$ gives $\Pr(D^+ = 0, Z_1 = 1 | X_1^*) = r^+(X_1^*)e_1^*(X_1^*)$. Hence (42) equals $\mathbb{E}[\mu_{X_1^*} r(X_1^*)]$, where $\mu_x = \mathbb{E}[Y | Z_1 = 1, D^+ = 0, X_1^* = x]$. Using that $Y = Y^E | Z_1 = 1$ and $D^+ = 1 - R^+ | Z_1 = 1$, and by Assumption 4, $\mu_x = \mathbb{E}[Y^E | R^+ = 1, X_1^* = x]$, which equals $\mathbb{E}[Y^E | R = 1, X_1^* = x]$ by Assumption 6. Applying Bayes rule for densities, (42) equals $\mathbb{E}[Y^E | R = 1] \Pr(R = 1)$, and thus $\tau_{ATR} = \mathbb{E}[m^{RND}(G, \eta^0)] / \mathbb{E}[r(X_1)]$. Following reasoning analogous to Theorem 2 also yields $\delta_{ANR} = \mathbb{E}[m^{RND}(G, \eta^0) - m^{RF}(G, \eta^0)] / \mathbb{E}[1 - r(X_1)]$.

The last step is to show that the two point-identified effects fall within the corresponding bounds. I demonstrate the argument for the upper bound on τ_{ATR} , the argument for the lower bound is symmetric. $m^{RND}(G, \eta^0)$ and $m^U(G, \eta^0)$ differ only in the first term. Theorem 1 establishes that the first term of $m^U(G, \eta^0)$ has expectation $\mathbb{E}[\gamma_{X_1^*} r(X_1^*)]$, where $\gamma_x = \mathbb{E}[Y | Z_1 = 1, D^+ = 0, Y \geq$

$q(1 - p(x), x), X_1^* = x]$, whereas the present proof establishes that the first term of $m^{RND}(G, \eta^0)$ has expectation $\mathbb{E}[\mu_{X_1^*} r(X_1^*)]$. By the definition of the truncated expectation, $\mu_x \leq \gamma_x$ for every x , and $r(x) \geq 0$. Hence, the point-identified effect cannot exceed the corresponding upper bound. The same argument applied symmetrically establishes the result for the lower bound, and for δ_{ANR} the same comparison applies since both moments are differenced with $m^{RF}(G, \eta^0)$ and scaled by $\mathbb{E}[1 - r(X_1)]$. Thus, the statement follows. $\square$

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Supplementary Appendix for “Parenthood Timing and Gender Inequality”

SA1 Dynamic Framework

This section maps the cross-sectional framework used throughout the paper into a dynamic framework, explains why explicitly specifying the dynamic framework aids little in the interpretation of timing effects, and presents identification when reliance is defined using per-period fertility rather than endline fertility.

Time starts at the moment of a woman’s first IUI and ticks up to period T . Without loss of generality, childless women can choose whether to undergo one IUI each period. Define $W_t \in \{1, \dots, \overline{w}\}$, $R_t \in \{0, 1\}$, and $R_t^+ \in \{0, 1\}$ as the period-specific counterparts of W , R , and R^+ , measured up to t . All three are irreversible. Taken together, W_t and R_t characterize the periods in which IUI and non-IUI conceptions occur, allowing arbitrary breaks between attempts and non-IUI conception. Let $Y'_t(k)$ denote the potential outcome in period t if treatment starts in period k , and $Y'_t(\infty)$ if the woman never receives treatment.

Let $Z'_j \in \{0, 1\}$ indicate whether a successful IUI occurs in period j , and define the treatment indicator in period t as $D_t = \max(\{Z'_j : j \leq t\} \cup \{1 - R_t\})$. For notational convenience, define $A_t = t$, as if all women underwent IUI in every period up to t , and let X'_t be covariates in period t , implicitly including an indicator for undergoing IUI.

Assumption 7 (Dynamic Conditional Sequential Unconfoundedness).

$(Y'_t(d), R_{t'}^+, R_{t'}, W_t) \perp\!\!\!\perp Z'_j \mid X'_j$ for all d, j, t, t' and $X'_j \in \mathcal{X}'_j$.

This assumption states that among women undergoing IUI in period j , success is independent of potential outcomes and latent types conditional on observed covariates; including the indicator for undergoing IUI ensures it holds trivially for women who do not undergo IUI this period. It differs from Assumption 4, which assumed independence among women sharing the same attempt number rather than the same period, but the two can be mapped by including both attempt number and time since the first IUI in the covariates.

Let $H = \min(\{j : R_j = 0\} \cup \{\infty\})$ denote the period in which a woman becomes a non-reliant, with $H = \infty$ if she remains a reliant through period T . Under Assumption 7 and adapted Assumptions 2 and 5, plugging period- t outcomes but endline fertility status into the moments of Theorem 2 yields bounds on $\mathbb{E}[Y'_t(H) - Y'_t(0) \mid H \leq T]$, the average timing effect for women who become non-reliant by the end of the sample period. These are the main timing estimates used throughout the paper.

Alternatively, using period- t outcomes and current fertility status in the corresponding moments yields bounds on $\mathbb{E}[Y'_t(H) - Y'_t(0) \mid H \leq t]$, the average timing effect for women who have become non-reliant by period t . This compares women who have already become mothers under both scenarios, but at different points in time, and is used in Section 5.3 to construct the window-adjusted timing estimates.

Since the timing of the first IUI and the timing of conception without IUI are observed, the distribution of H is identified, and hence so is the average shift in timing up to period t , $\mathbb{E}[H \mid H \leq t]$. However, among women whose first IUI succeeds, it is not possible to determine how much each would have delayed conception had the IUI failed. Consequently, the bounds are only informative about the effect of an average timing shift induced by IUI failure, highlighting the limited additional usefulness of a fully dynamic framework for interpretation.

Finally, assuming no anticipation, $Y'_t(k) = Y'_t(\infty)$ for $t < k$, bounds on the parenthood effect follow from Theorem 2 for both women who remain reliers in period t , $\mathbb{E}[Y'_t(0) - Y'_t(\infty) \mid H > t]$, and women who remain reliers through the end of the data window, $\mathbb{E}[Y'_t(0) - Y'_t(\infty) \mid H > T]$. The mapping into the cross-sectional potential outcomes in period t is $Y'_t(0) = Y^E$, $Y'_t(\infty) = Y^N$, and $Y'_t(H) = Y^L$.

SA2 Robustness and Extensions

Appendix SA2.1 reports bounds for the window-matched timing estimates. Appendix SA2.2 compares my estimates with conventional methods period by period and gives estimation details for each method. Appendix SA2.3 bounds the effects for women who would remain partnered and not begin antidepressant use after procedure failure, and reports effects on antidepressant uptake and partnership stability directly. Appendix SA2.4 tests the monotonicity assumption, and Appendix SA2.5 reports the main estimates allowing the direction of monotonicity to vary with covariates. Appendix SA2.6 varies the cross-fitting folds and forest size and Appendix SA2.7 the number of procedures used in estimation. Appendix SA2.8 presents estimates of work hours adjusted for potential parental leave. Appendix SA2.9 reports bounds under controls for the clinical protocol and Appendix SA2.10 bounds under the selection adjustment of Section 5.6. Appendix SA2.11 shows that the paths in Figure 15 are insensitive to the level anchor.

SA2.1 Bounds for the Window-Matched Timing Estimates

Section 5.3 reports point estimates for the window-matched timing estimates, which re-estimate the timing effect from year 3 onward using parenthood status in each period rather than status at the end of the window. Figure SA1 reports the corresponding bounds. The results remain relatively close to the baseline estimates.

SA2.2 Conventional Methods

The estimator comparisons use two reduced forms. The first, $\psi^{RF}(G, \xi^0)$, is the orthogonal moment for $m^{RF}(G, \eta^0)$ defined in Section 3.4 and compares first-IUI success with failure throughout the full procedure sequence. The second, $\psi^{RF,1}(G, \xi^0)$, uses only the first procedure and is the orthogonal moment for $m^{RF,1}(G, \eta^0) = YZ_1/e_1(X_1) - Y(1 - Z_1)/(1 - e_1(X_1))$. The *IV* rows use $\psi^{RF,1}$, whereas the *sequential IV* rows use ψ^{RF} . I denote the relevant reduced form in year t by $\hat{\psi}_t^{RF}$ and the corresponding counterfactual level by $\hat{C}_t$.

Each reduced form is paired with a corresponding first stage: $\pi^1(G, \xi^0)$ for the *IV* rows and $\pi(G, \xi^0)$ for the *sequential IV* rows. I denote the relevant first stage in year t by $\hat{\pi}_t$. The *endline* estimator is $\hat{\beta}^{end} = (\hat{\psi}_T^{RF}/\hat{\pi}_T)/\hat{C}_T$, so the reduced form accumulated over the outcome window is

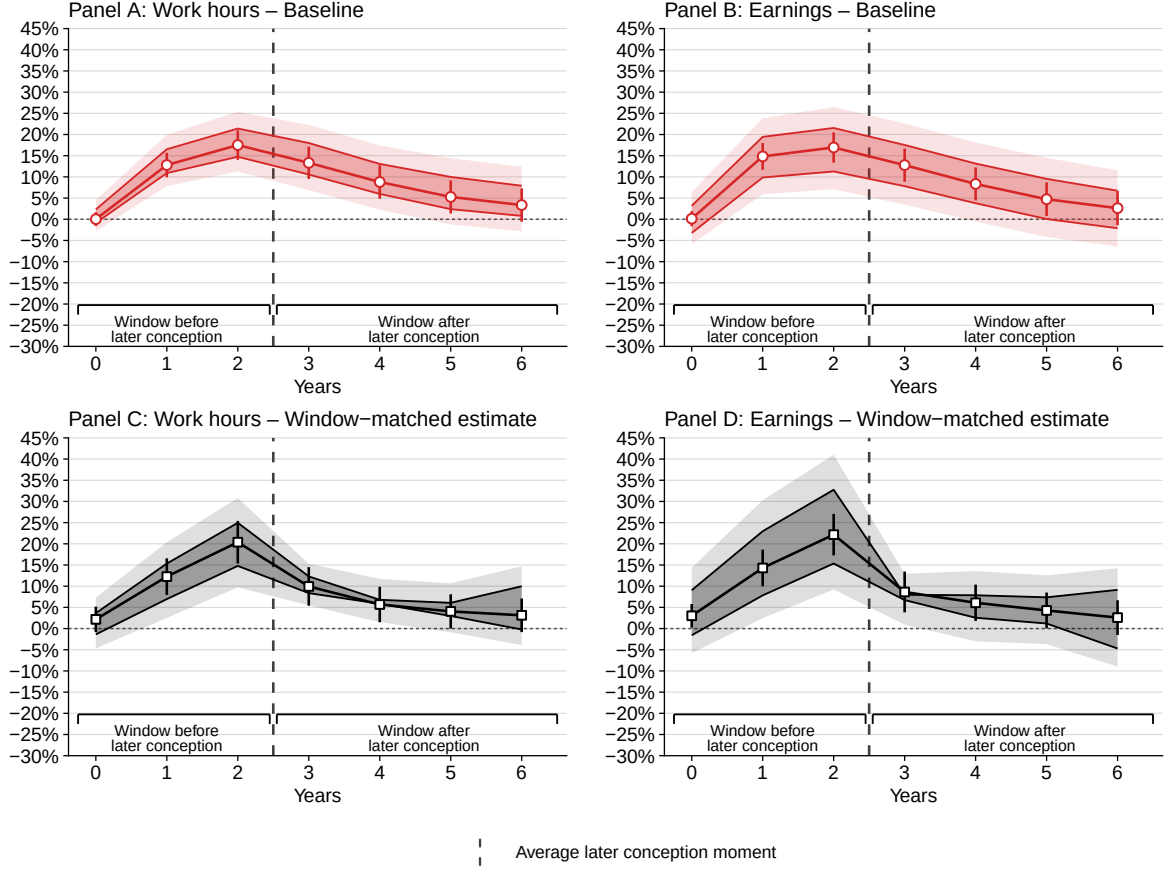


Figure SA1: Bounds for the Window-Matched Timing Estimates

Note: Effects of later parenthood on women's work hours and earnings by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level. *Baseline* uses parenthood status at the end of the observation window; *window-matched* uses the negative of the parenthood effect for years 0–2 and re-estimates the timing effect using parenthood status in each period from year 3 onward. Translucent bands give the identified set and the fainter outer band its 95% confidence interval. Figure 9 reports the corresponding point estimates.

scaled by the first stage in the final year. The *per-period* estimator instead applies the contemporaneous first stage in each year and then aggregates, $\hat{\beta}^{pp} = \sum_t (\hat{\psi}_t^{RF} / \hat{\pi}_t) / \sum_t \hat{C}_t$.

The recursive estimator adjusts for births following IUI failure, following [Gallen et al. \(2024\)](#) and [Bensnes et al. \(2025\)](#), under the restriction in equation (4). Let g_j denote the share of the control arm becoming mothers in year j . For $t = 1, \dots, T$, the estimator recursively computes

$$\hat{d}_t = \left(\hat{\psi}_t^{RF} + \sum_{j=2}^t g_j \hat{d}_{t-j+1} \right) / \hat{\pi}_1, \quad (43)$$

$$\hat{\mu}_t = \hat{C}_t^{obs} - \sum_{j=1}^t g_j \hat{d}_{t-j+1}. \quad (44)$$

and reports $\sum_t \hat{d}_t / \sum_t \hat{\mu}_t$. Here, $\hat{C}_t^{obs}$ is the observed mean in the control arm, which includes women who have become mothers by year t , while $\hat{\mu}_t$ estimates the never-treated counterfactual. I apply the recursive estimator to both reduced forms.

The event-study implementation follows [Kleven, Landais, & Søgaaard \(2019\)](#). I define the event as the year of conception, measured as the year of first birth minus one for births occurring in the first nine months of the year, and estimate $Y_{it} = \sum_{k \neq -1} \alpha_k 1_{\{t=k\}} + \gamma_{c(it)} + \delta_{a(it)} + \varepsilon_{it}$ over the twelve years from five years before to six years after conception. The specification includes calendar-year fixed effects γ and age fixed effects δ , with the year before conception omitted, and standard errors clustered at the woman level. The counterfactual is the fitted value net of the event-time indicators, and the reported estimate is the sum of $\hat{\alpha}_k$ for $k \geq 1$ divided by the counterfactual summed over the same years. I estimate this specification for the IUI sample restricted to conceptions between 2013 and 2016 and for the representative sample of couples conceiving a first child over the same period.

The [Zhang & Rubin \(2003\)](#) and [Lee \(2009\)](#) bounds use the same moment functions as my baseline bounds but discard all information after the first procedure and do not exploit monotonicity across the procedure sequence.

The last-procedure IV uses the last procedure observed up to and including the year in question.

Standard errors are analytic for the bounds and endline estimators. For the per-period and recursive estimators, I use 500 bootstrap draws, resampling the sample once in each draw and recomputing all components. For the event-study estimates, I use the standard error of the summed numerator while holding the denominator fixed.

Figure [SA2](#) presents the per-period estimates. In most years, conventional IV and sequential IV estimates lie near the largest negative effects included in the bounds or are more negative. Recursive IV estimates lie closer to the smallest negative effects included in the bounds, while their sequential counterpart tracks my point estimates closely. While most estimates converge to the 10%–20% range for both work hours and earnings, the event-study estimates diverge, exceeding 30% in both the IUI and representative samples by year 6. They also display substantial pre-trends in both samples.

SA2.3 Mental Health and Relationship Stability

Women who remain childless after trying to achieve pregnancy may experience mental health deterioration or relationship breakdowns, potentially affecting labor market outcomes. On the one hand, these consequences may be an integral part of the effect of not having children, making it important to understand the extent to which these mechanisms drive the labor market impacts. On the other hand, such issues may also stem not from childlessness itself but from fertility treatments or from the experience of trying and failing to conceive, which could limit the generalizability of the estimates to settings where women voluntarily remain childless.

Before assessing these factors, it is useful to clarify when the side effects of failing to conceive matter for generalizability. They are arguably less concerning for my estimates of timing effects: I find that women who experience delays perform better, and doing so despite potential side effects reinforces that delaying childbearing mitigates losses. By contrast, they are more important for my estimates of the effects of parenthood, where I find that large gender gaps persist even when couples fail to conceive. If these patterns reflect such side effects, the estimates may understate the

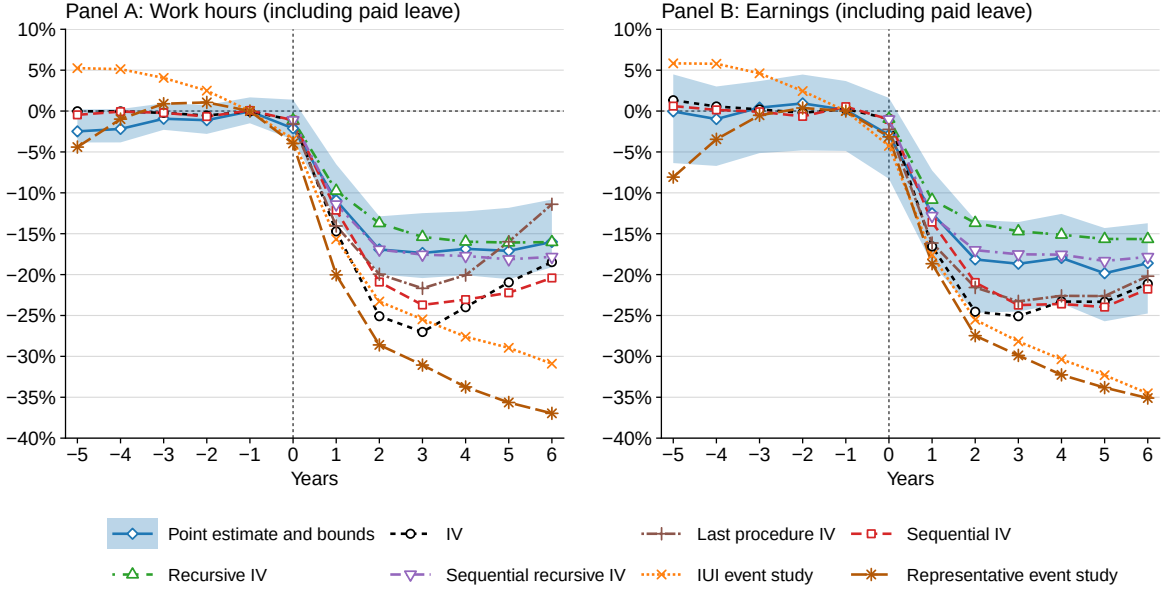


Figure SA2: Estimators of the Effect of Parenthood on Women's Hours and Earnings, by Year

Note: Effect of parenthood on women's annual work hours and earnings by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level in the absence of children. The shaded band gives the identified set and the solid line with markers the point estimate under conditionally random reliance. *IV* scales each period's reduced form by its contemporaneous first stage; *last procedure IV* defines the instrument as success at the last procedure a woman undergoes; *sequential IV* targets reliers rather than compliers, as described in Section 3.3.1; *recursive* variants additionally adjust for control units who subsequently become treated, following [Gallen et al. \(2024\)](#) and [Bensnes et al. \(2025\)](#). Event-study estimates follow [Kleven, Landais, & Sogaard \(2019\)](#) and are reported for the intrauterine-insemination sample and for a representative sample of women having a first child between 2013 and 2016, normalized to zero in the year before the first procedure. Section 5.7 reports the corresponding window totals.

role of parenthood and overstate selection beyond my sample.

To provide suggestive evidence, one might estimate the effect of failing to conceive, relative to successful conception, on antidepressant uptake or separation, which I do at the end of this section. However, this does not directly address the generalizability concern, since both conceiving and failing to conceive may affect mental health and relationship stability relative to not attempting conception. The contrast between failure and successful conception therefore does not reveal the effect of failure relative to not attempting.

To directly assess the role of mental health and relationship impacts, I modify the baseline approach to bound the labor market effects for women who, in the event of procedure failure, would (i) remain childless, (ii) be cohabiting with or registered to a partner, and (iii) not initiate antidepressant use. If the adjusted bounds remain close to the baseline, it indicates that these issues, independent of the source, have limited empirical relevance for the labor market impacts.

The formal procedure classifies women who meet condition (i) but not (ii) or (iii) as having conceived naturally, thereby removing them from the group used to construct the bounds. This group accounts for 10% of the sample and 15% of non-reliers. The adjustment also addresses potential monotonicity violations by excluding women who would have conceived naturally had IUI

succeeded but did not after failure because of subsequent mental health problems or relationship breakdowns; the formal argument amounts to adapting Theorem 1 by redefining R and D .

The procedure is conservative because it also excludes women who would have experienced such issues irrespective of parenthood, as well as those whose labor market outcomes would have remained unaffected even if the issues arose due to childlessness or procedure failure. This occurs because it is not possible to identify which women who conceived at the first IUI fall into these categories, so the trimming step must discard more observations from the tails, which in turn produces wider bounds. If such women are primarily the ones experiencing these issues, the bounds will widen without shifting systematically in either direction, much as if some childless women had been excluded at random.

Panels C and D of Figure SA4 present the results under the label *broadened selection*. The point estimates change little: the effect of parenthood on work hours moves from -15.9% to -16.3% , while the earnings effect remains at -17.7% . The timing effects increase from 10.5% to 12.0% for work hours and from 10.0% to 11.7% for earnings. The main change is instead in the width of the identified sets. For parenthood effects, the width increases from 6.9 to 30.0 percentage points for work hours and from 10.1 to 40.0 points for earnings. This widening is expected mechanically, since the broader definition increases the group over which outcomes must be trimmed. The fact that the bounds widen roughly symmetrically in both directions is also what one would expect if these shocks were as good as random and did not affect labor-market outcomes. Overall, the exercise suggests that severe mental health problems and separation do not drive the point estimates, although a conservative adjustment that holds these mechanisms fixed makes the bounds less informative.

Figure SA3 presents estimates of the effect of conceiving at the first IUI, relative to failure throughout the sequence, on antidepressant use and remaining partnered. Effects on antidepressant use are precisely estimated and indistinguishable from zero in every year. Effects on remaining partnered are zero before and immediately after the first procedure but become positive and significant from year 2, reaching 6.8 percentage points in year 5 against a control mean of 88.7%. Conceiving at the first IUI therefore has no detectable effect on antidepressant use but makes couples more likely to remain together.

SA2.4 Testing Monotonicity

Monotonicity states that all reliers are subsequent reliers, implying that the subsequent relier share is at least as large as the relier share: $\Pr(R^+ = 1) \geq \Pr(R = 1)$. Monotonicity further implies that the subsequent relier share is at least as large as the relier share at each covariate value: $r^+(X_1^*) \geq r(X_1^*)$. Since the conditional shares are estimated nonparametrically, formally testing whether their differences allow rejecting monotonicity is not trivial, but comparing them offers some insight. Panel A in Figure SA4 plots the empirical distribution of the difference between estimated conditional subsequent relier and relier shares in year 6. For 34.8% of observations, the difference is below zero. While this would contradict monotonicity if observed in the true shares, such deviations can result from estimation error when the shares are close. Namely, when all subsequent reliers are reliers $\Pr(R^+ = R) = 1$, the difference should be below zero for 50% of observations. Consistent

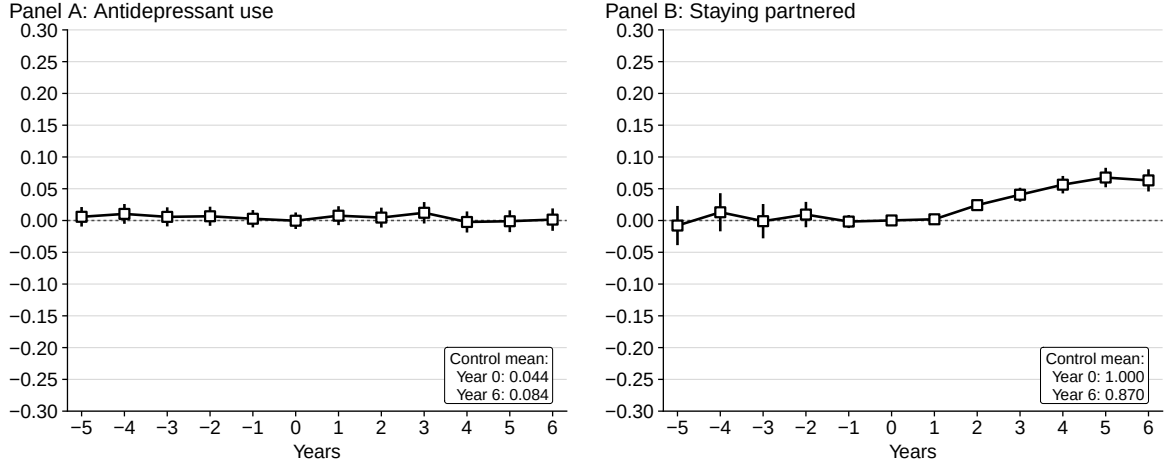


Figure SA3: Effects on Antidepressant Uptake and Partnership Stability

Note: Reduced-form effects of conceiving at the first intrauterine insemination on antidepressant uptake and on remaining cohabiting with or registered to a partner, by year relative to the first procedure, in percentage points. Markers give point estimates and vertical segments their 95% confidence intervals; the control mean is reported beneath each year.

with this, the differences are generally small: the median is 0.034, and only 6.4% of observations fall below -0.1 , suggesting no clear monotonicity violations.

To formally test monotonicity using covariates, I adapt the approach of [Semenova \(2025\)](#). I partition the sample into $J = 25$ discrete cells $\mathcal{X}_{1,j}^*$ based on quintiles of women's work hours and age in the year prior to their first IUI. Since these two covariates are highly predictive of the remaining covariates used in the analysis, including additional ones results in small cells (e.g., almost no women have extremely high work hours while having extremely low earnings). Monotonicity implies that the subsequent relier share is at least as large as the relier share in each cell, meaning that each value in the vector $\mu = (\mathbb{E}[r^+(X_1^*) - r(X_1^*) \mid X_1^* \in \mathcal{X}_{1,j}^*])_{j=1}^J$ must be non-negative. The null hypothesis is $-\mu \leq 0$, and the test statistic is $\max_{j \in J} \frac{-\hat{\mu}_j}{\hat{\sigma}_j}$. The critical value is the self-normalized critical value of [Chernozhukov et al. \(2019\)](#). σ_j are estimated using multiplier bootstrap with 100 draws and weights $w_i \sim \exp(1)$ to account for the uncertainty in the estimation of propensity scores (results remain unchanged when treating scores as fixed). The p -value for the test statistic is 0.93, so the differences are not statistically significant and monotonicity is not rejected. Using women's earnings instead of hours to partition the sample yields the same conclusion.

Panel B in Figure SA4 repeats the above when restricting focus to reliers who would remain cohabiting with their partners and not take up antidepressants after failing to conceive, treating those who separate from their partners or take up antidepressants after procedure failure as if they had children and excluding them from the population covered by the bounds. The estimated difference between the subsequent relier and relier shares is below zero for only 7.0% of observations and below -0.1 for just 0.3%, with a median of 0.129 and a p -value of 0.999, providing stronger support for monotonicity in this group.

Panels C and D report what the assumption costs. Alongside the baseline they show estimates

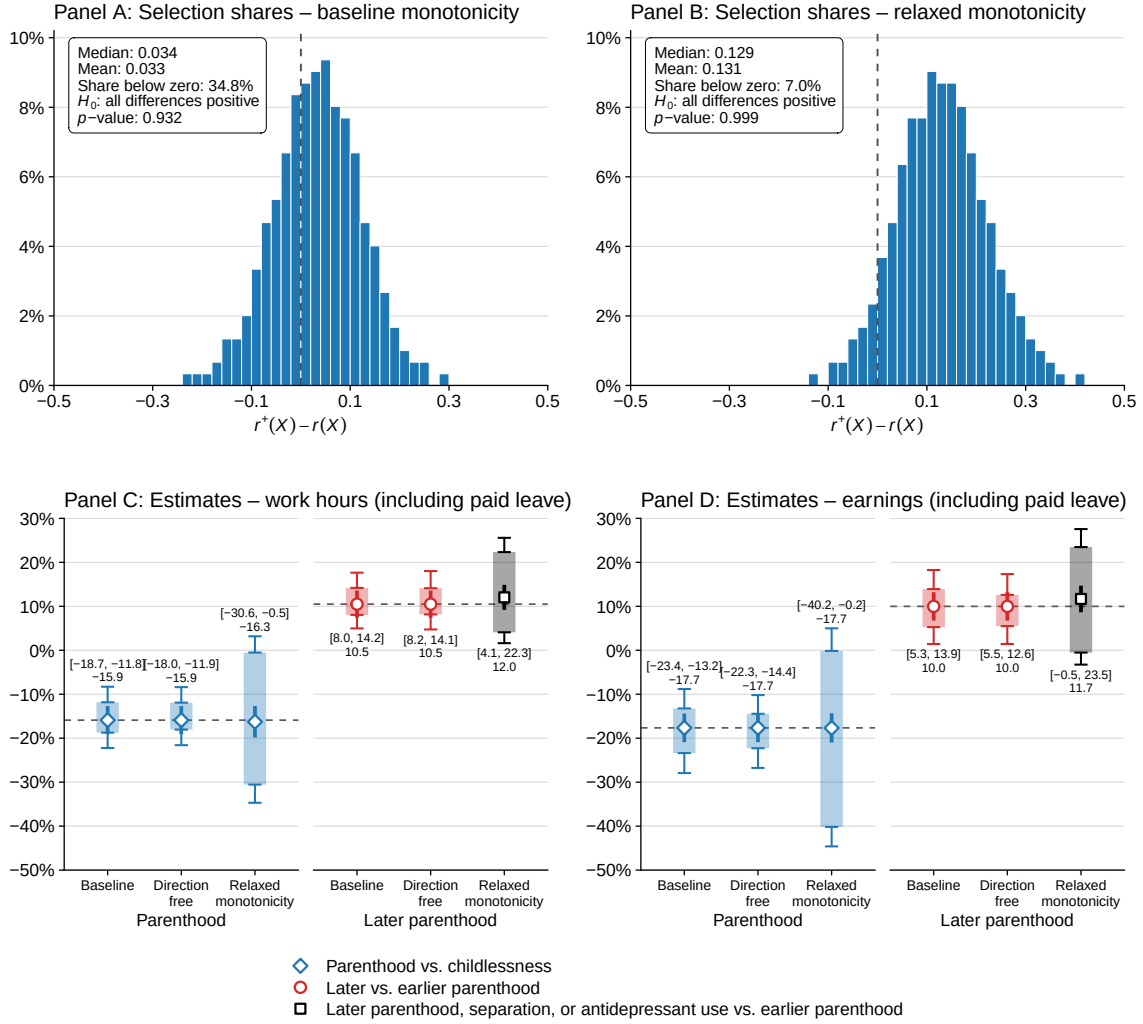


Figure SA4: Testing and Relaxing Monotonicity

Note: Panels A and B give the empirical distribution of the difference between the covariate-conditional subsequent relier and relier shares six years after the first intrauterine insemination, under the baseline selection definition and under the broadened definition described in Appendix SA2.3. Panels C and D give the effect of parenthood and of later parenthood on work hours and earnings, cumulated over years 1–6 and expressed as a percentage of the counterfactual level, under three specifications: the baseline, which imposes one direction of monotonicity; *direction free*, which leaves it unrestricted and defines selection at the current horizon rather than the final one; and *broadened selection*, which applies the definition of Panels B and Appendix SA2.3. Translucent bars give the identified set, whiskers its 95% confidence interval, and markers the point estimate under conditionally random reliance.

leaving the direction of monotonicity free, and estimates using the broadened selection definition of Panel B. None of the three moves the point estimate: for work hours the effect of parenthood is -15.9% in all three, and for earnings -17.7% in the baseline and in the broadened definition. What changes is the width of the identified set, discussed further in Appendix SA2.3.

SA2.5 Relaxing Monotonicity

To address violations of monotonicity in the Lee (2009), Semenova (2025) relaxes the assumption by allowing its direction to vary with X_1^* . To test the sensitivity of my results, I adopt a similar

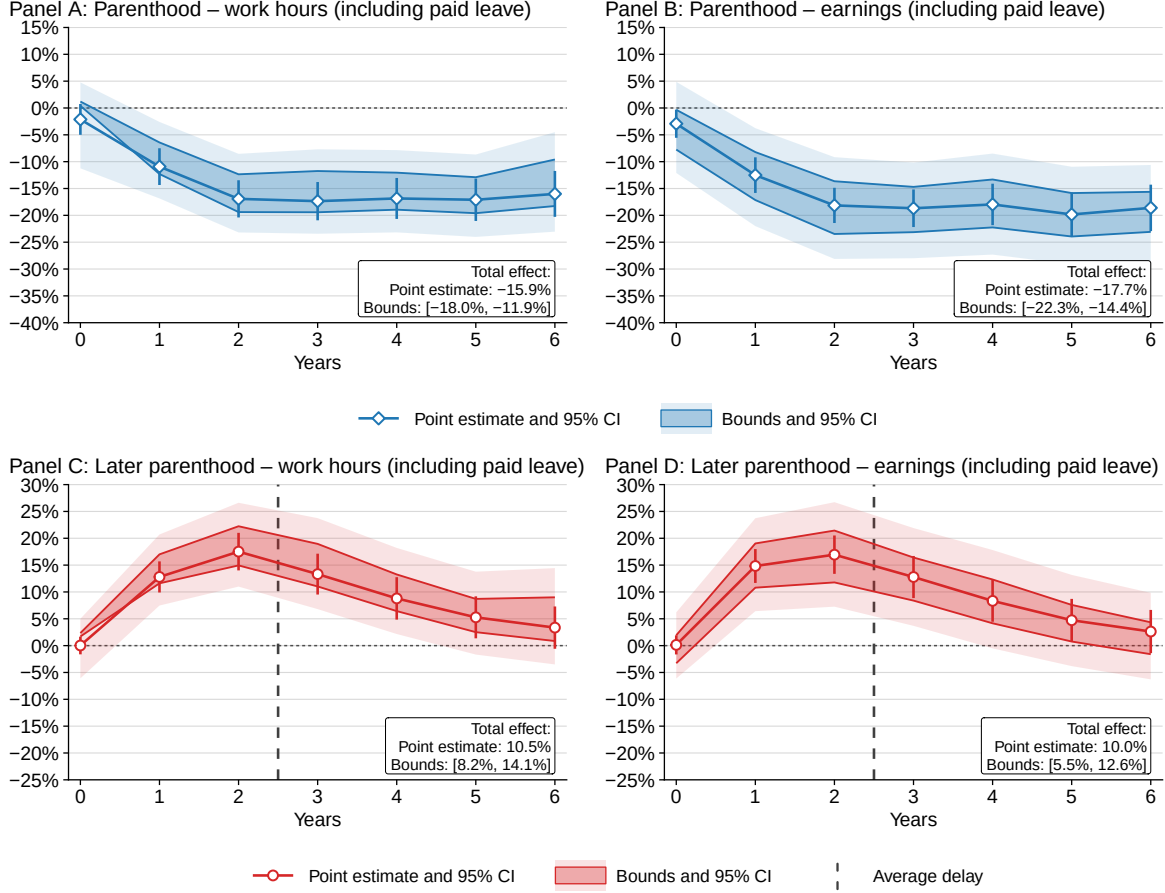


Figure SA5: Effects on Women Under Covariate-Conditional Monotonicity

Note: Effects of motherhood on women's annual work hours and earnings (in EUR), estimated allowing the direction of monotonicity to vary with covariates; see Appendix SA2.5. Time relative to first intrauterine insemination. Sample includes all women who underwent intrauterine insemination for their first child between 2013 and 2016 and were cohabiting with a male partner in the year before the first procedure.

approach. If the reversal of the estimated relifer and subsequent relifer shares occurs because the true shares are close, this method and the baseline approach should yield similar results.

Define $\mathcal{X}_{help}^* = \{x : r^+(x) \geq r(x)\}$ and $\mathcal{X}_{hurt}^* = \mathcal{X}_1^* \setminus \mathcal{X}_{help}^*$. The relaxed monotonicity assumption is that $\forall x \in \mathcal{X}_{help}^* R^+ \geq R$ a.s., and $\forall x \in \mathcal{X}_{hurt}^* R^+ < R$ a.s.. Table SA1 presents orthogonal moments for the case when $X_1^* \in \mathcal{X}_{hurt}^*$. The estimator for θ_b , for $b \in \{L, U\}$, is then given by:

$$\hat{\theta}_b = \frac{\sum_i \left(\psi^b(G_i, \hat{\zeta}_i) 1_{\{p(X_1^*) \leq 1\}} + \psi^{b-}(G_i, \hat{\zeta}_i) 1_{\{p(X_1^*) > 1\}} \right)}{\sum_i \left(\psi^R(G_i, \hat{\zeta}_i) 1_{\{p(X_1^*) \leq 1\}} + \psi^{R+}(G_i, \hat{\zeta}_i) 1_{\{p(X_1^*) > 1\}} \right)}. \quad (45)$$

I implement it following the baseline approach. Since a weighted generalized quantile forests estimator is not available, I estimate q^0 using quantile regressions. Figure SA5 presents the estimates for women's outcomes, which resemble the baseline results. Over years 1–6 the point estimates are identical to the baseline to one decimal place—15.9% for the effect of parenthood on work hours, -17.7% on earnings, 10.5% and 10.0% for the timing effects—and the identified sets are slightly narrower, by 0.8 percentage points for work hours and 2.3 points for earnings.

Table SA1: Moment Functions for Covariate-Conditional Monotonicity

Moment functions	
$\psi_L^-(W, \zeta_0)$	$\begin{aligned} & \frac{Z_1}{e_1^*(X_1^*)} (1 - D^+) Y - \Pi_{j=1}^{\bar{w}} \frac{1 - Z_j}{1 - e_j^*(X_j^*)} (1 - D^+) Y 1_{\{Y > q^0(1 - 1/p(X_1^*), X_1^*)\}} \\ & - q^0(1 - 1/p(X_1^*), X_1^*) \left[\frac{Z_1}{e_1^*(X_1^*)} (1 - D^+ - r^+(X_1^*)) \right. \\ & \quad - \Pi_{j=1}^{\bar{w}} \frac{1 - Z_j}{1 - e_j^*(X_j^*)} \frac{1}{p(X_1^*)} (1 - D^+ - r_1(X_1^*)) \\ & \quad \left. - \Pi_{j=1}^{\bar{w}} \frac{1 - Z_j}{1 - e_j^*(X_j^*)} (1 - D^+) (1_{\{Y > q^0(1 - 1/p(X_1^*), X_1^*)\}} - 1/p(X_1^*)) \right] \\ & \quad - \frac{Z_1 - e_1^*(X_1^*)}{e_1^*(X_1^*)} \beta^+(1, X_1^*) r^+(X_1^*) \\ & \quad + \sum_{k=1}^{\bar{w}} 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{1 - D_j}{1 - e_j^*(X_j^*)} \frac{e_k^*(X_k^*) - D_k}{1 - e_k^*(X_k^*)} \\ & \times \left[\left(r_k(X_1^*) r_k^L(X_k^*) z_k^{L-}(X_k^*) + \frac{q^0(1 - 1/p(X_1^*), X_1^*)}{p(X_1^*)} (r_1(X_1^*) - r_k(X_1^*)) \right) \right. \\ & \quad \left. + q^0(1 - 1/p(X_1^*), X_1^*) r_k(X_1^*) (1/p(X_1^*) - r_k^L(X_k^*)) \right] \end{aligned}$
$\psi_U^-(W, \zeta_0)$	$\begin{aligned} & \frac{Z_1}{e_1^*(X_1^*)} (1 - D^+) Y - \Pi_{j=1}^{\bar{w}} \frac{1 - Z_j}{1 - e_j^*(X_j^*)} (1 - D^+) Y 1_{\{Y < q^0(1/p(X_1^*), X_1^*)\}} \\ & - q^0(1/p(X_1^*), X_1^*) \left[\frac{Z_1}{e_1^*(X_1^*)} (1 - D^+ - r^+(X_1^*)) \right. \\ & \quad - \Pi_{j=1}^{\bar{w}} \frac{1 - Z_j}{1 - e_j^*(X_j^*)} \frac{1}{p(X_1^*)} (1 - D^+ - r_1(X_1^*)) \\ & \quad \left. - \Pi_{j=1}^{\bar{w}} \frac{1 - Z_j}{1 - e_j^*(X_j^*)} (1 - D^+) (1_{\{Y < q^0(1/p(X_1^*), X_1^*)\}} - 1/p(X_1^*)) \right] \\ & \quad - \frac{Z_1 - e_1^*(X_1^*)}{e_1^*(X_1^*)} \beta^+(1, X_1^*) r^+(X_1^*) \\ & \quad + \sum_{k=1}^{\bar{w}} 1_{\{A \geq k\}} \Pi_{j=1}^{k-1} \frac{1 - D_j}{1 - e_j^*(X_j^*)} \frac{e_k^*(X_k^*) - D_k}{1 - e_k^*(X_k^*)} \\ & \times \left[\left(r_k(X_1^*) r_k^U(X_k^*) z_k^{U-}(X_k^*) + \frac{q^0(1/p(X_1^*), X_1^*)}{p(X_1^*)} (r_1(X_1^*) - r_k(X_1^*)) \right) \right. \\ & \quad \left. + q^0(1/p(X_1^*), X_1^*) r_k(X_1^*) (1/p(X_1^*) - r_k^U(X_k^*)) \right] \end{aligned}$
$\psi^{R+}(G, \zeta^0)$	$r^+(X_1^*) + (1 - D^+ - r^+(X_1^*)) \frac{Z_1}{e_1^*(X_1^*)}$
Nuisance functions	
$\zeta^0(x_1^*, \dots, x_{\bar{w}}^*)$	$\{e_1^*(x_1^*), \dots, e_{\bar{w}}^*(x_{\bar{w}}^*), r_1(x_1^*), \dots, r_{\bar{w}}^*(x_{\bar{w}}^*), r^+(x_1^*), q(p(x_1^*), x_1^*), q(1 - p(x_1^*), x_1^*), \\ \beta^1(x_1^*), \dots, \beta_{\bar{w}}^*(x_{\bar{w}}^*), \beta^+(x_1^*), z^{U+}(x_1^*), z^{L+}(x_1^*), z^{U-}(x_1^*), \dots, z_{\bar{w}}^{U-}(x_{\bar{w}}^*), q^0(1/p(x_1^*), x_1^*), \\ q^0(1 - 1/p(x_1^*), x_1^*), z_1^{L-}(x_1^*), \dots, z_{\bar{w}}^{L-}(x_{\bar{w}}^*), r_1^L(x_1^*), \dots, r_{\bar{w}}^L(x_{\bar{w}}^*), r_1^U(x_1^*), \dots, r_{\bar{w}}^U(x_{\bar{w}}^*)\}$
$q^0(u, x)$	$\inf\{q : u \leq \mathbb{E}[1_{\{Y \leq q\}} / \Pi_{j=2}^{\bar{w}} (1 - e_j^*(X_j^*)) \mid X_1^* = x, D = 0] / \mathbb{E}[\Pi_{j=2}^{\bar{w}} (1 - e_j^*(X_j^*)) \mid X_1^* = x, D = 0]\}$
$z_k^{L-}(x)$	$\mathbb{E}[Y / \Pi_{j=k+1}^{\bar{w}} (1 - e_j^*(X_j^*)) \mid Y \geq q^0(1 - 1/p(X_1^*), X_1^*), D = 0, X_k^* = x]$
$z_k^{U-}(x)$	$\mathbb{E}[\Pi_{j=k+1}^{\bar{w}} (1 - e_j^*(X_j^*)) \mid Y \geq q^0(1 - 1/p(X_1^*), X_1^*), D = 0, X_k^* = x]$
$r_k^L(x)$	$\mathbb{E}[Y / \Pi_{j=k+1}^{\bar{w}} (1 - e_j^*(X_j^*)) \mid Y \leq q^0(1/p(X_1^*), X_1^*), D = 0, X_k^* = x]$
$r_k^U(x)$	$\mathbb{E}[\Pi_{j=k+1}^{\bar{w}} (1 - e_j^*(X_j^*)) \mid Y \leq q^0(1/p(X_1^*), X_1^*), D = 0, X_k^* = x]$

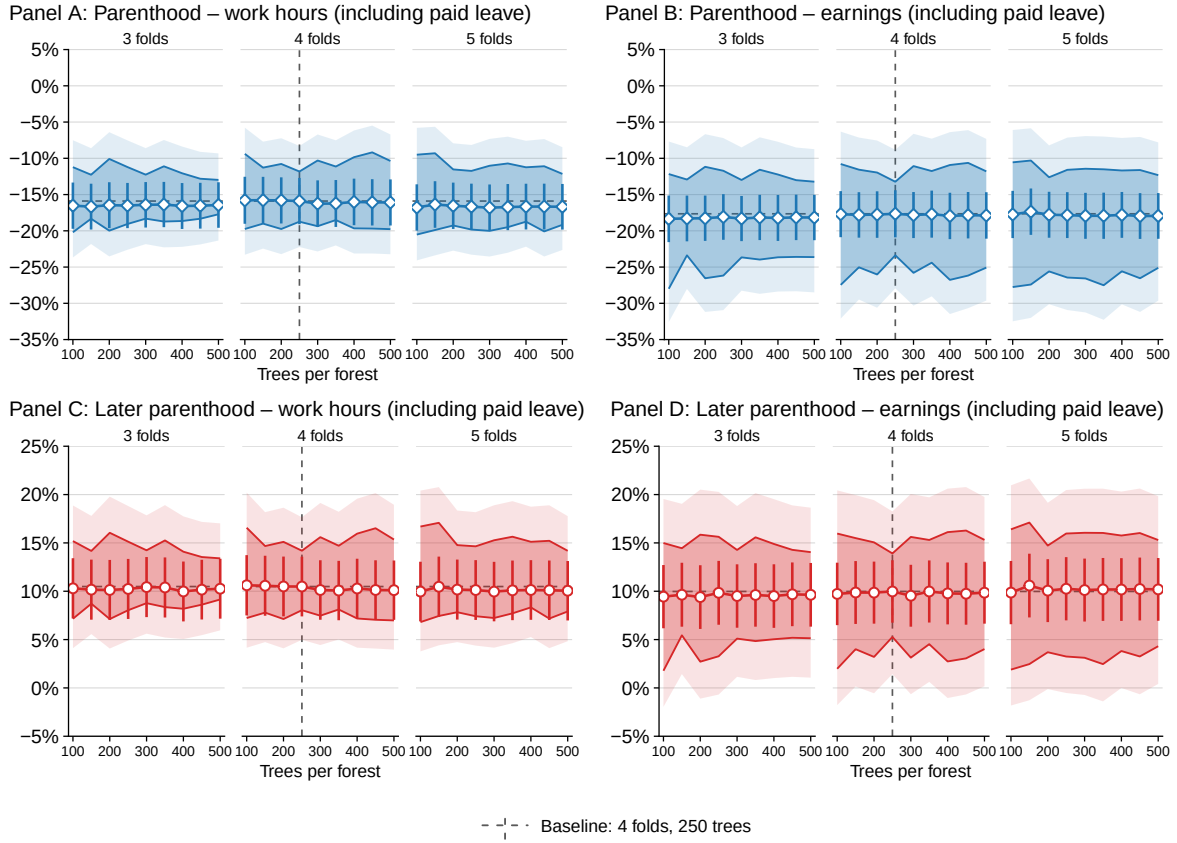


Figure SA6: Sensitivity to Cross-Fitting Folds and Forest Size

Note: Point estimates under conditionally random reliance, cumulated over years 1–6 and expressed as a percentage of the counterfactual level, over a grid of three cross-fitting fold counts and nine forest sizes. The baseline uses 4 folds and 250 trees and is marked separately.

SA2.6 Cross-Fitting Folds and Forest Size

The baseline uses 4-fold cross-fitting and generalized random forests with 250 trees. Figure SA6 re-estimates the effects over a grid of three fold counts and nine tree counts. The point estimate for the effect of parenthood on work hours ranges from -16.8% to -15.8% across the grid, against -15.9% in the baseline, and on earnings from -18.3% to -17.4% , against -17.7% . The timing point estimates range from 10.0% to 10.6% for work hours and from 9.4% to 10.6% for earnings, against 10.5% and 10.0% . The largest deviation from the baseline anywhere on the grid is 0.9 percentage points.

SA2.7 Number of Procedures Used

The baseline uses data on the first ten procedures. Figure SA7 re-estimates the effects using between nine and thirteen. For the effect of parenthood on work hours, the point estimate moves from -15.9% at ten procedures to -16.0% at nine and -17.5% at thirteen, and for earnings from -17.7% to -17.8% and -18.2% . The timing point estimates are similarly stable, moving from 10.5% to between 8.8% and 10.5% for work hours and from 10.0% to between 9.0% and 10.3% for earnings. The largest deviation from the baseline in any of the four panels is 1.7 percentage

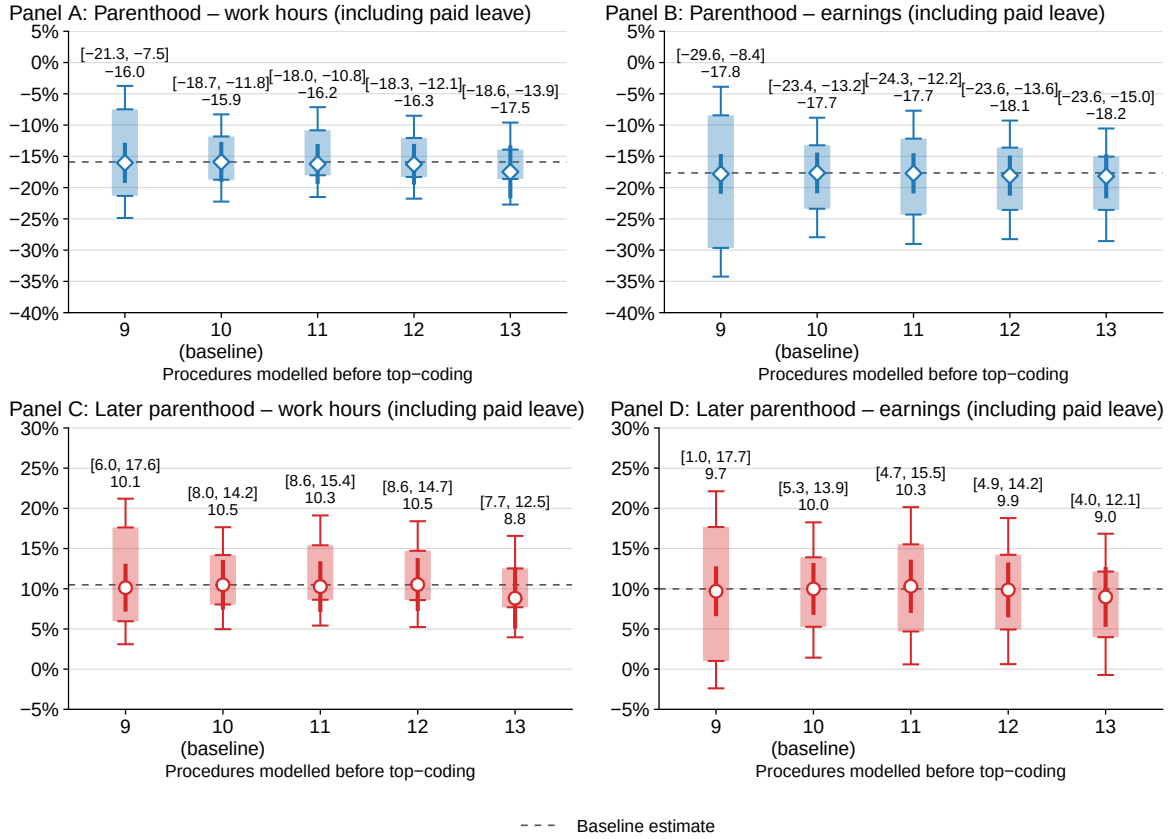


Figure SA7: Sensitivity to the Number of Procedures Used

Note: Effects cumulated over years 1–6 and expressed as a percentage of the counterfactual level, estimated using data on between nine and thirteen procedures. Ten is the baseline. Translucent bars give the identified set, whiskers its 95% confidence interval, and markers the point estimate under conditionally random reliance.

points, at thirteen procedures, where the nuisance functions are estimated on small samples. Using fewer procedures mechanically widens the bounds because it defines a smaller group of reliers and therefore requires more trimming. For example, truncating at nine procedures widens the identified set for work hours to $[-21.3\%, -7.5\%]$, compared with $[-18.7\%, -11.8\%]$ at ten. This is why I use ten procedures throughout.

SA2.8 Work Hours for Unobserved Leave

I define maximum-leave-adjusted hours by scaling women's reported work hours in each child-birth year by $36/52$, accounting for up to 16 weeks of leave. Figure SA8 shows results for female work hours. The estimates change little beyond the first year of parenthood: the effect of parenthood moves by 11.6 percentage points in year 0 and 17.4 points in year 1, but by at most 2.7 points in years 2 through 5 and not at all in year 6. The timing estimates behave the same way, moving by 9.9 and 27.0 points in years 0 and 1 and by at most 3.8 points thereafter.

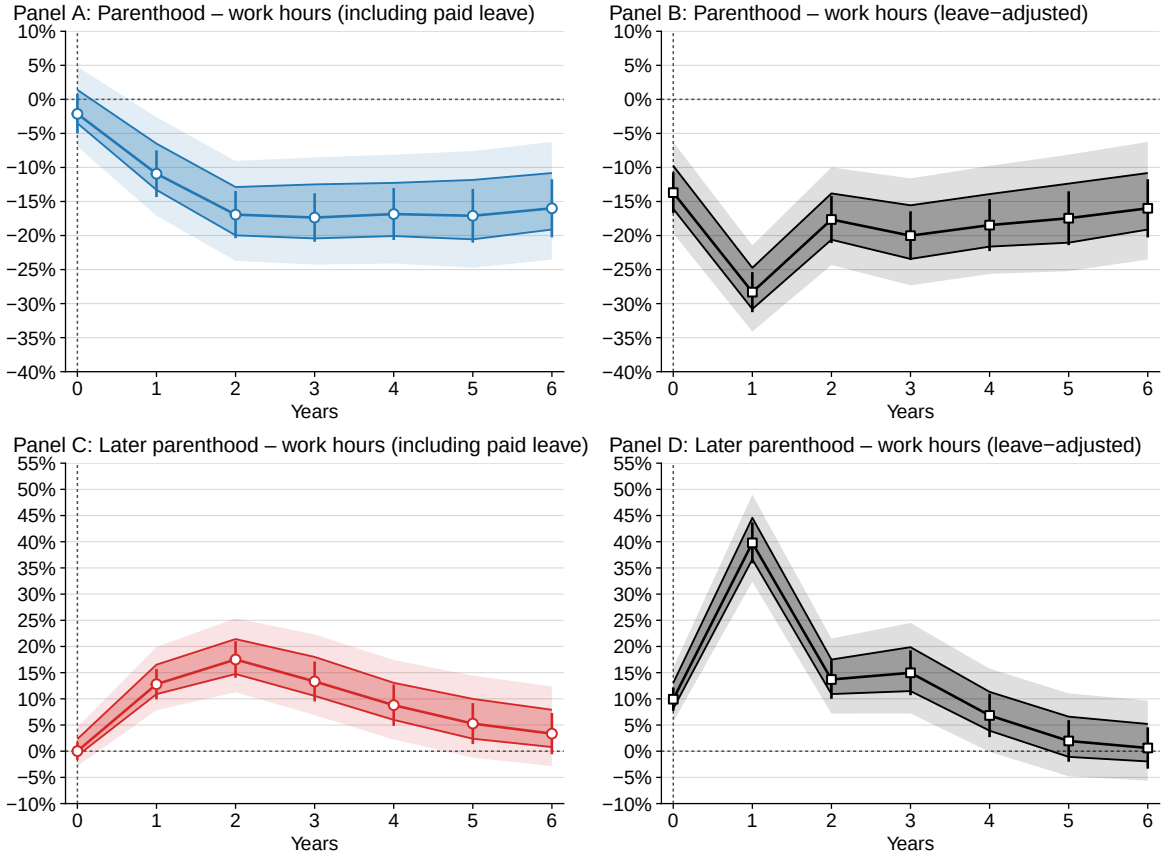


Figure SA8: Work Hours Adjusted for Unobserved Parental Leave

Note: Effects of parenthood and of later parenthood on women's annual work hours, by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level. *Reported hours* are as recorded in the data and include paid leave; *leave adjusted* scales reported hours in each childbirth year by 36/52, accounting for up to sixteen weeks of leave. Since the control group consists of women without children, effects on actual hours lie within the union of the two sets of bounds. Translucent bands give the identified set and the fainter outer band its 95% confidence interval; markers give the point estimate under conditionally random reliance.

SA2.9 Controls for the Clinical Protocol

Section 5.6 reports per-period point estimates under three progressively richer sets of controls for the clinical protocol. Figure SA9 reports the corresponding bounds on the window totals. For the effect of parenthood on work hours the point estimate moves from -15.9% in the baseline to between -16.1% and -15.7% , and the identified set from $[-18.7\%, -11.8\%]$ to at widest $[-20.3\%, -9.7\%]$. For earnings the point estimate moves from -17.7% to between -17.8% and -17.3% . The timing point estimates move from 10.5% to between 10.0% and 10.2% for work hours and from 10.0% to between 9.8% and 10.3% for earnings. No point estimate moves by more than 0.5 percentage points.

SA2.10 Bounds under the Selection Adjustment

Section 5.6 reports per-period point estimates under the selection adjustment. Figure SA10 reports the corresponding bounds on the window totals. For the effect of parenthood on work hours the identified set moves from $[-18.7\%, -11.8\%]$ in the baseline to $[-20.0\%, -13.2\%]$ under mild

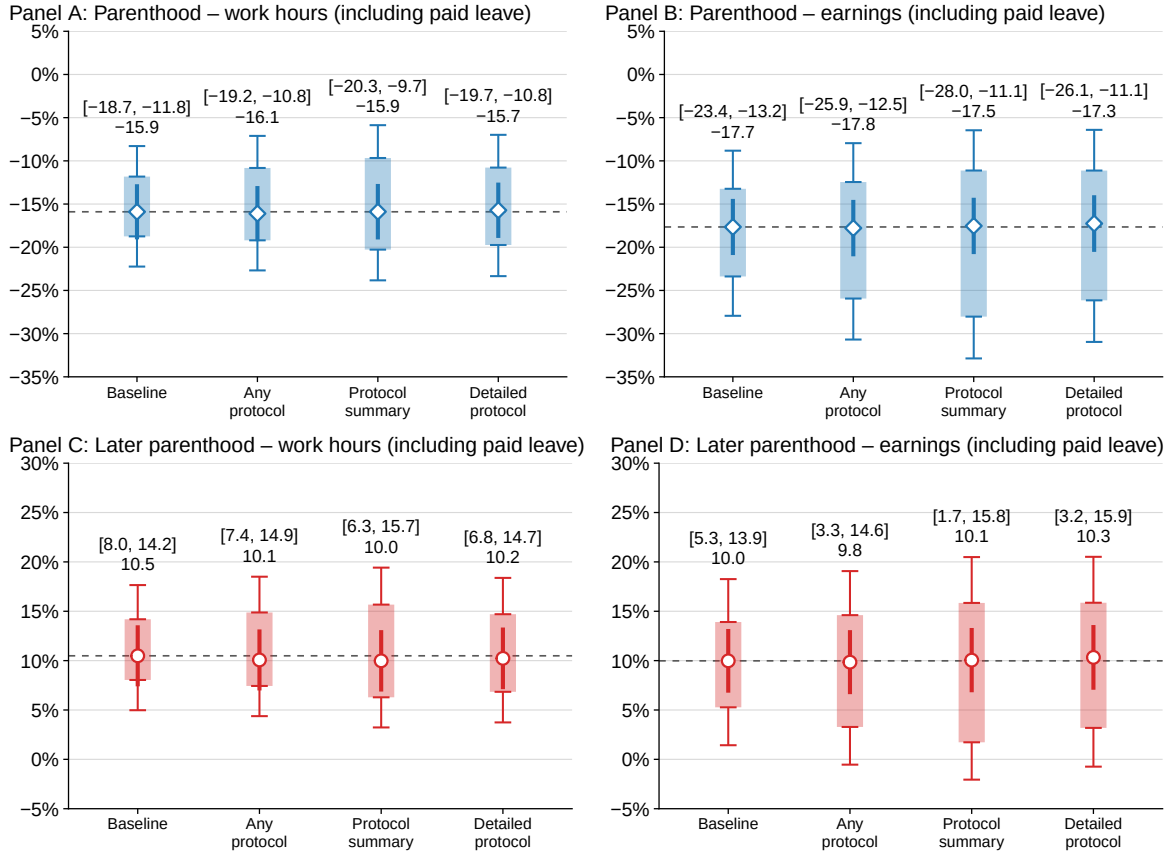


Figure SA9: Bounds under Controls for the Clinical Protocol

Note: Effects cumulated over years 1–6 and expressed as a percentage of the counterfactual level. *Baseline* predicts success from both partners' age and education and the procedure type; *any protocol* adds an indicator for whether the stimulation protocol is recorded, *protocol summary* a coarse classification of the regime, and *detailed protocol* the full regime and its intensity. Translucent bars give the identified set, whiskers its 95% confidence interval, and markers the point estimate under conditionally random reliance. Figure 12 reports the corresponding per-period estimates.

selection and $[-22.5\%, -15.9\%]$ under strong selection, and for earnings from $[-23.4\%, -13.2\%]$ to $[-24.6\%, -14.6\%]$ and $[-26.9\%, -17.2\%]$. The timing effects move from $[8.0\%, 14.2\%]$ to $[9.7\%, 15.9\%]$ and $[13.2\%, 19.6\%]$ for work hours, and from $[5.3\%, 13.9\%]$ to $[7.0\%, 15.8\%]$ and $[10.5\%, 19.6\%]$ for earnings. The adjustment shifts the sets rather than widening them, and in every case toward larger effects.

SA2.11 Anchoring the Estimated Paths

The three paths in Figure 15 are constructed by anchoring on the estimated childless relier level series and scaling it by the estimated effects. Figure SA11 repeats the exercise using three alternatives: the relier childless level used in the main analysis, the full-sample early-motherhood level, and the non-relier early-motherhood level. The results are virtually unchanged.

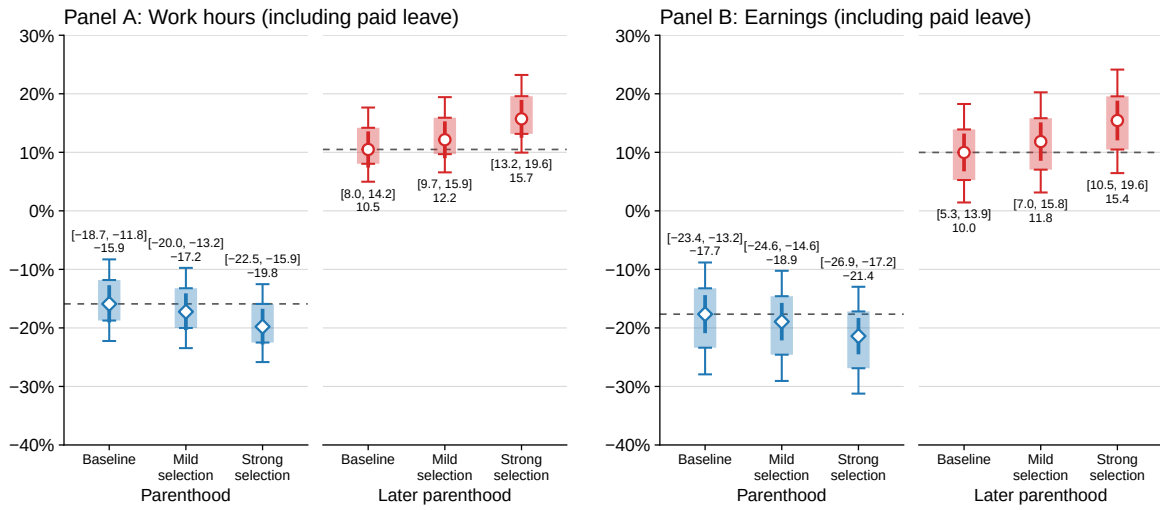


Figure SA10: Bounds under the Selection Adjustment

Note: Effects cumulated over years 1–6 and expressed as a percentage of the counterfactual level, in the baseline and under the two selection-adjusted outcome series described in Section 5.6. Translucent bars give the identified set, whiskers its 95% confidence interval, and markers the point estimate under conditionally random reliance. Figure 13 reports the corresponding per-period estimates.

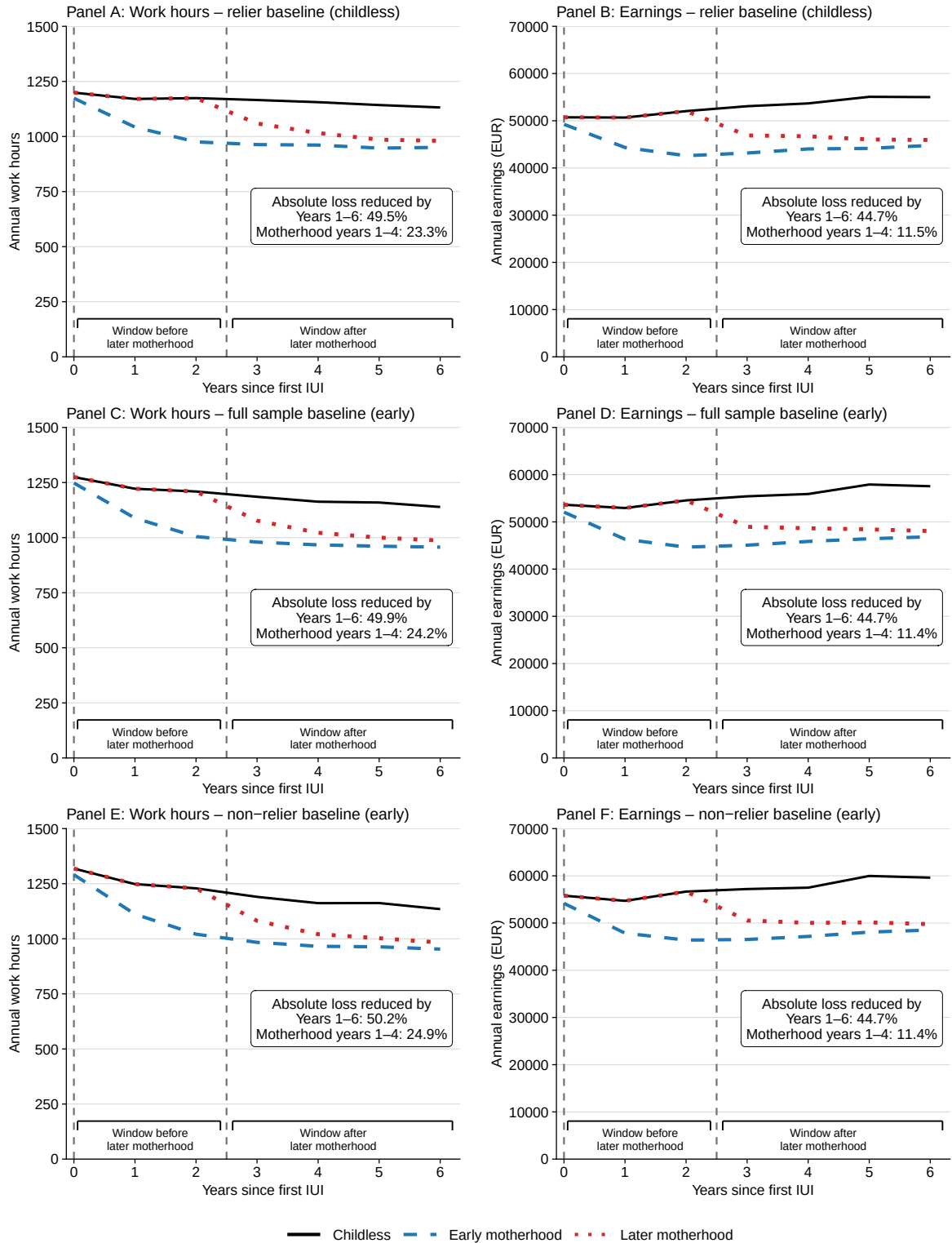


Figure SA11: Estimated Paths under Alternative Level Anchors

Note: The childless, early-motherhood and later-motherhood paths of Figure 15, constructed under three alternative level anchors: the relier childless level, which is used in the main figure, the full-sample early-motherhood level, and the non-relier early-motherhood level. In each case the anchor is scaled by the estimated effects of parenthood and of its timing, so the three paths compose by construction. Boxes report the shortfall of each mother's path against the childless path, cumulated over years 1–6 and over each woman's own first four years of motherhood, together with the implied reduction in the absolute loss.

SA3 Implementation of the Selection Test

Let X_j^P denote the covariates used by the baseline specification to predict success at procedure j : both partners' age and education and the procedure type. Let $p_j(X_j^P)$ denote the fitted success probability. Let X^B denote the balance-test covariates: employment, earnings, and hours for the woman and her partner, measured in the year before the first procedure. Let H_j indicate an unobserved binary type affecting success propensity. The outcome Y is cumulative earnings over the full observation window beginning with the year of the first IUI.

I use two residualized outcomes, both defined among women whose procedure succeeds. First, $\tilde{Y}_j^P$ is Y residualized with respect to the baseline propensity score $p_j(X_j^P)$ and determines type assignment. Selection is therefore introduced conditional on the baseline success model. Second, $\tilde{Y}_j^{PB}$ is Y residualized with respect to both X_j^P and X^B and is used to measure potential-outcome imbalances. The exercise thus asks whether selection beyond that captured by $p_j(X_j^P)$ can reproduce balance in X^B and, if so, how much imbalance in Y remains after both X_j^P and X^B are accounted for. This mirrors the observable balance test in Section 4.6 and the conditioning used in the main estimation, respectively.

A selection model is indexed by two parameters: the log odds ratio δ between the two unobserved types and the share q of the low type among women whose procedure succeeds. Type membership shifts the log odds of success by $+\delta q$ for the high type and $-\delta(1 - q)$ for the low type, so the two types differ by δ . I add a common constant to the log odds so that the implied weights reproduce the observed number of failures. This pins down the model-implied propensities $p_j(X_j^P, H_j)$. Among women whose procedure succeeds, I assign the share q with the lowest values of $\tilde{Y}_j^P$ to the low type and the remainder to the high type.

Under a candidate model, inverse-probability weighting of successful procedures must reproduce the covariate distribution among all women who reach procedure j . Hence, $\mathbb{E}[X^B(1 - Z_j/p_j(X_j^P, H_j))] = 0$. I test these six moment conditions jointly. In the pooled specification, the covariance matrix is clustered at the woman level because each woman may contribute several procedures.

The implied outcome imbalance is recovered from women whose procedure succeeds. Weighting these women by $(1 - p_j(X_j^P, H_j))/p_j(X_j^P, H_j)$, normalized to mean one, gives the model-implied distribution of the potential outcome among women whose procedure fails. The difference between the unweighted and weighted means of $\tilde{Y}_j^{PB}$ is therefore the implied success–failure gap. The overall sample mean weights the success and failure means by $\mathbb{E}[p_j(X_j^P)]$ and $1 - \mathbb{E}[p_j(X_j^P)]$.

The pooled specification assigns types within each procedure, then pools all procedures to test balance in observables and quantify the implied imbalance in potential outcomes.

Because Y is measured over the full window beginning with the year of the first IUI, outcomes for women who succeed at later procedures may include some time before motherhood. Most procedures occur only a few months apart, however, so these women spend nearly the entire window as mothers. The outcome window is also defined identically for all women.

The same exercise could also be applied to the later-motherhood and childless potential out-

comes, relevant for non-reliers and reliers, respectively. Extending the exercise to these outcomes would require varying success propensities jointly over the full procedure sequence. For tractability, I therefore restrict the analysis to potential outcomes observed following successful procedures.

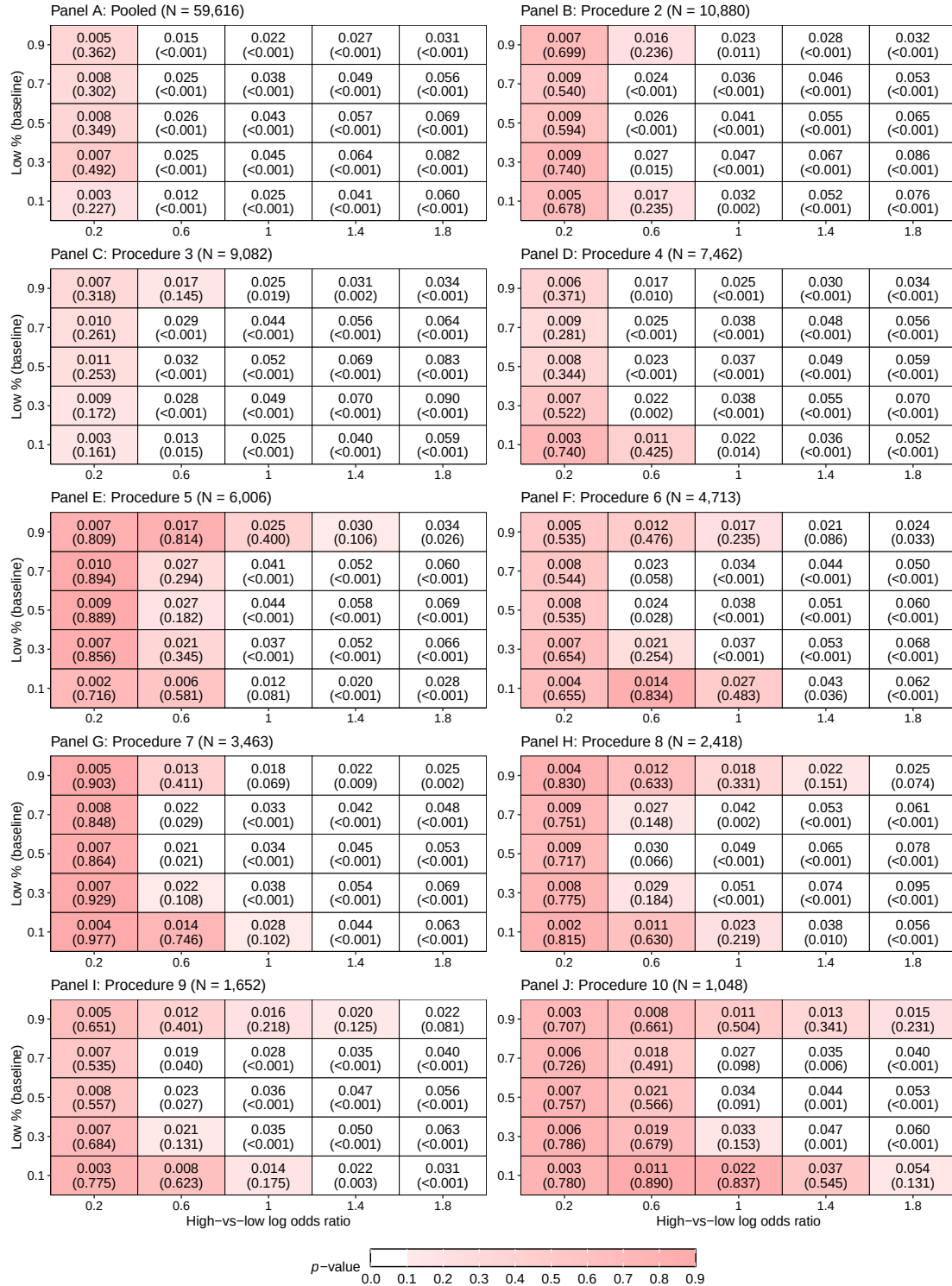


Figure SA12: Test for Selection on Unobservables Over the Procedure Sequence

Note: The outcome is cumulative earnings over the first six years following the first IUI. Cells and axes are defined as in Figure 3, Panel A which reports the first procedure; Section 3.6 describes the test. Panel A pools all ten procedures; Panels B–J report procedures 2 through 10 separately. In each panel the top number is the implied difference in potential outcomes between women whose procedure succeeds and the sample mean at that procedure, relative to that mean. The number in parentheses is the p -value from a joint test of whether the model is consistent with observed mean pre-IUI employment, earnings, and hours for the woman and her partner. The pooled p -value accounts for within-woman correlation across attempts.